

Minimisation

On numérote d'abord les sommets du graphe de x_1 à x_n où x_1 est le sommet origine et x_n le sommet final.

On pose $\lambda_1 = 0$ et $E_1 = \{ x_1 \}$;

A tout étape k ($k \geq 1$) du problème, on considère l'ensemble E_k des k sommets marqués (λ_i sont déjà déterminés) et, à tout sommet $x_i \in E_k$ on associe le sommet $x_i^* \notin E_k$, tel que $v(x_i, x_i^*)$ soit minimal ;

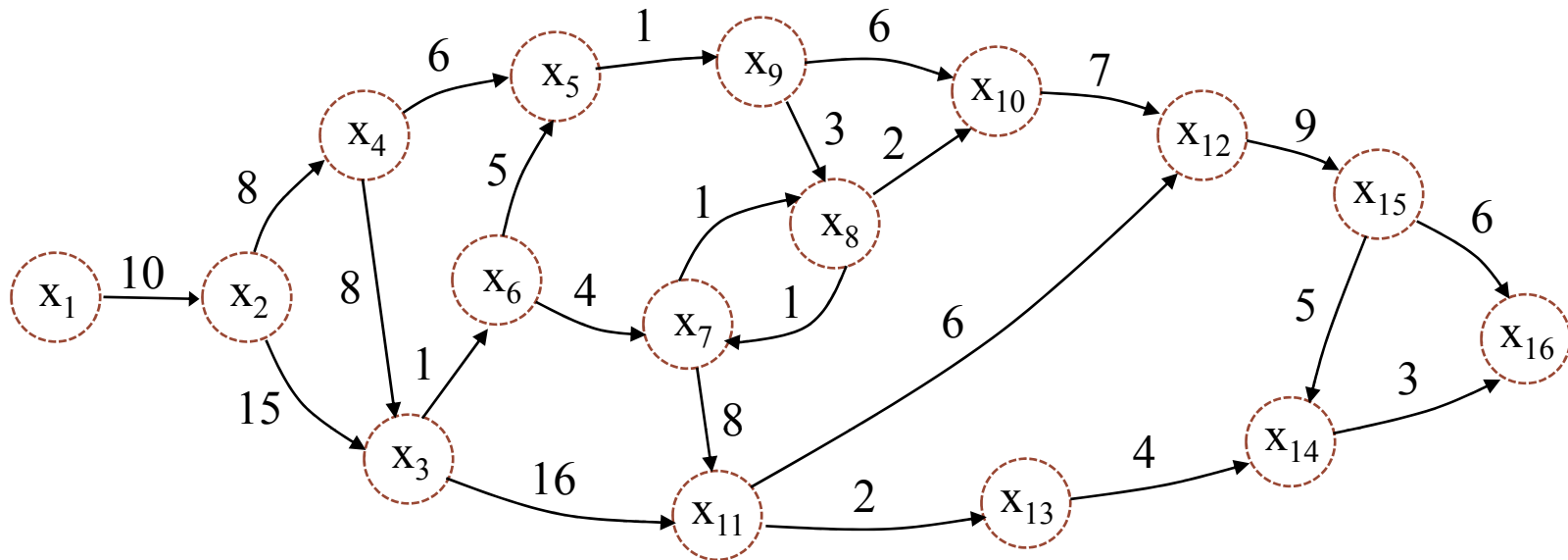
On détermine le sommet à marquer x_p^* ,

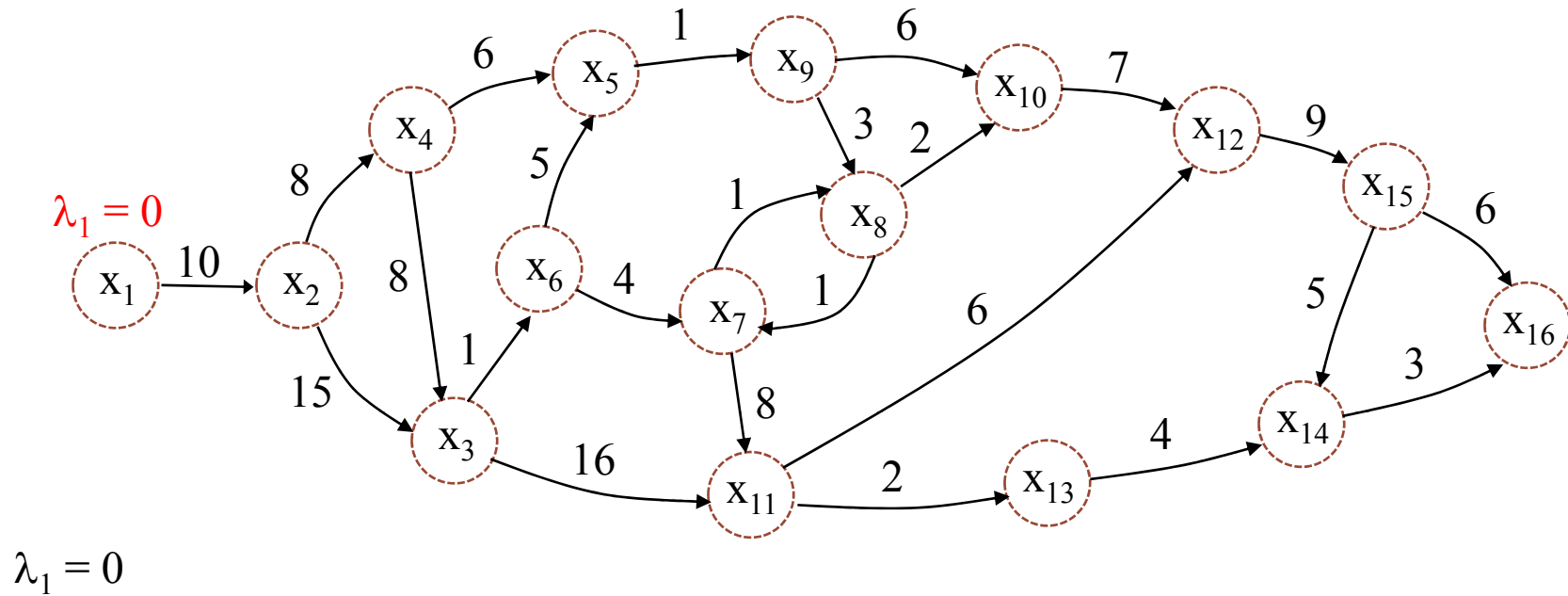
tel que $\lambda_p + v(x_p, x_p^*) = \min [\lambda_i + v(x_i, x_i^*)]$

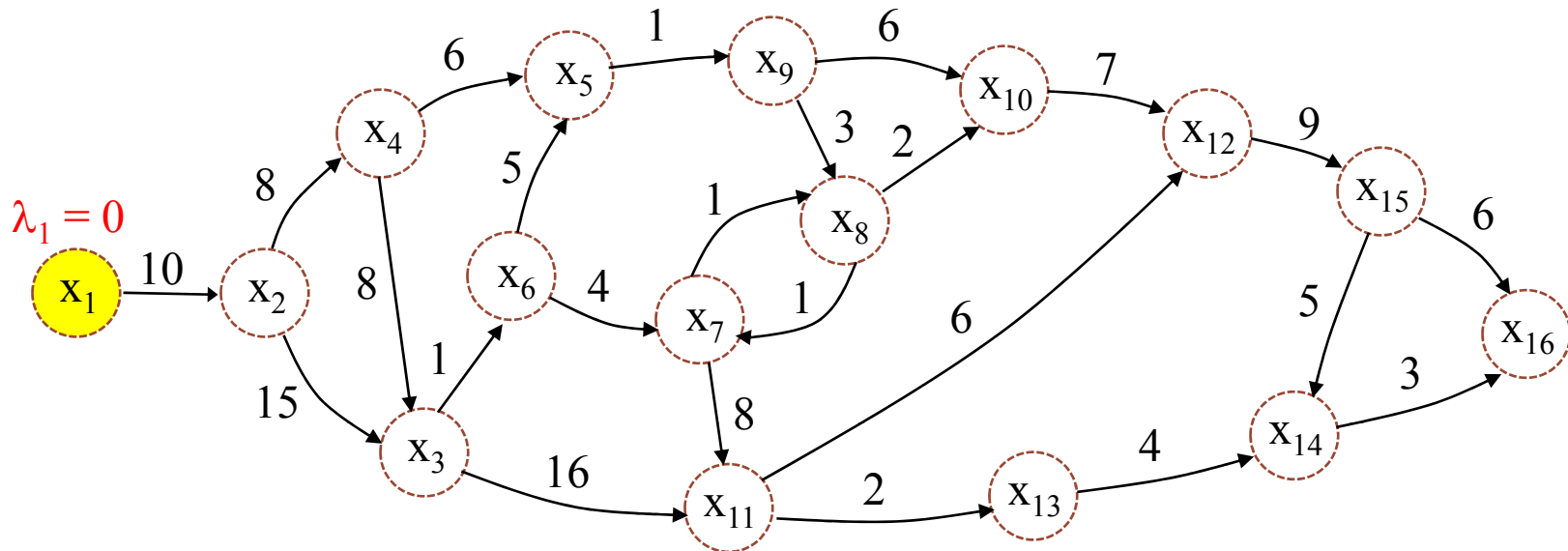
On pose : $E_{k+1} = E_k \cup \{ x_p^* \}$

Maximisation

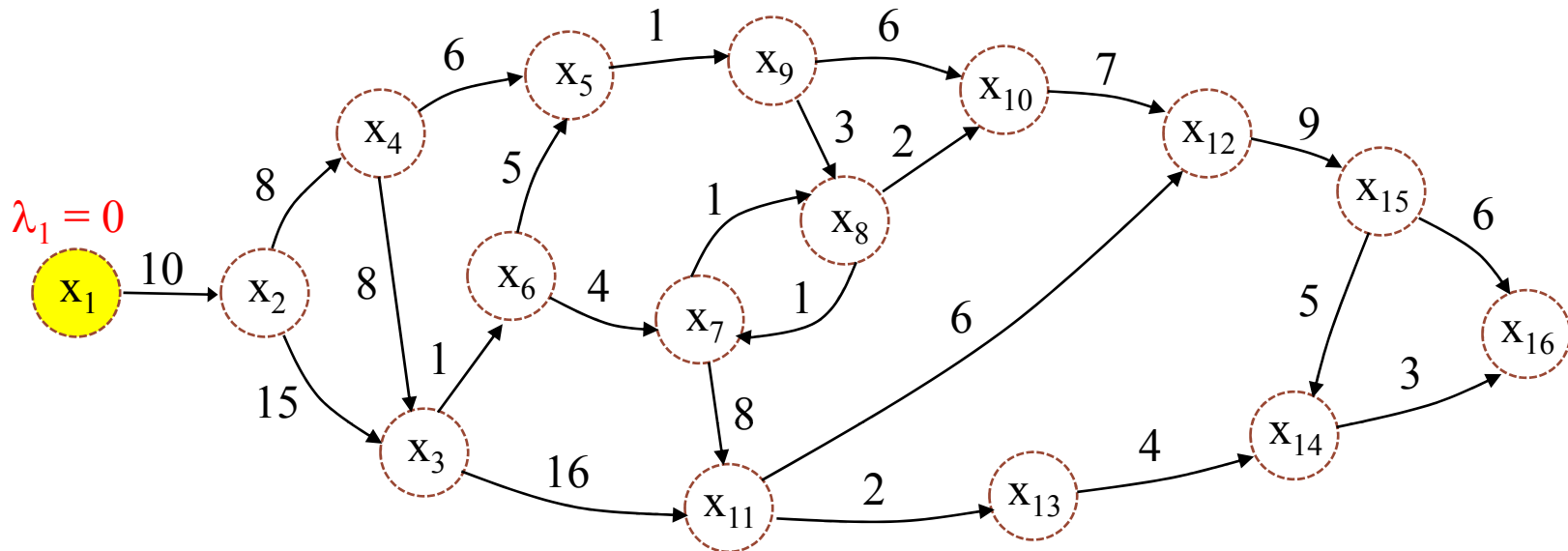






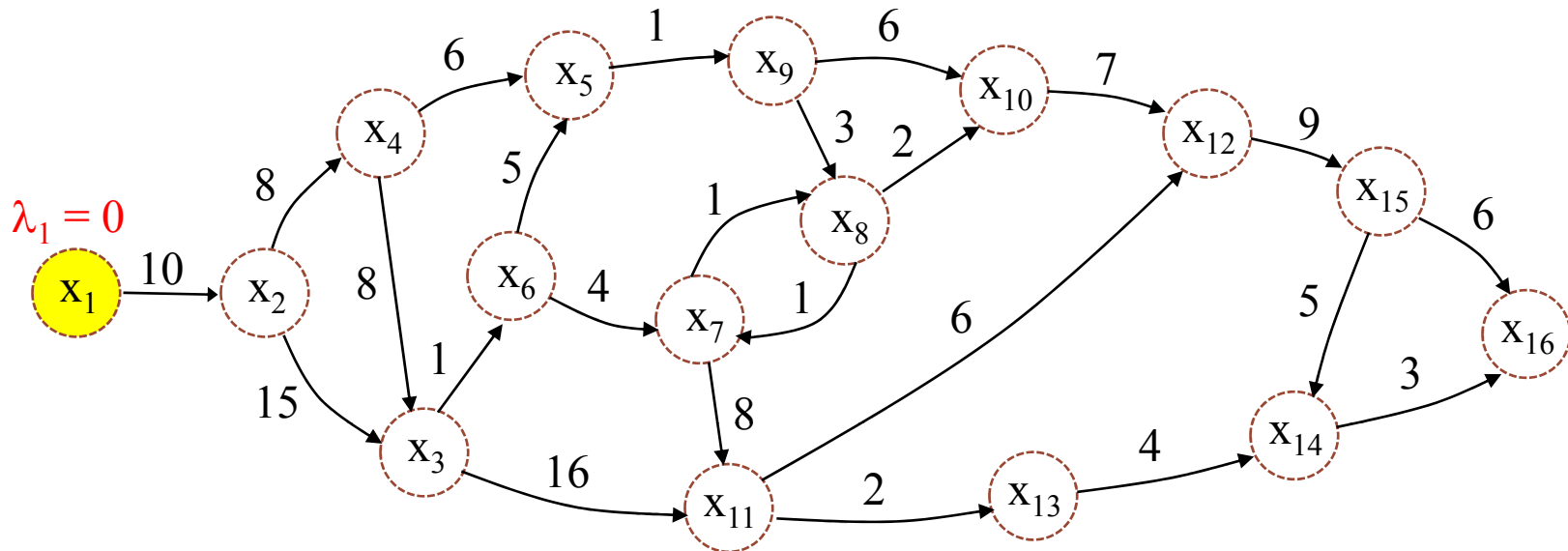


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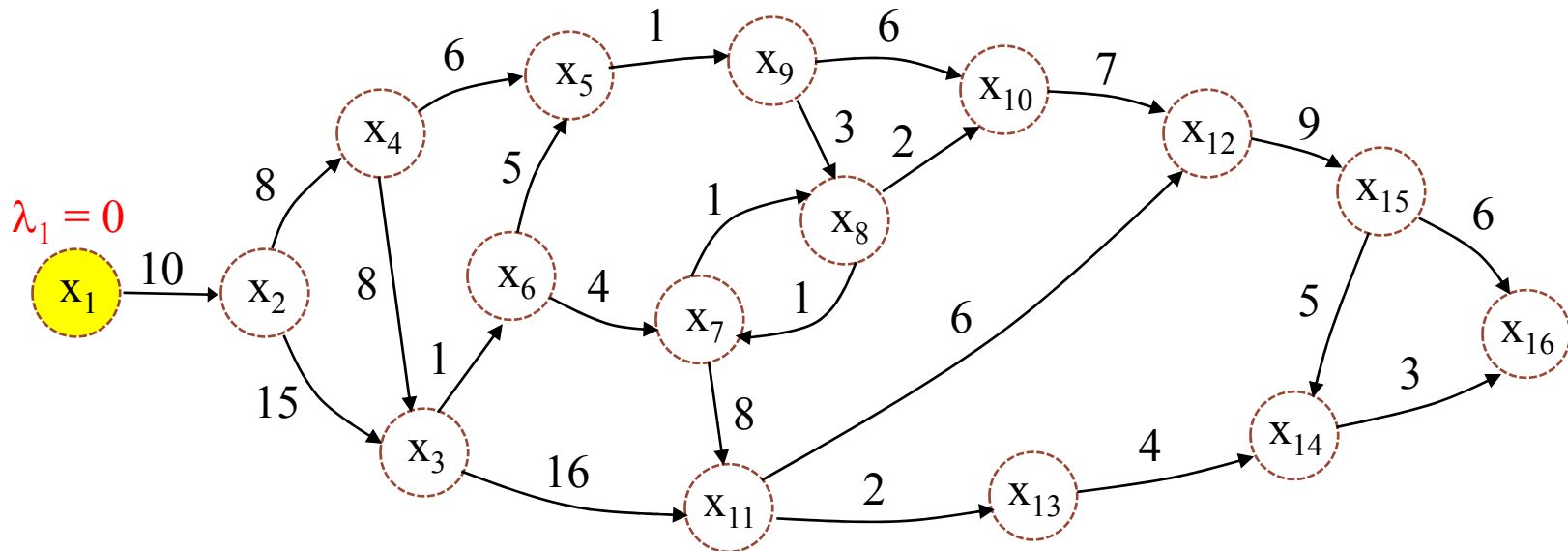
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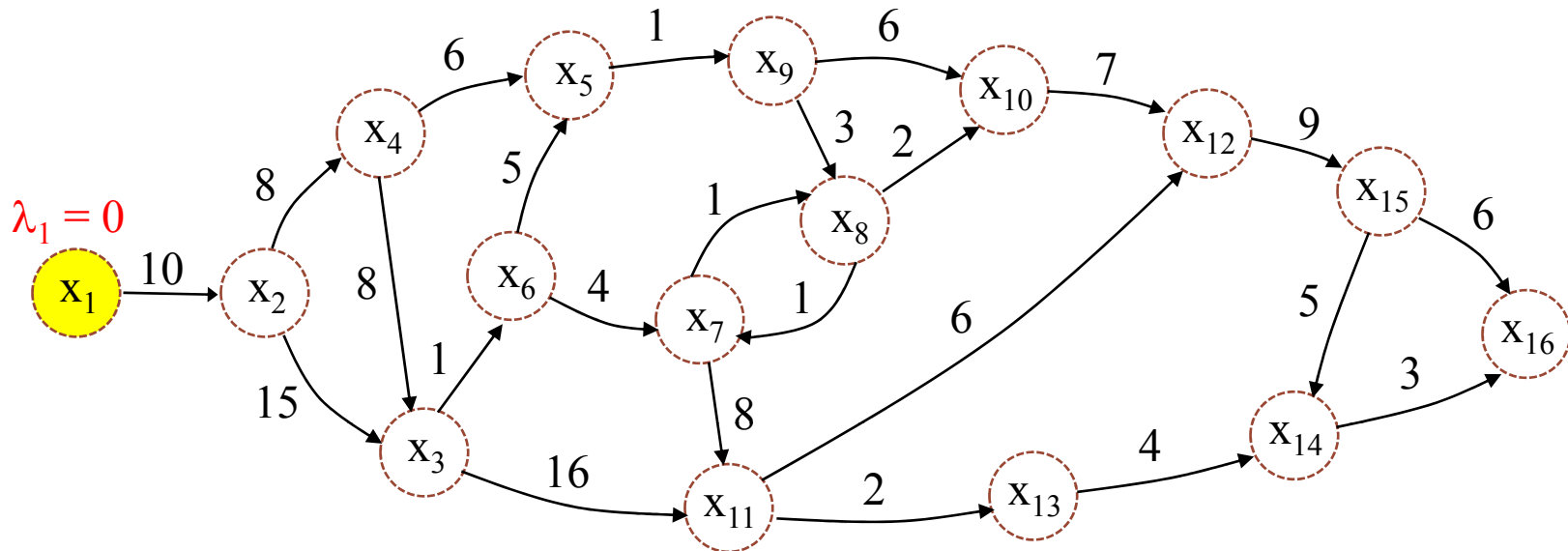
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$$\underline{k=1} \quad v(x_1, x_2)$$

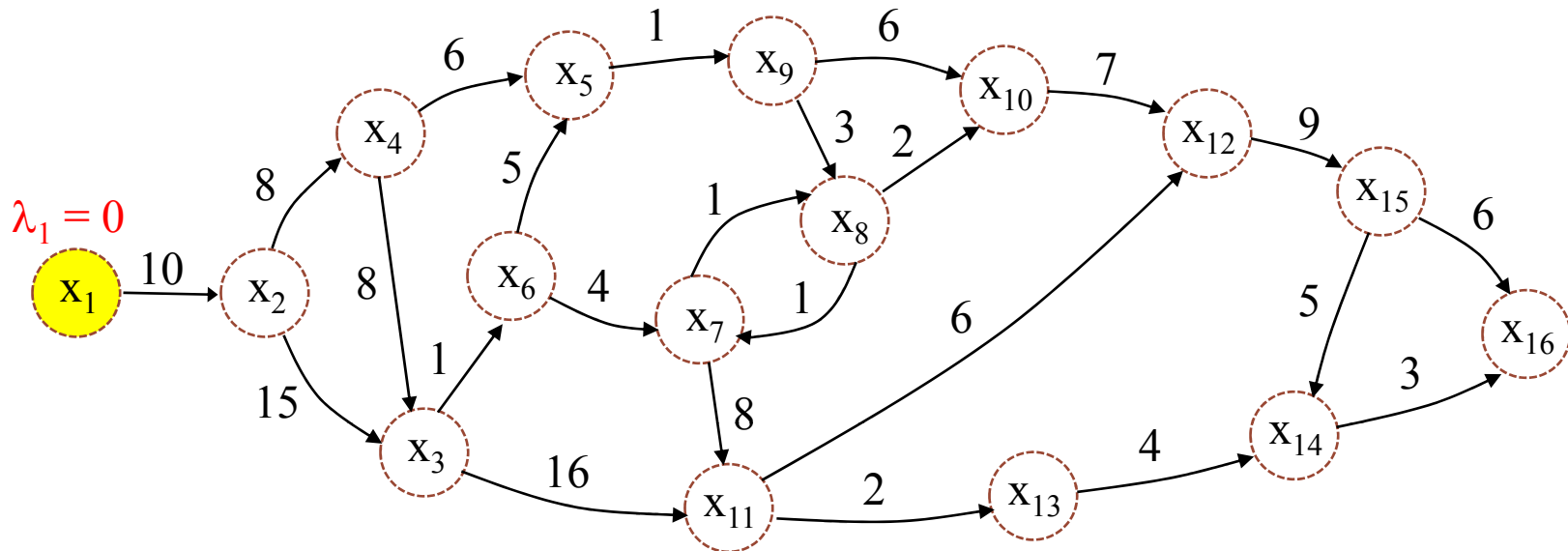
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$$\lambda_1 = 0 \quad E_1 = \{x_1\}$$

$$\underline{k=1} \quad v(x_1, x_2) \quad \lambda_2 = \lambda_1 + v(x_1, x_2) = 10$$

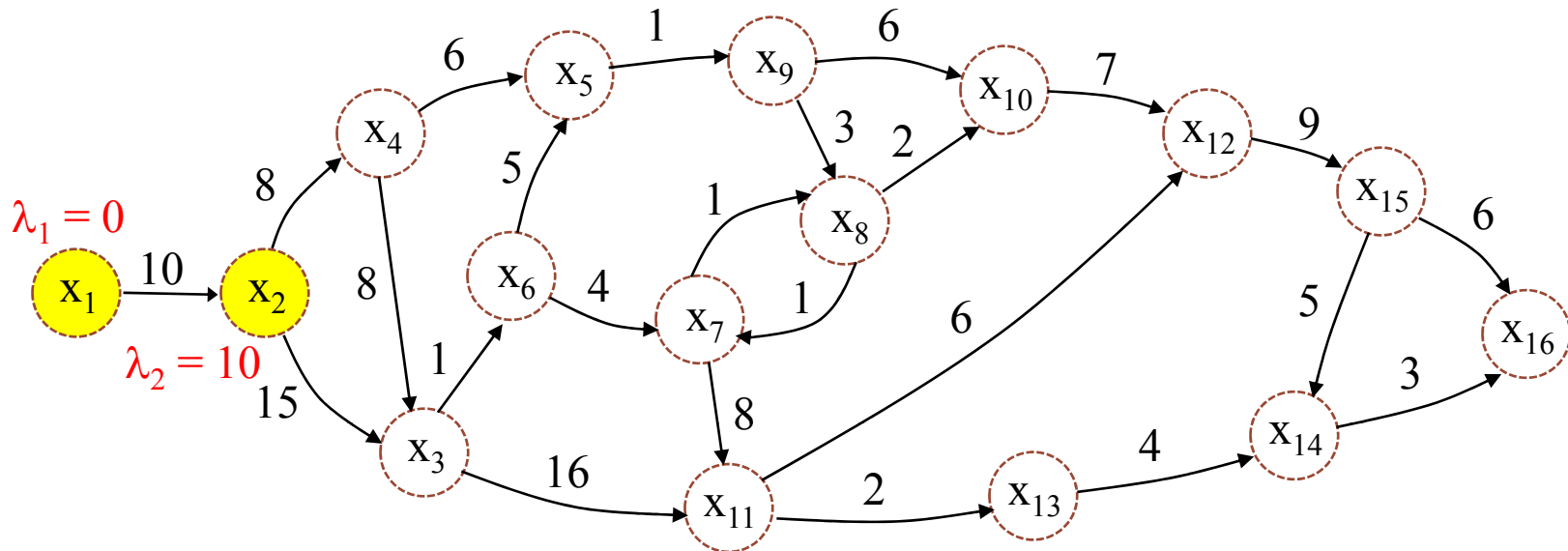
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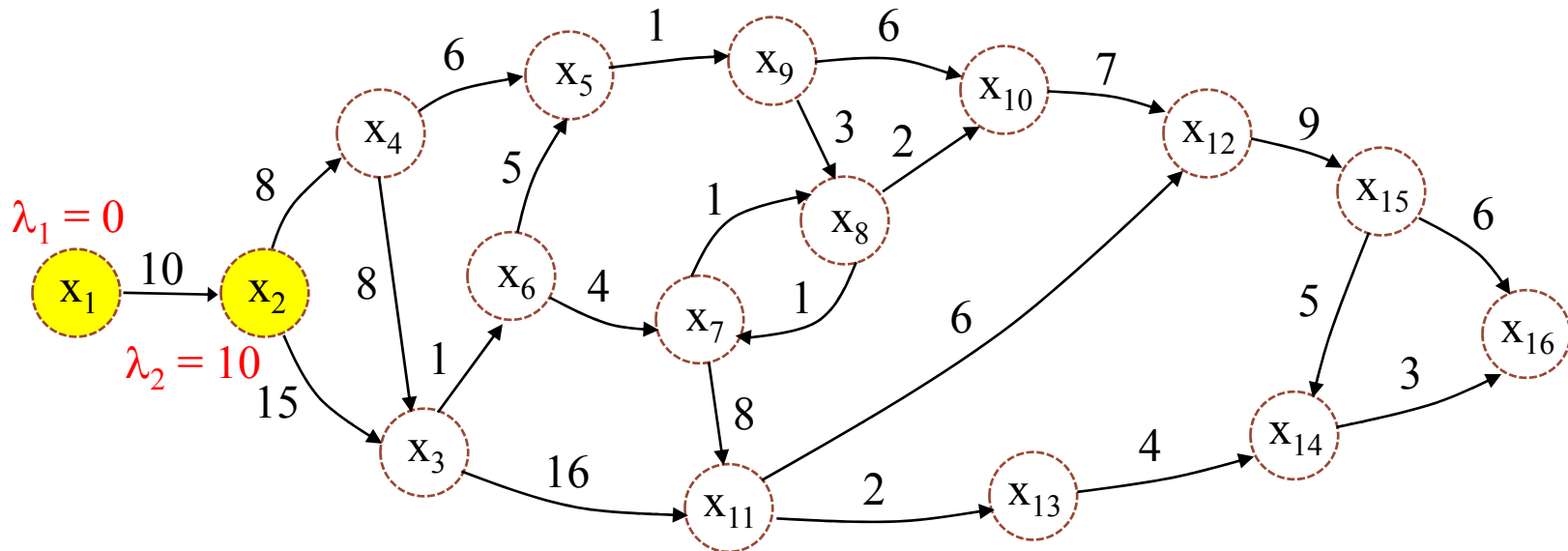
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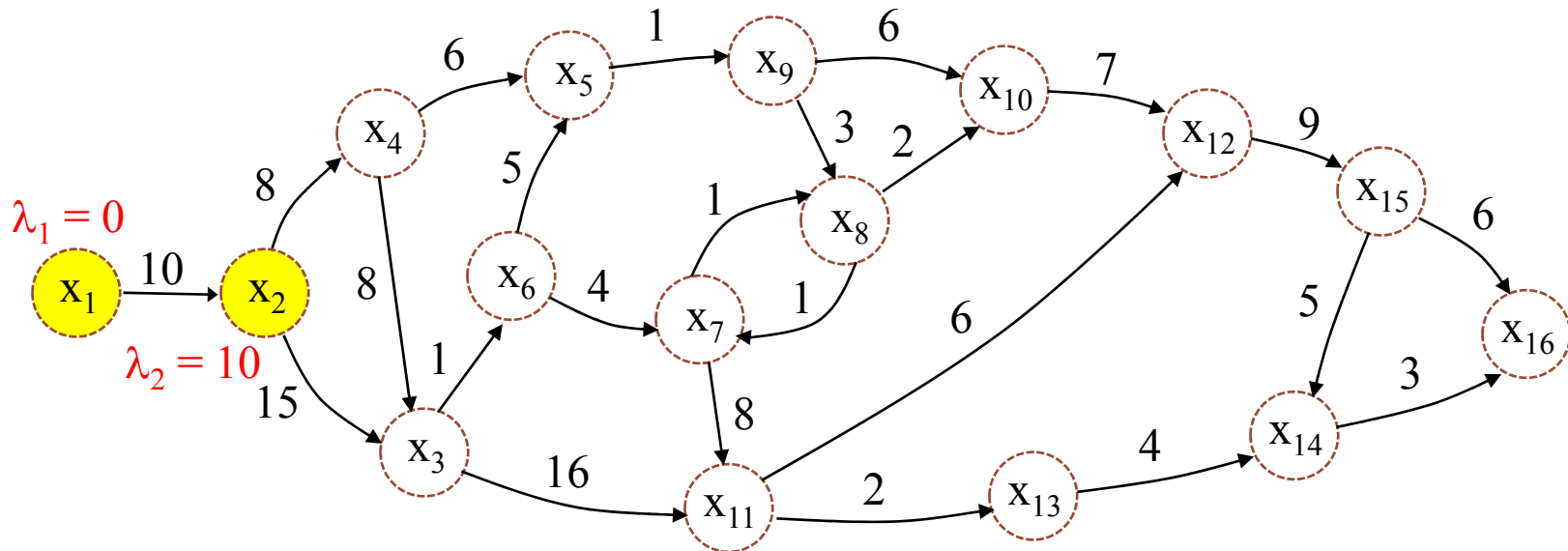


$$\lambda_1 = 0 \quad E_1 = \{ x_1 \}$$

$$\underline{\mathbf{k} = 1} \quad v(x_1, x_2) \quad \lambda_2 = \lambda_1 + v(x_1, x_2) = 10 \quad E_2 = \{ x_1, x_2 \}$$

$$\underline{\mathbf{k} = 2}$$

à tout sommet $x_i \in E_k$ on associe le sommet $x_i^* \notin E_k$, tel que $v(x_i, x_i^*)$ soit minimal ;
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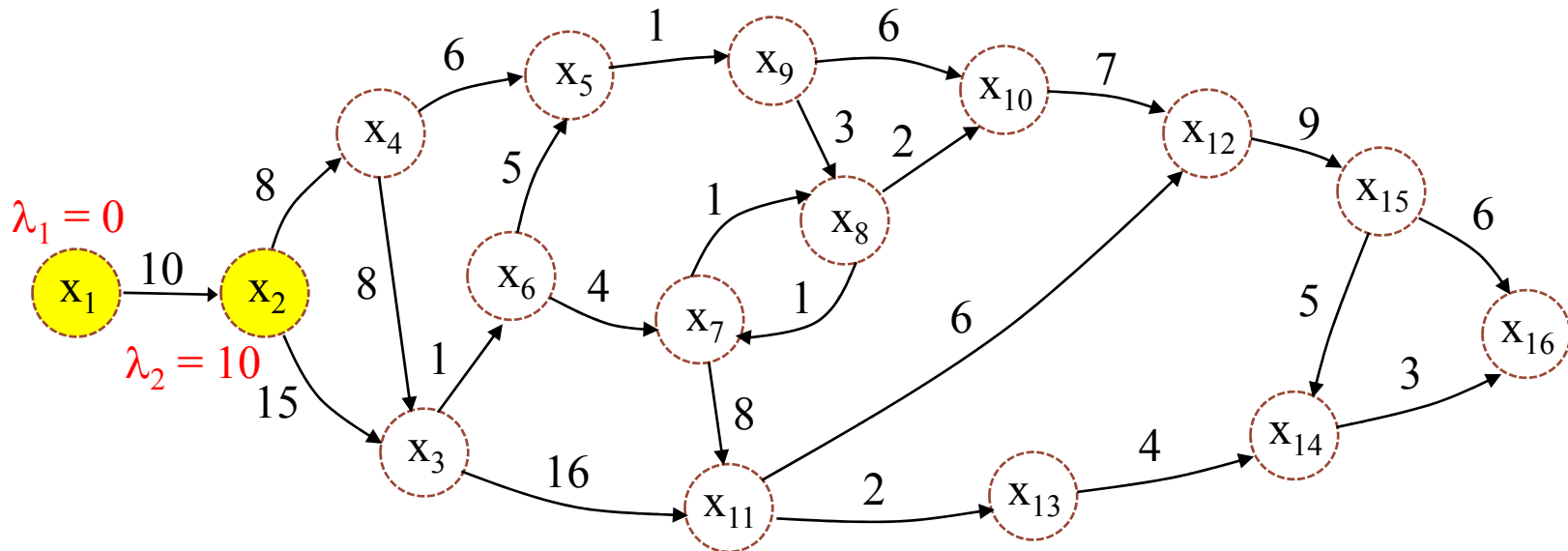


$$\lambda_1 = 0 \quad E_1 = \{ x_1 \}$$

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$$\underline{\mathbf{k}} = 2 \quad v(x_2, x_4)$$

à tout sommet $x_i \in E_k$ on associe le sommet $x_i^* \notin E_k$, tel que $v(x_i, x_i^*)$ soit minimal ;
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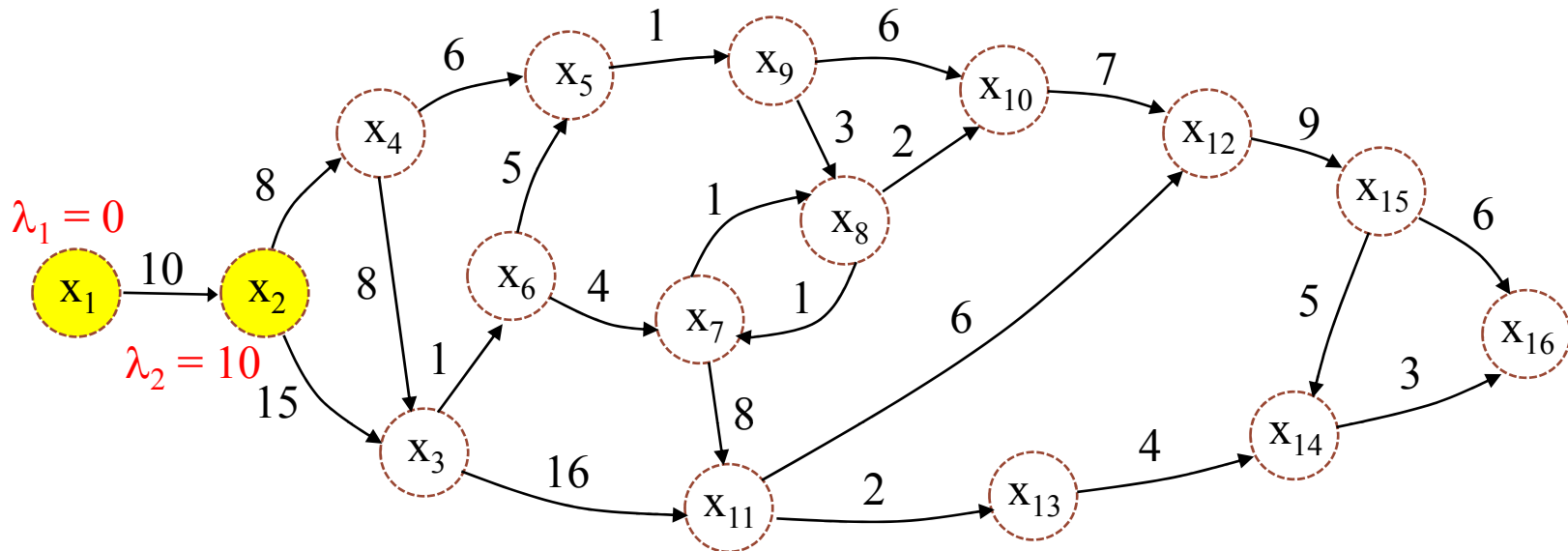


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$$\underline{\mathbf{k}} = 2 \quad v(x_2, x_4) \quad \lambda_4 = \lambda_2 + v(x_2, x_4) = 18$$

à tout sommet $x_i \in E_k$ on associe le sommet $x_i^* \notin E_k$, tel que $v(x_i, x_i^*)$ soit minimal ;
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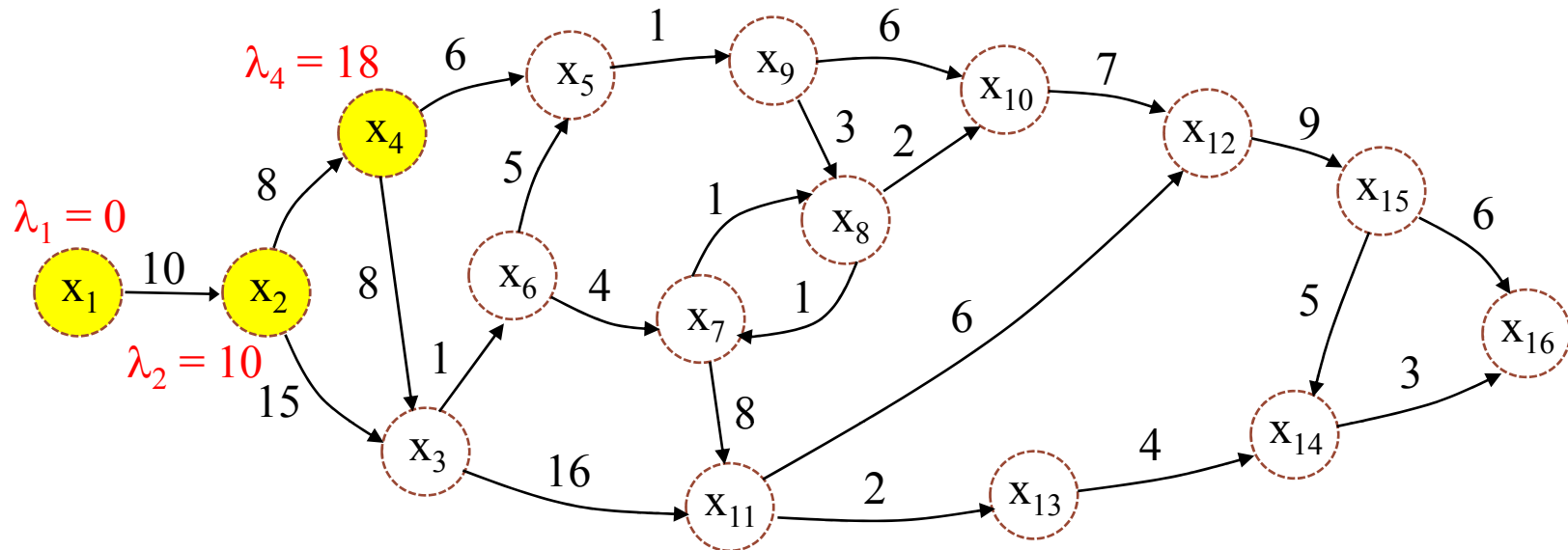


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$$\underline{k=2} \quad v(x_2, x_4) \quad \lambda_4 = \lambda_2 + v(x_2, x_4) = 18 \quad E_3 = \{ x_1, x_2, x_4 \}$$

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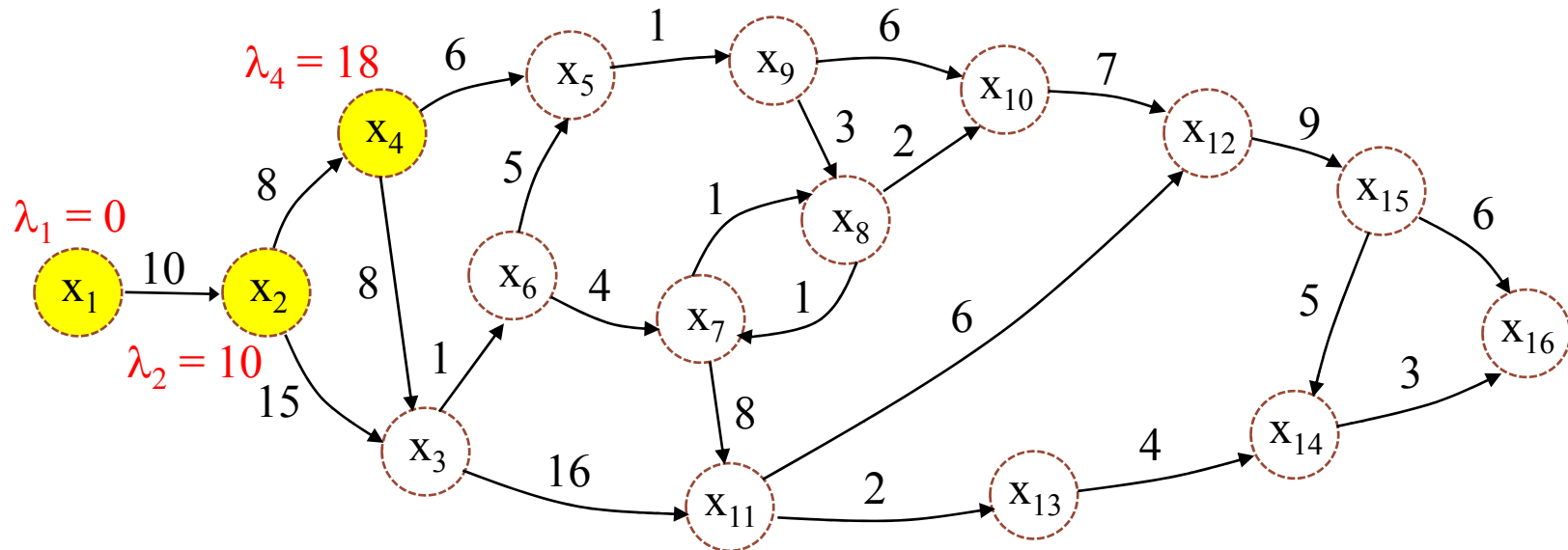


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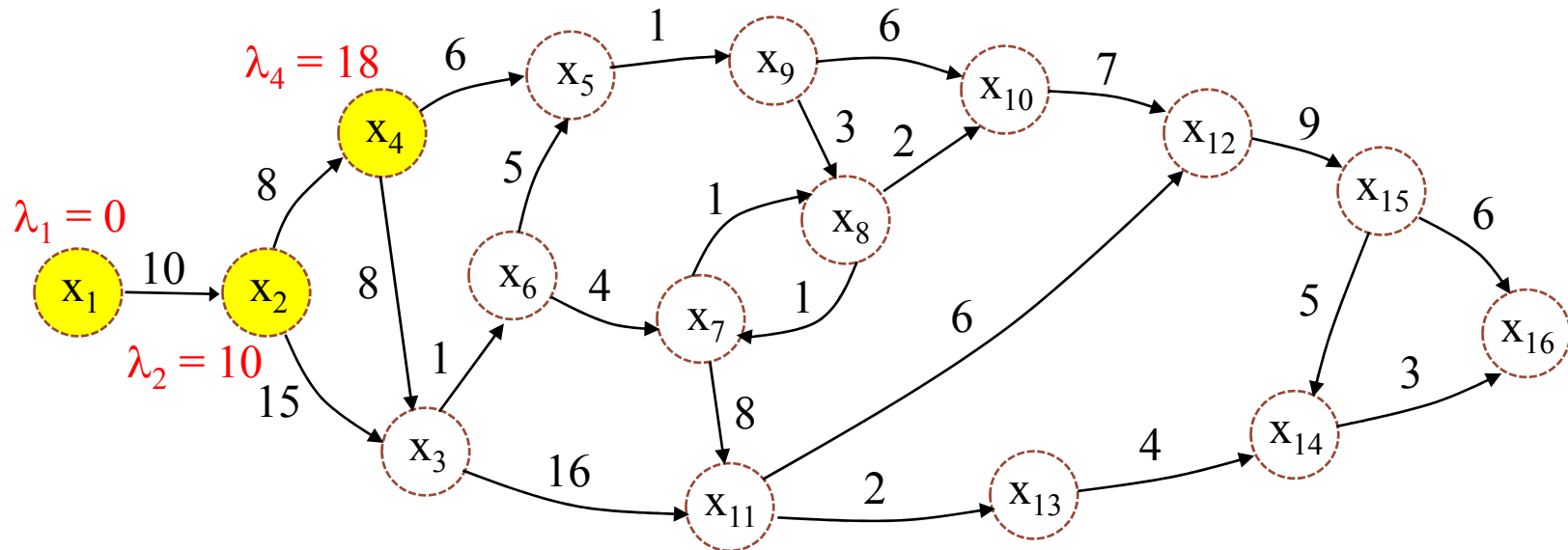
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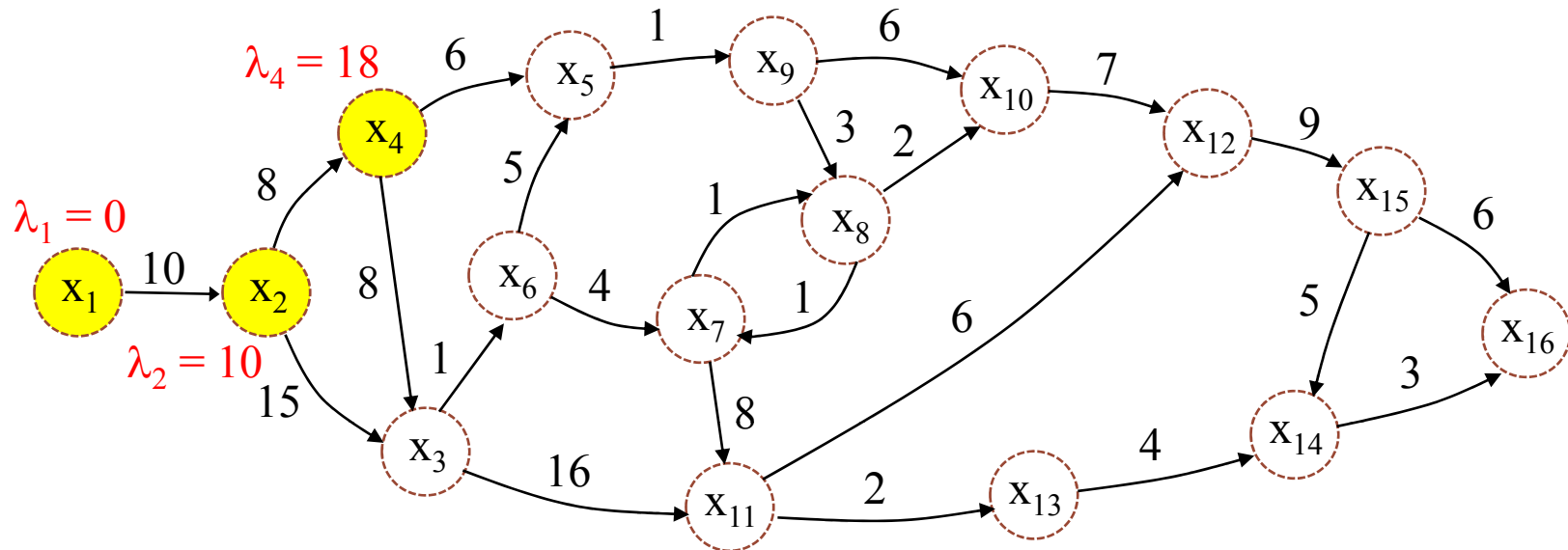
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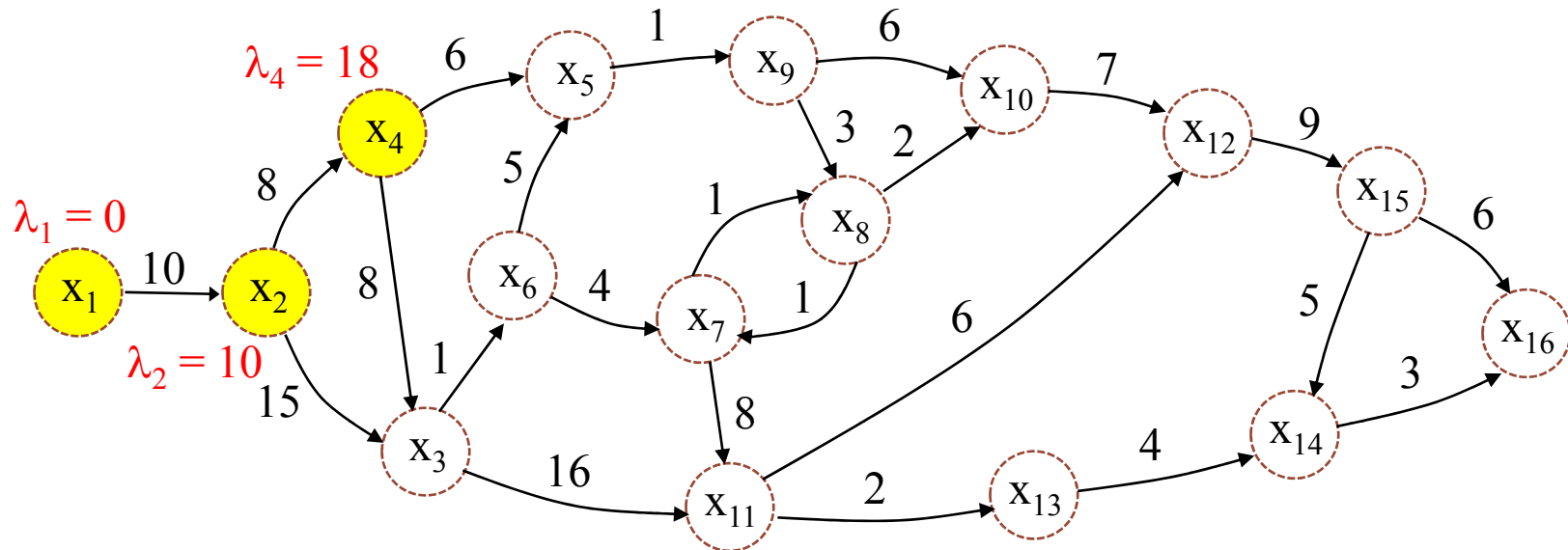
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$$\underline{k=3} \quad v(x_2, x_3) \\ v(x_4, x_5)$$

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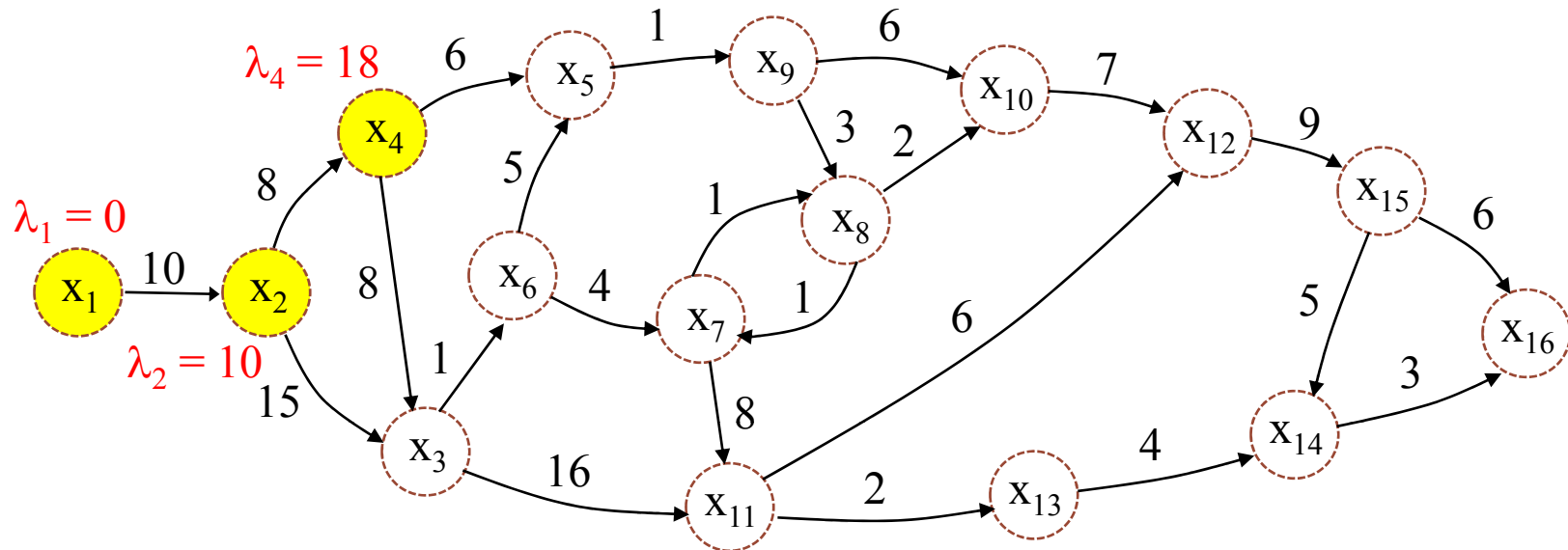
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$$\underline{k=3} \quad \begin{aligned} &v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25 \\ &v(x_4, x_5) \end{aligned}$$

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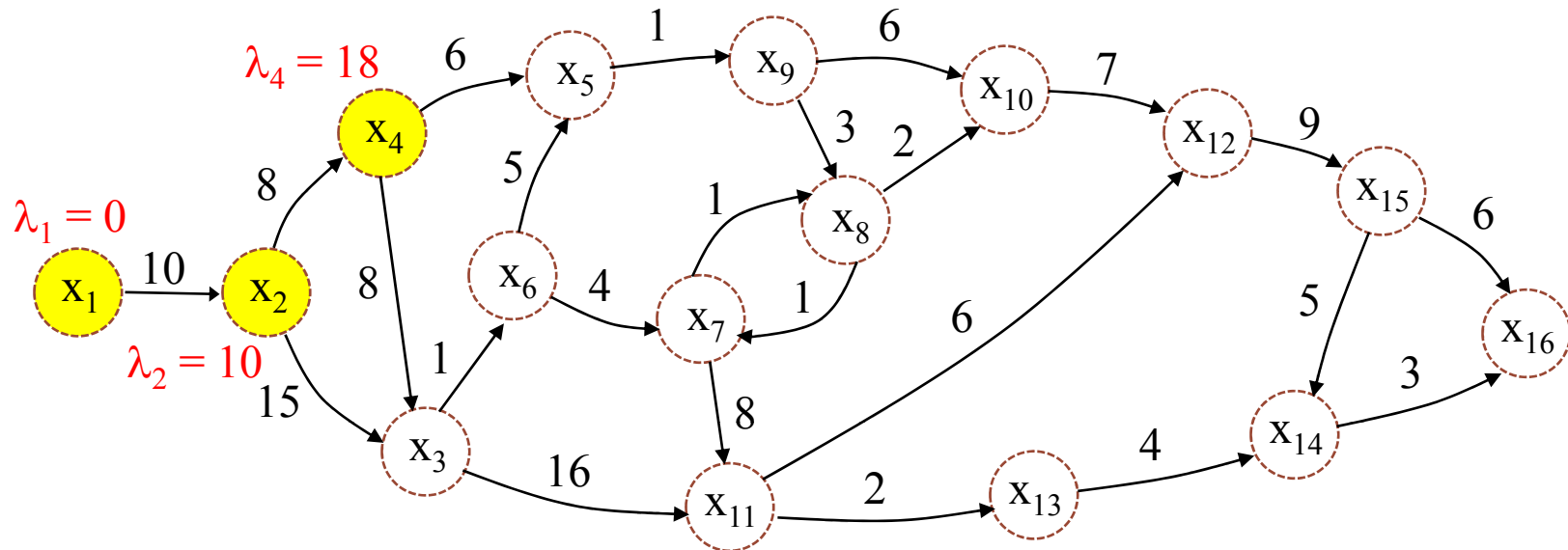
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$$\underline{k=3} \quad \begin{aligned} v(x_2, x_3) \quad \lambda_3 &= \lambda_2 + v(x_2, x_3) = 25 \\ v(x_4, x_5) \quad \lambda_5 &= \lambda_4 + v(x_4, x_5) = 24 \end{aligned}$$

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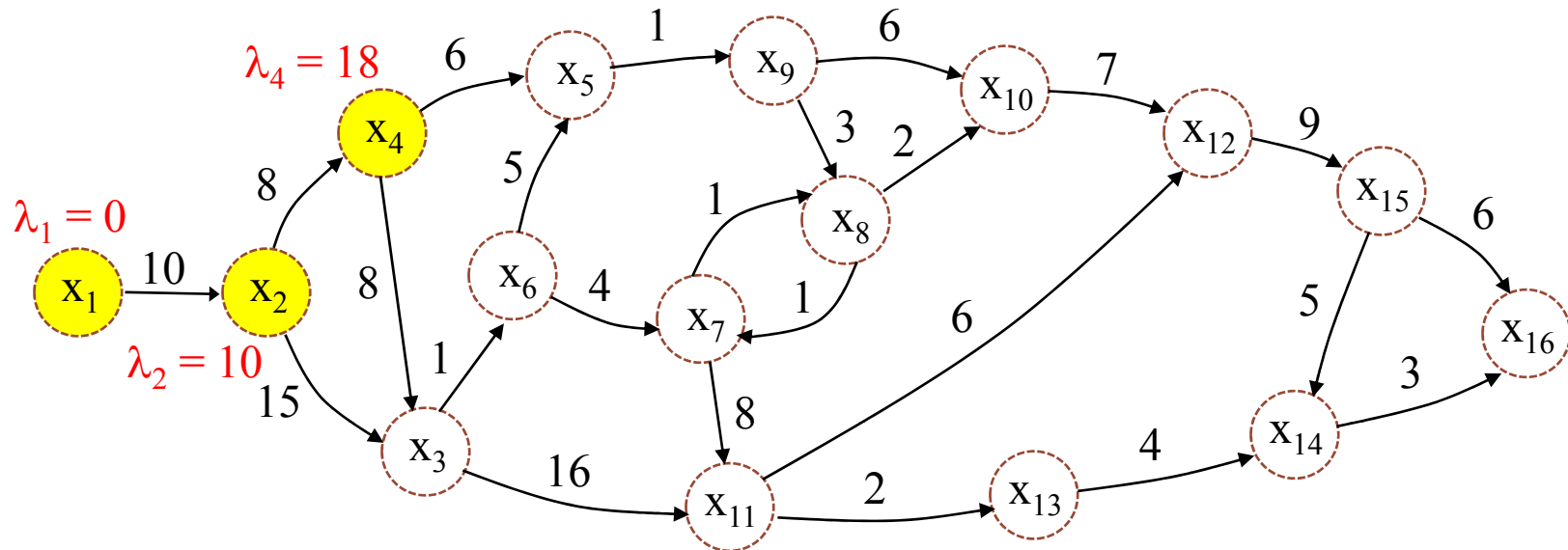
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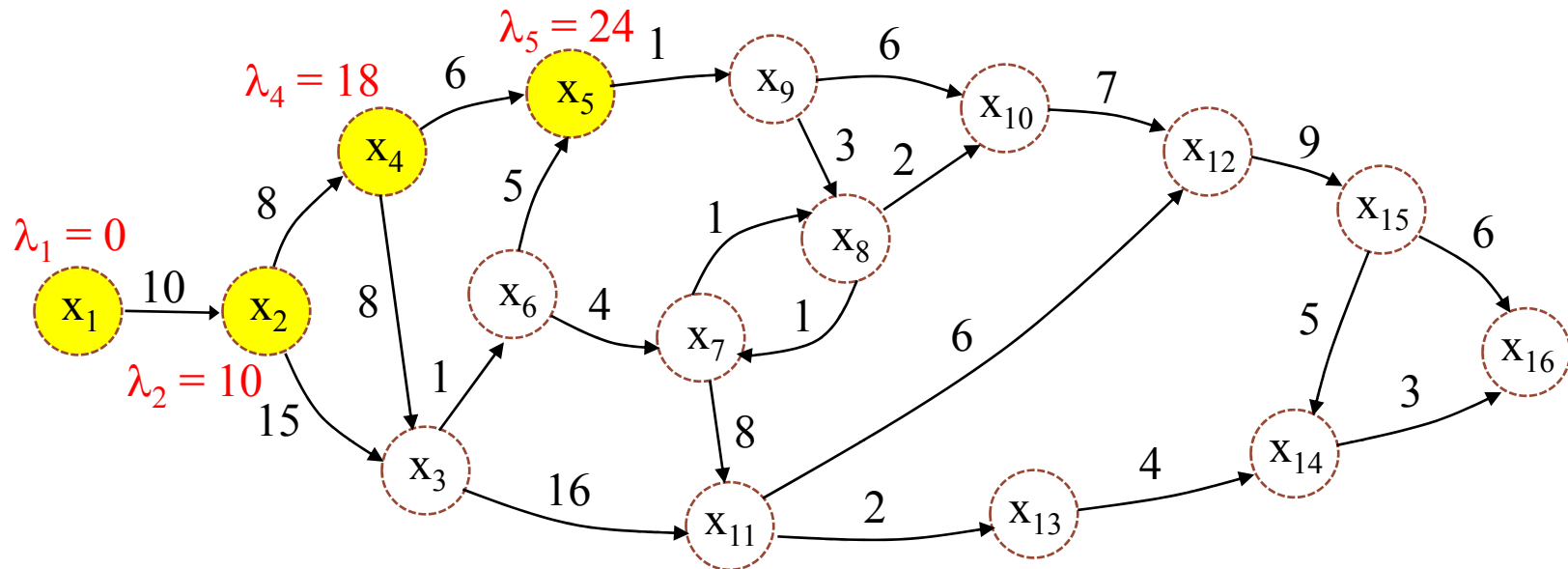
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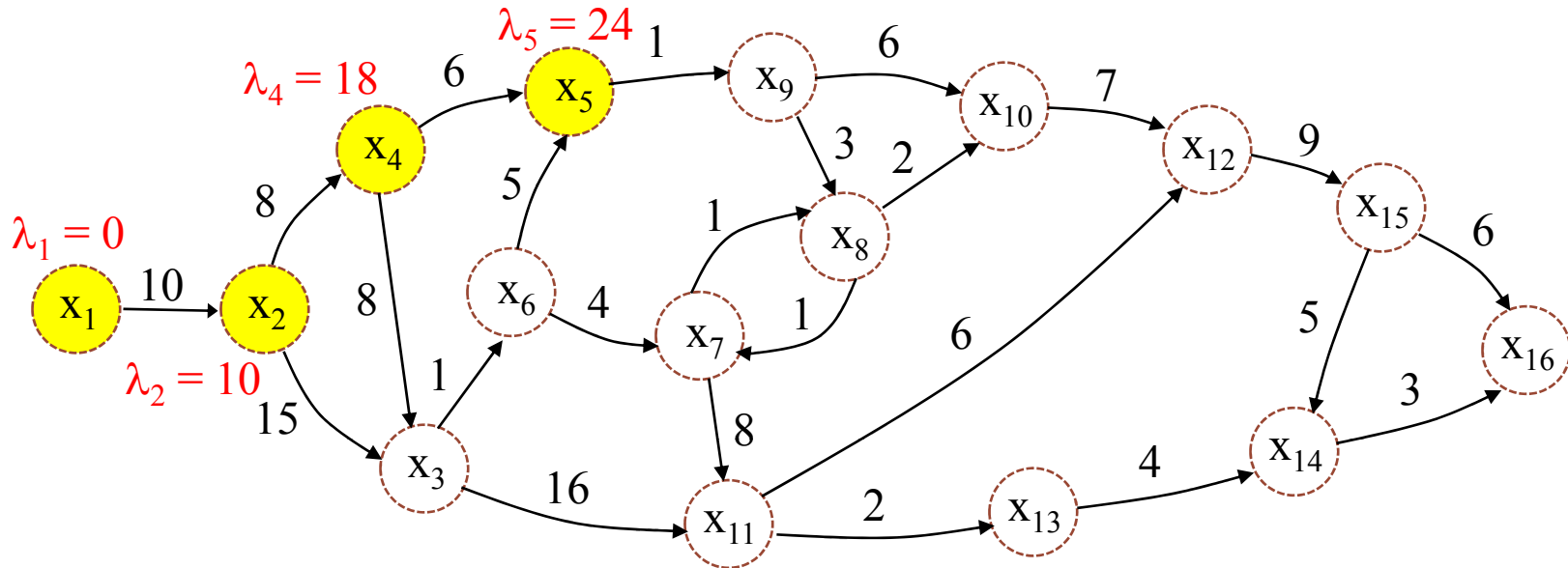
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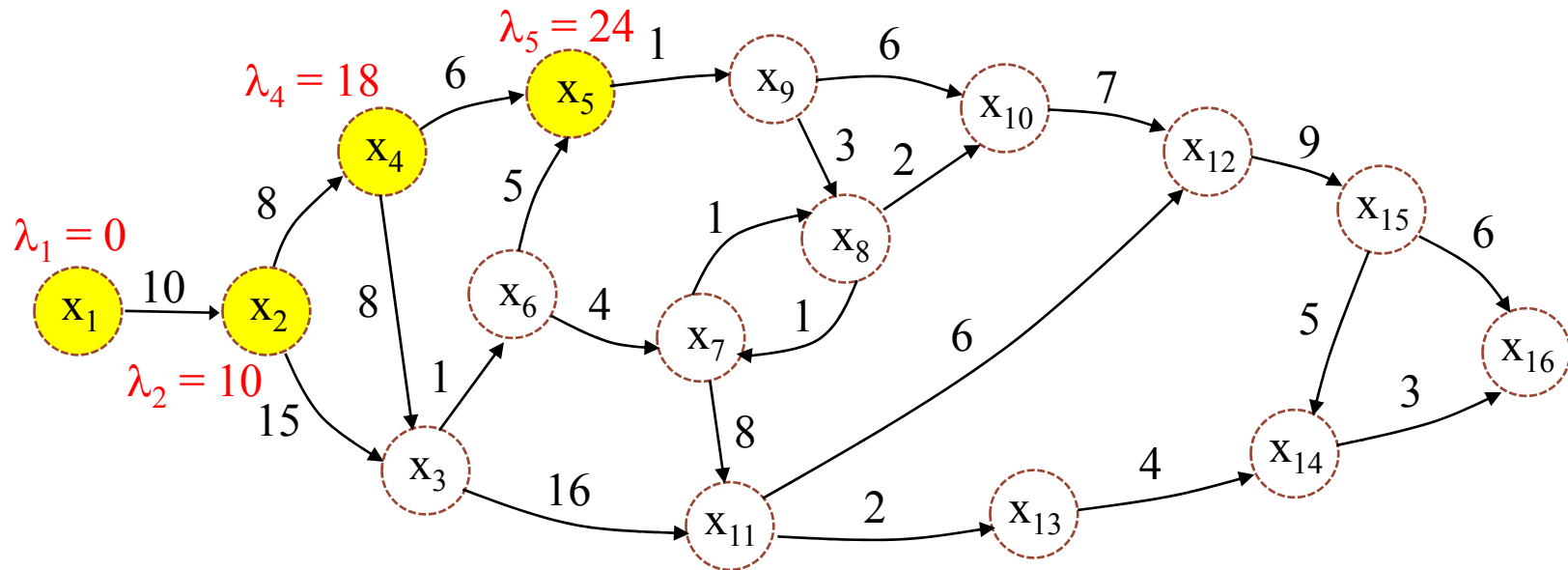
$$\underline{k=1} \quad v(x_1, x_2) \quad \lambda_2 = \lambda_1 + v(x_1, x_2) = 10 \quad E_2 = \{ x_1, x_2 \}$$

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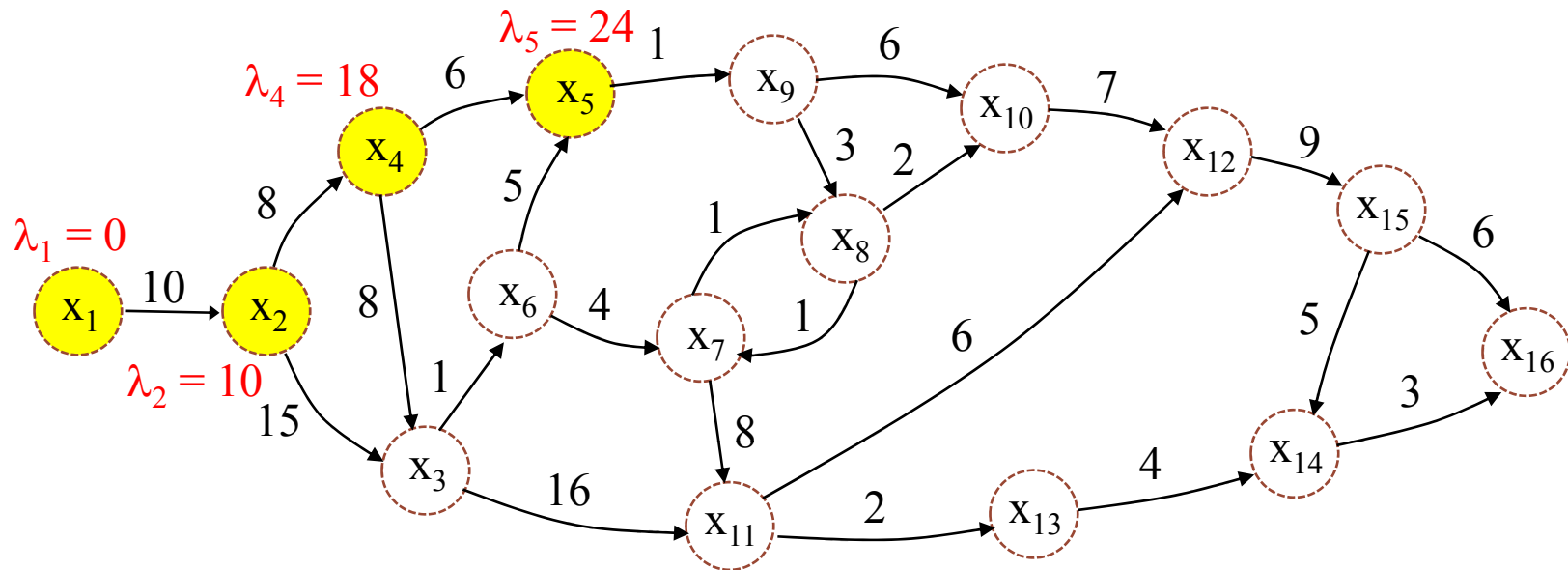
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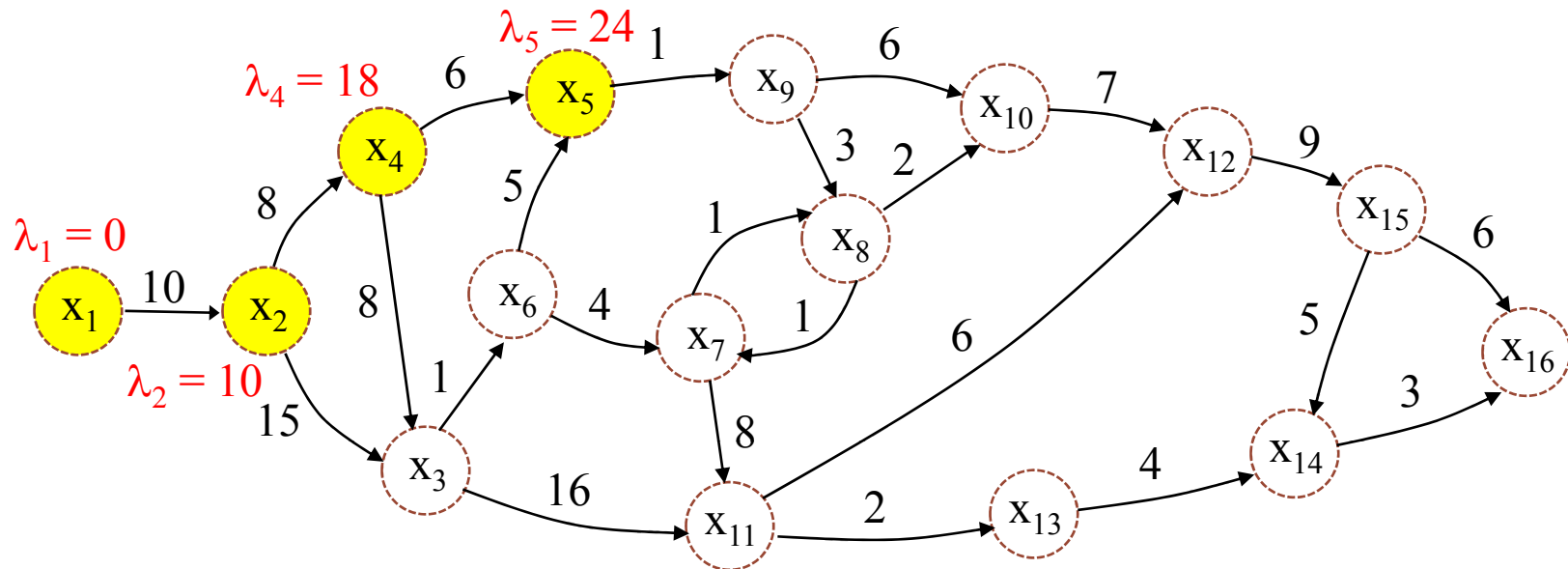
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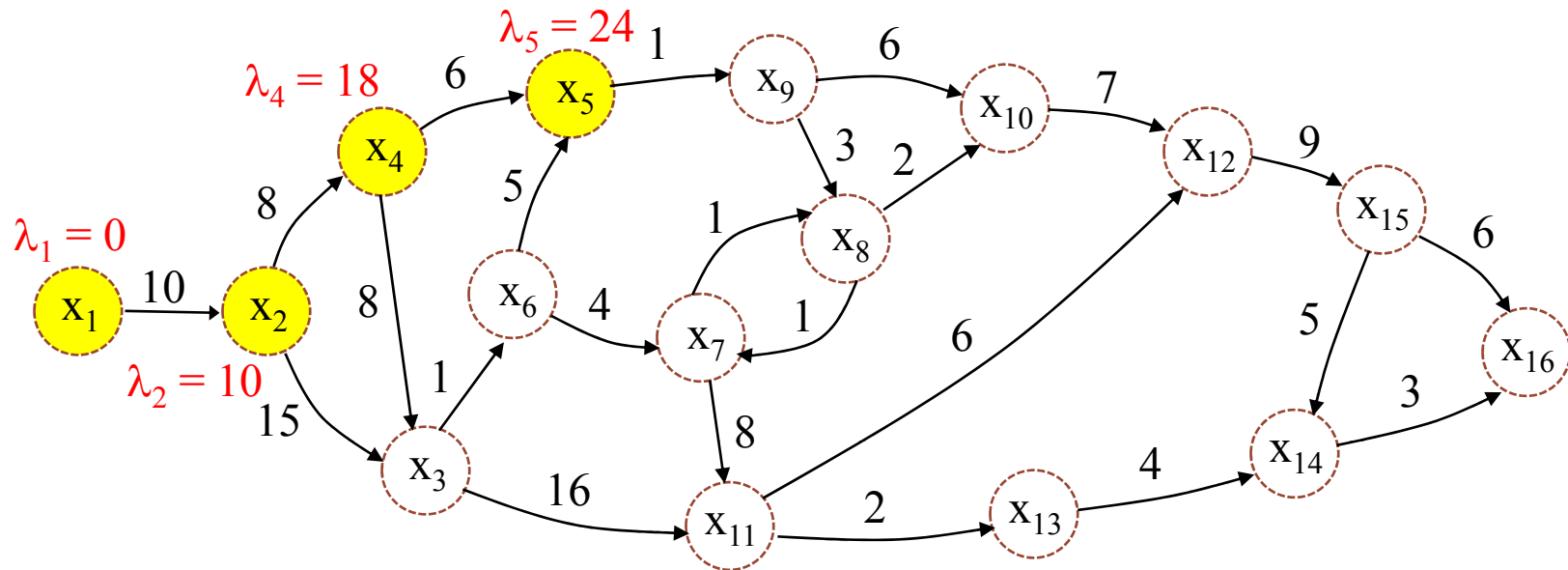
 $k = 4$



$k = 4$ $v(x_2, x_3)$

**k = 4** $v(x_2, x_3)$ $v(x_4, x_3)$

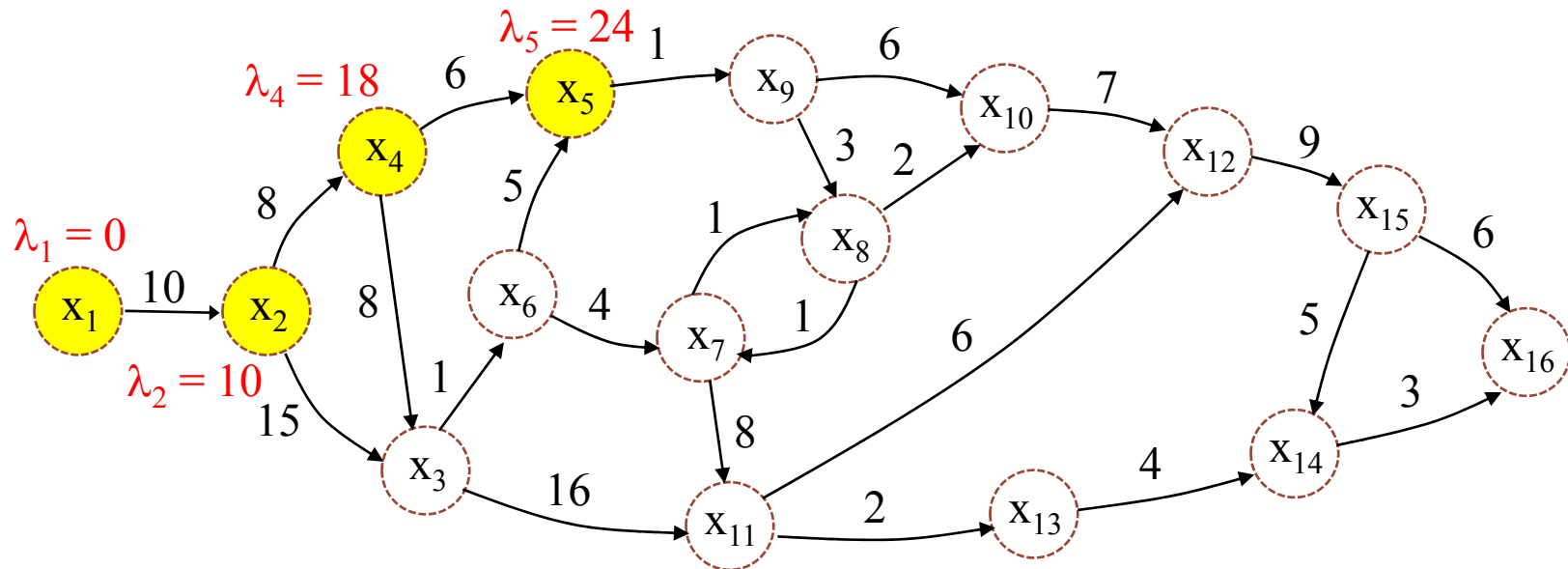
**k = 4** $v(x_2, x_3)$ $v(x_4, x_3)$ $v(x_5, x_9)$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3)$$

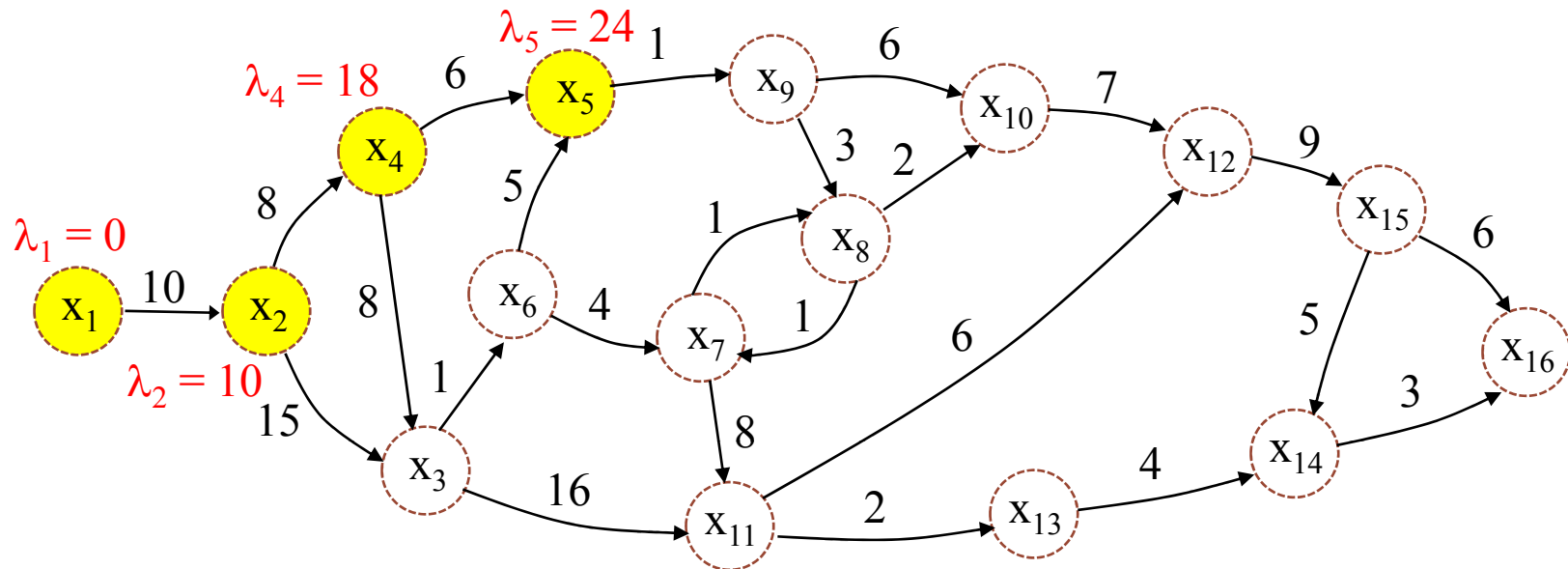
$$v(x_5, x_9)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

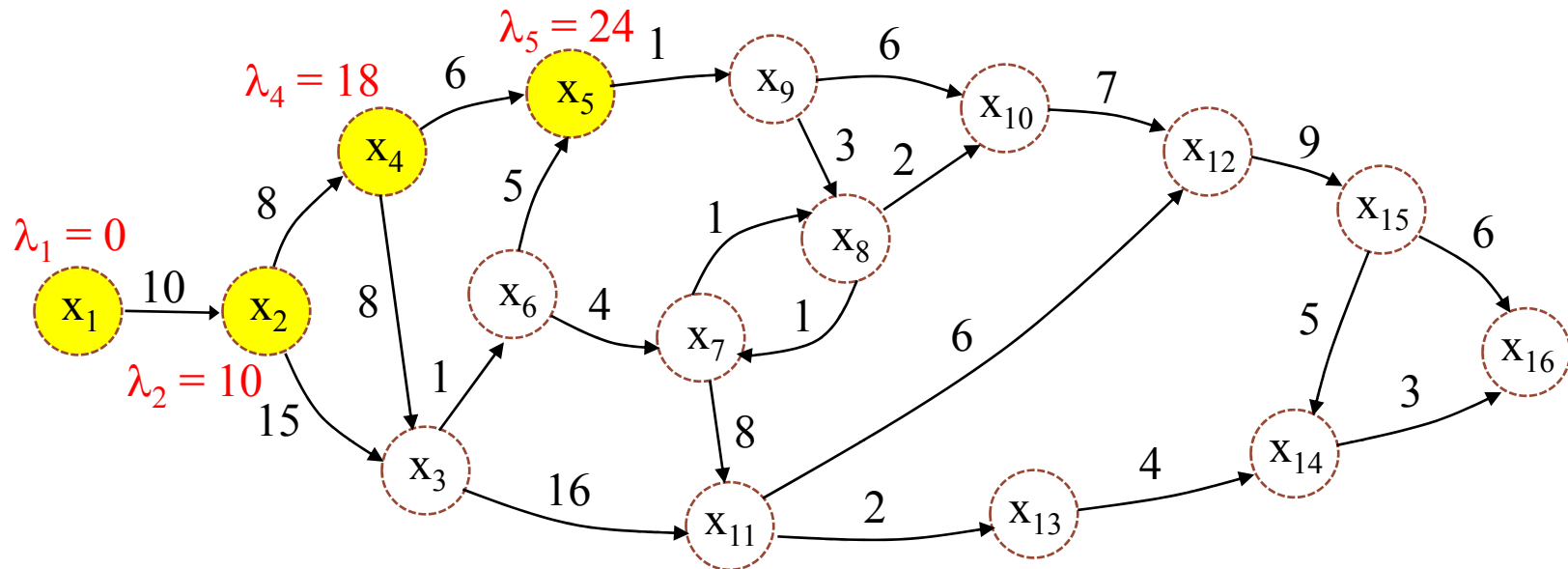
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$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

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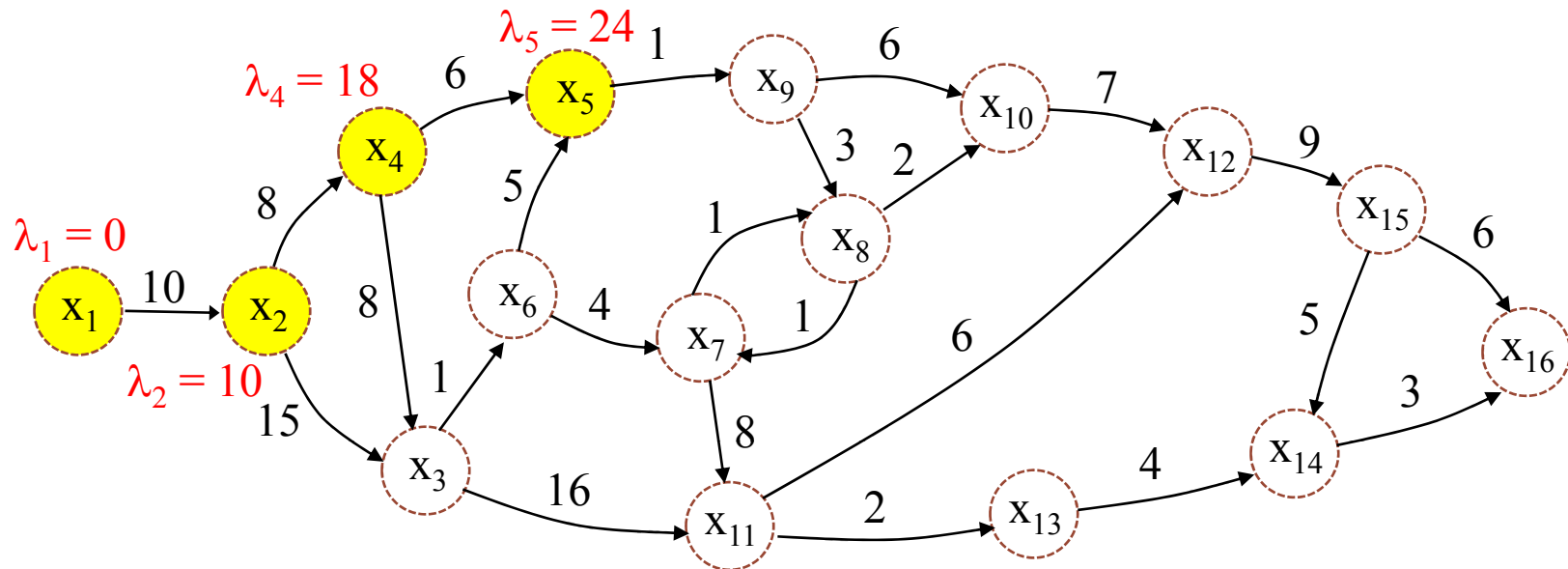
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

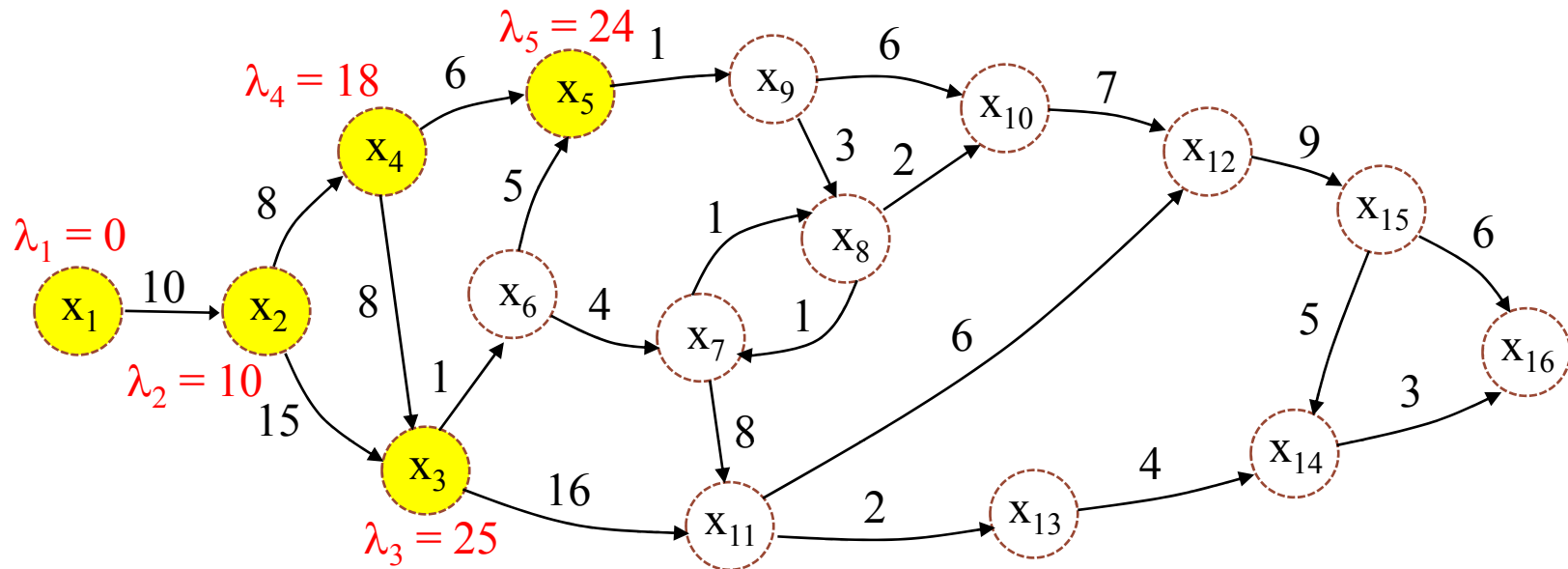
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$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

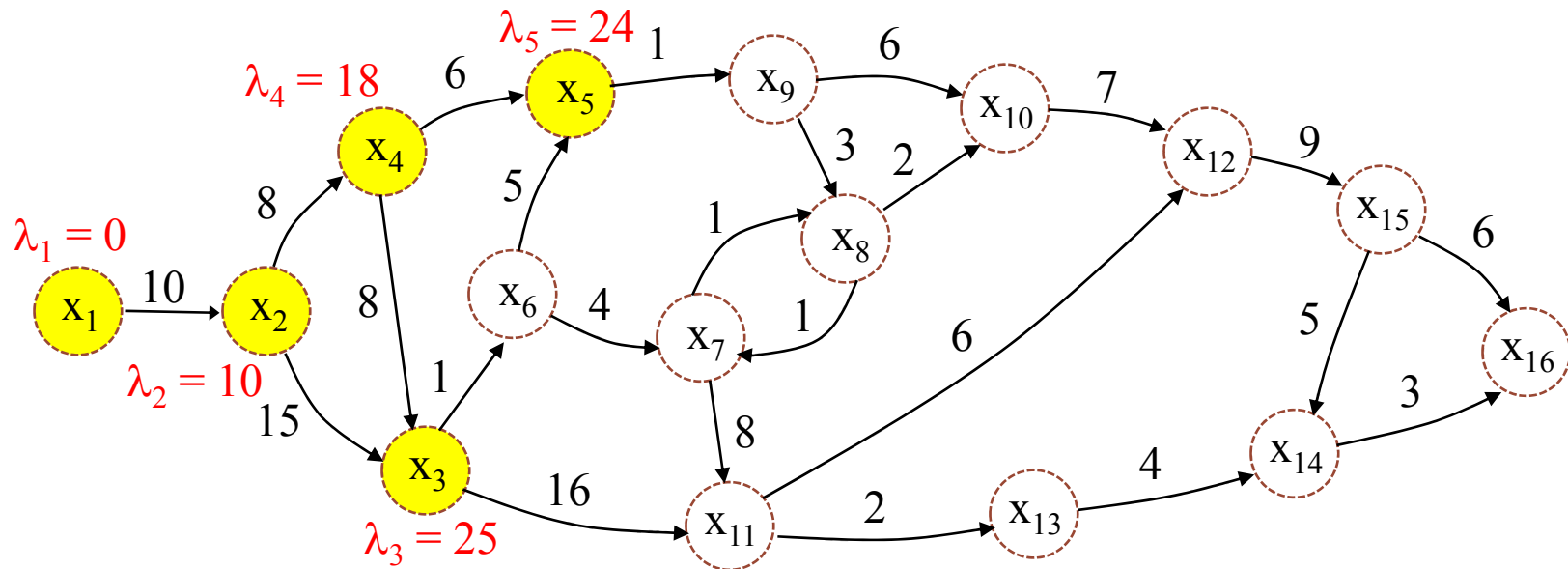
**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

**k = 4**

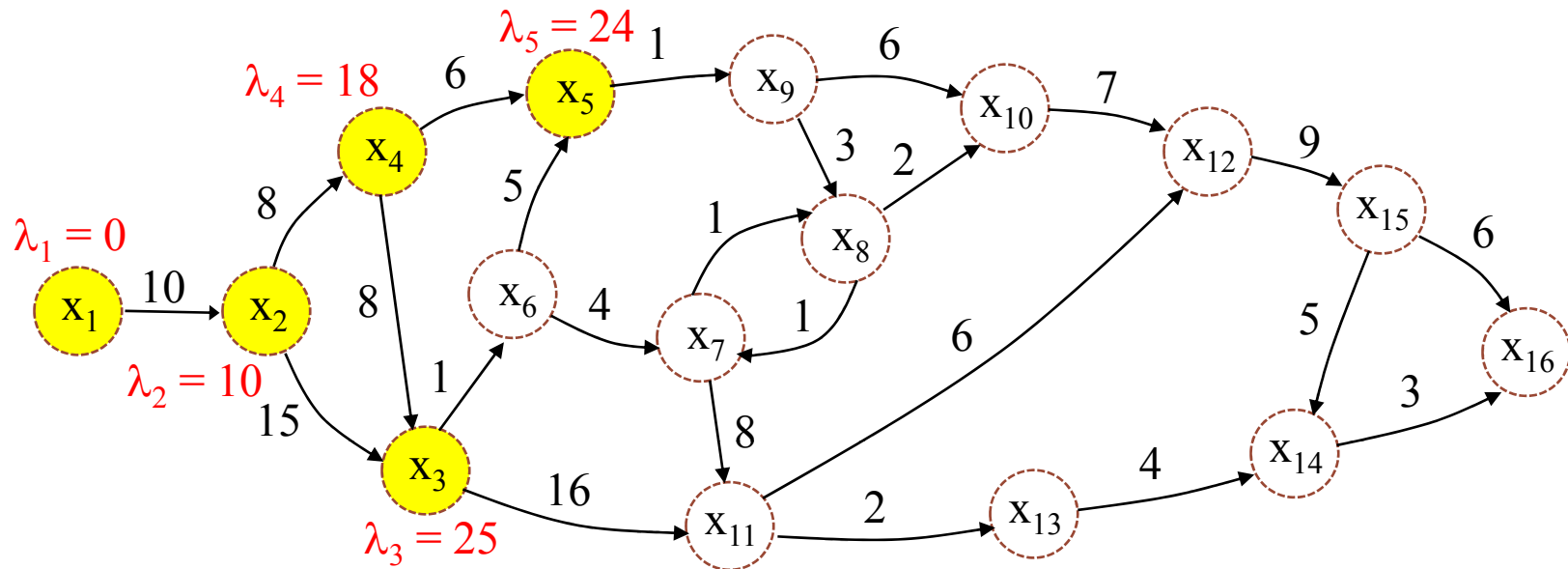
$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

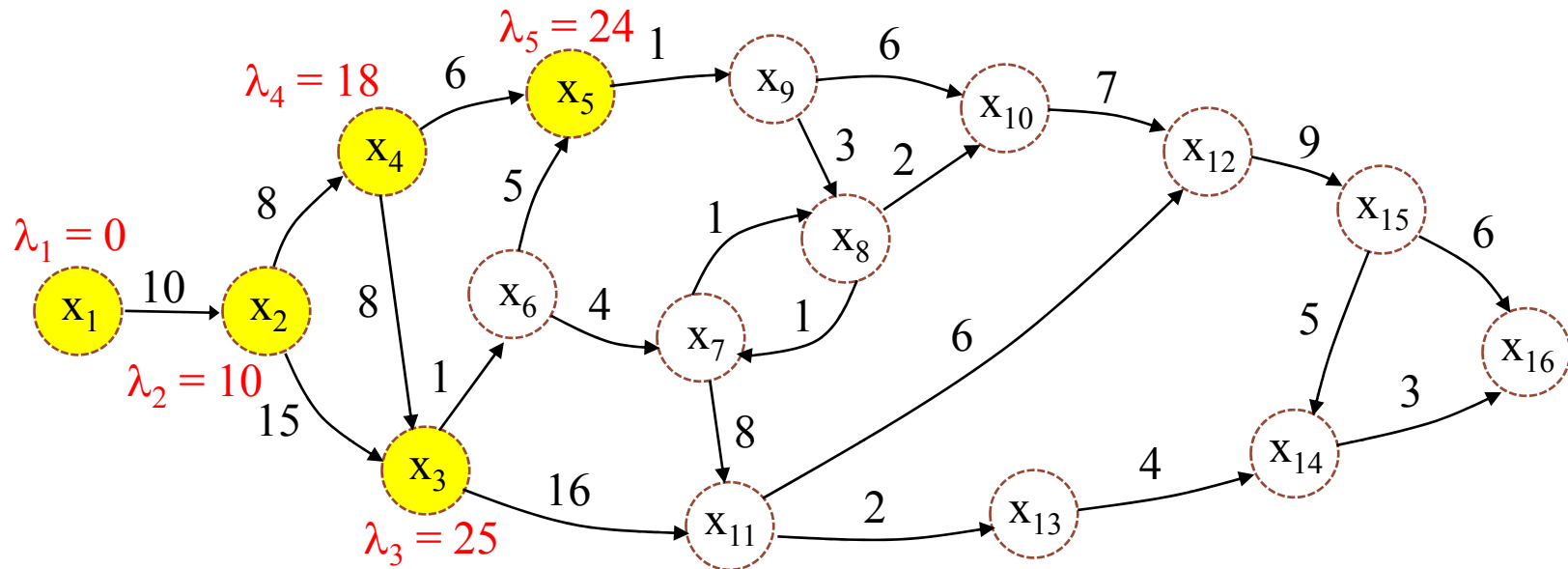
$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

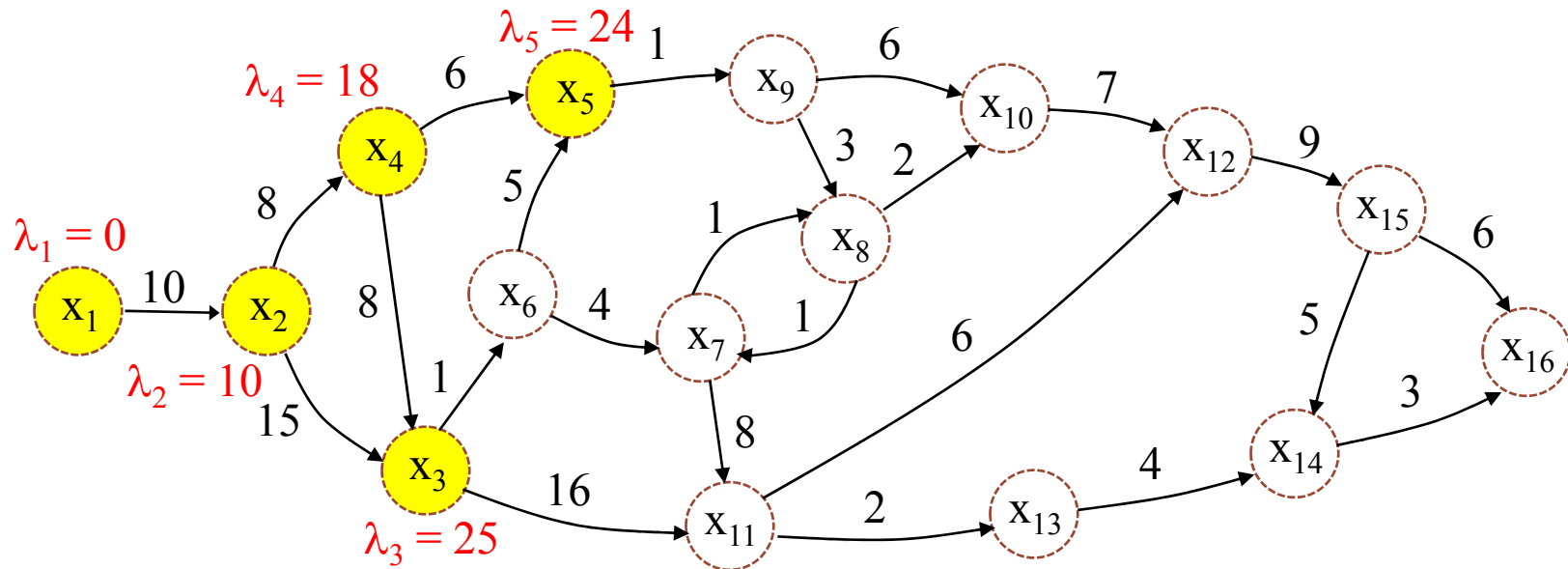
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6)$$

$$v(x_5, x_9)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

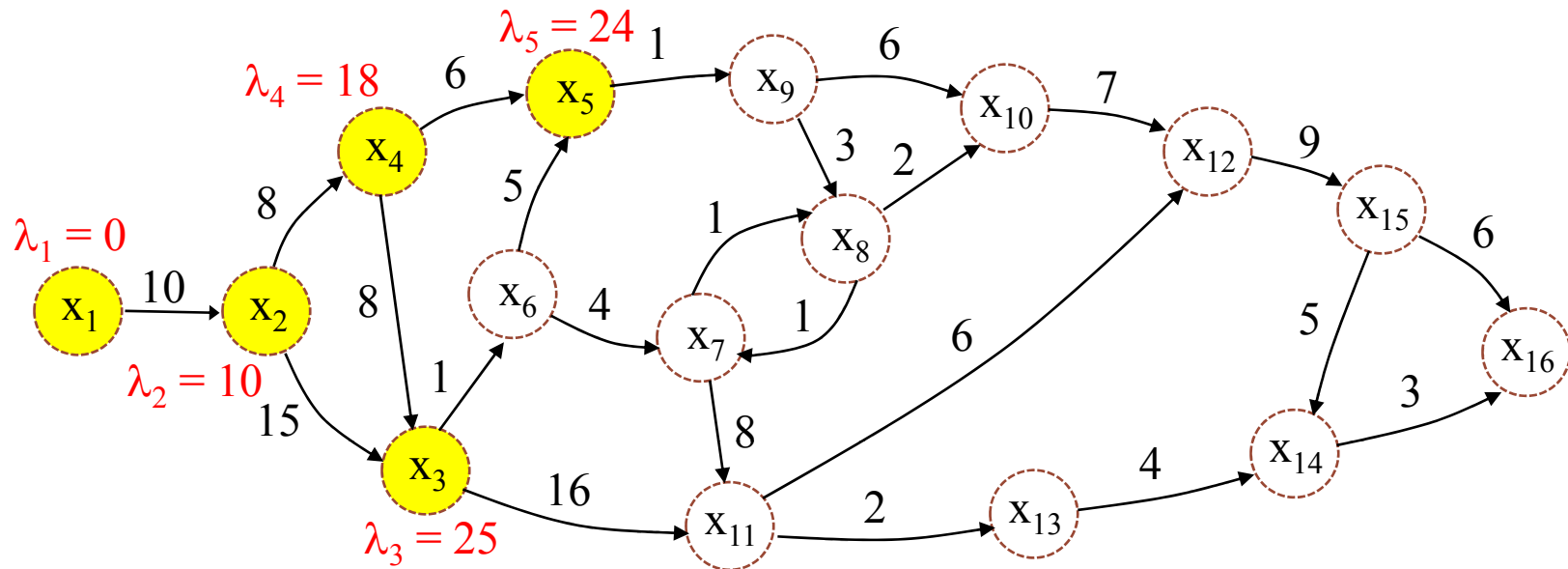
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

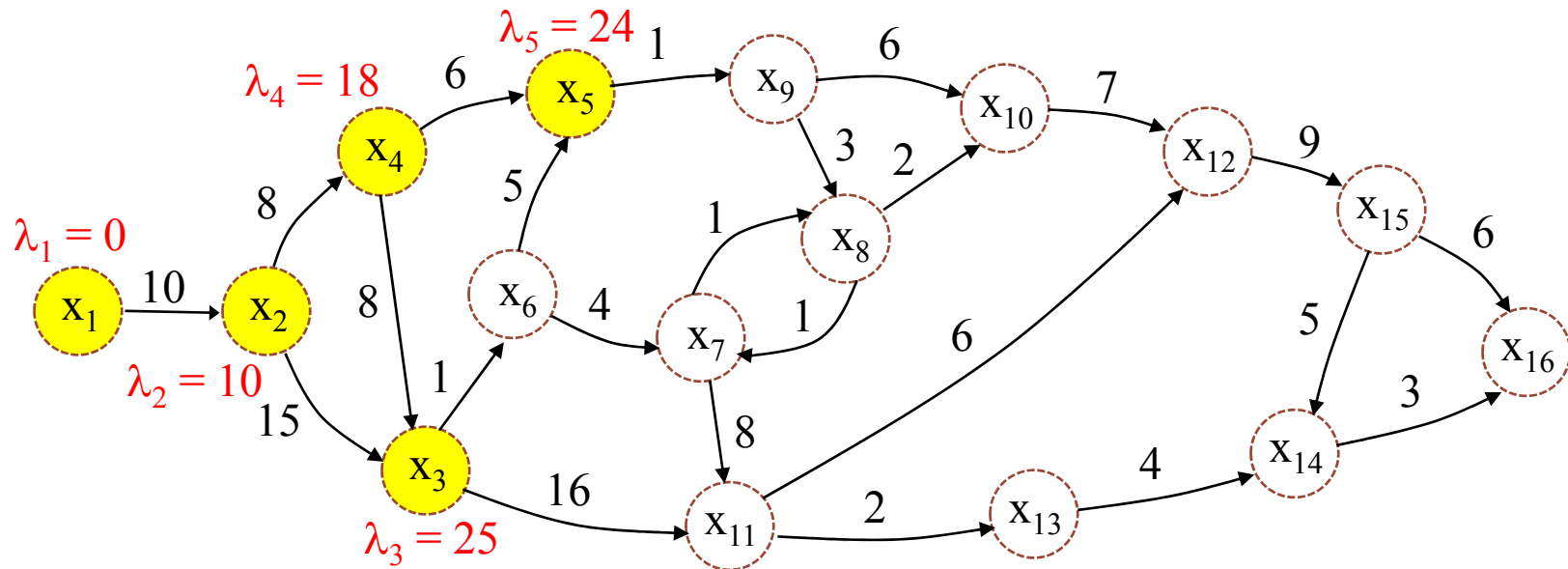
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

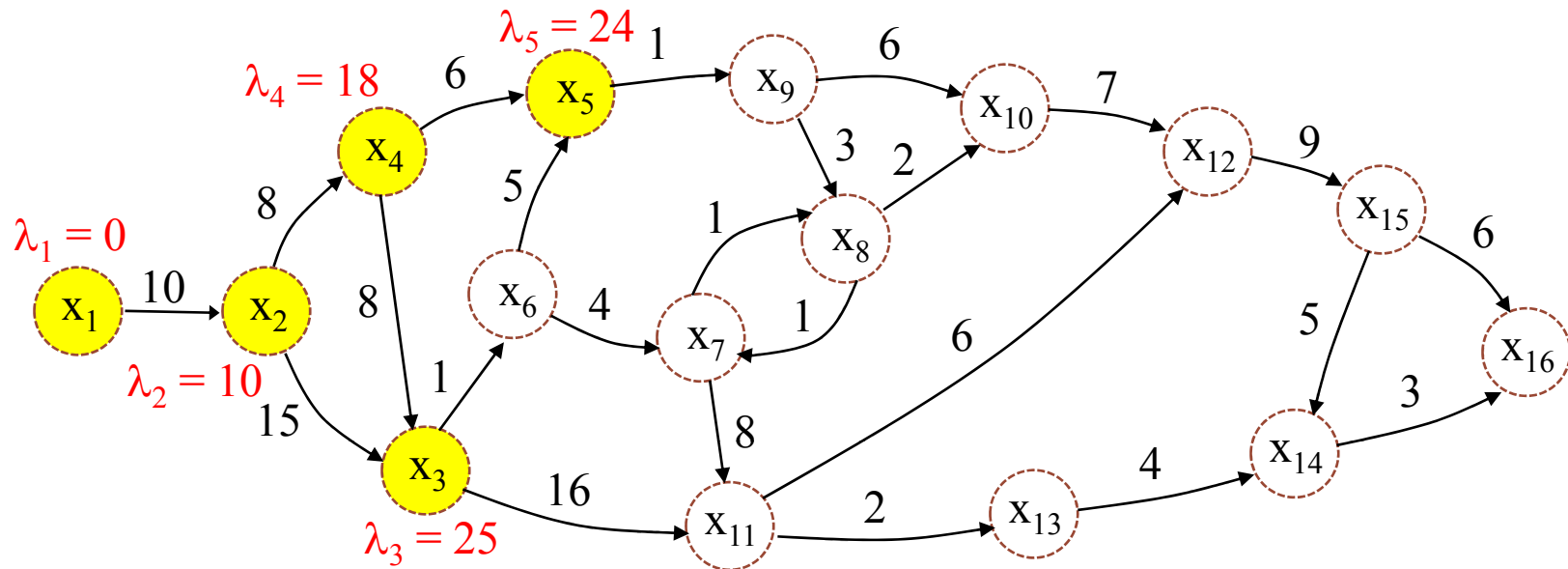
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

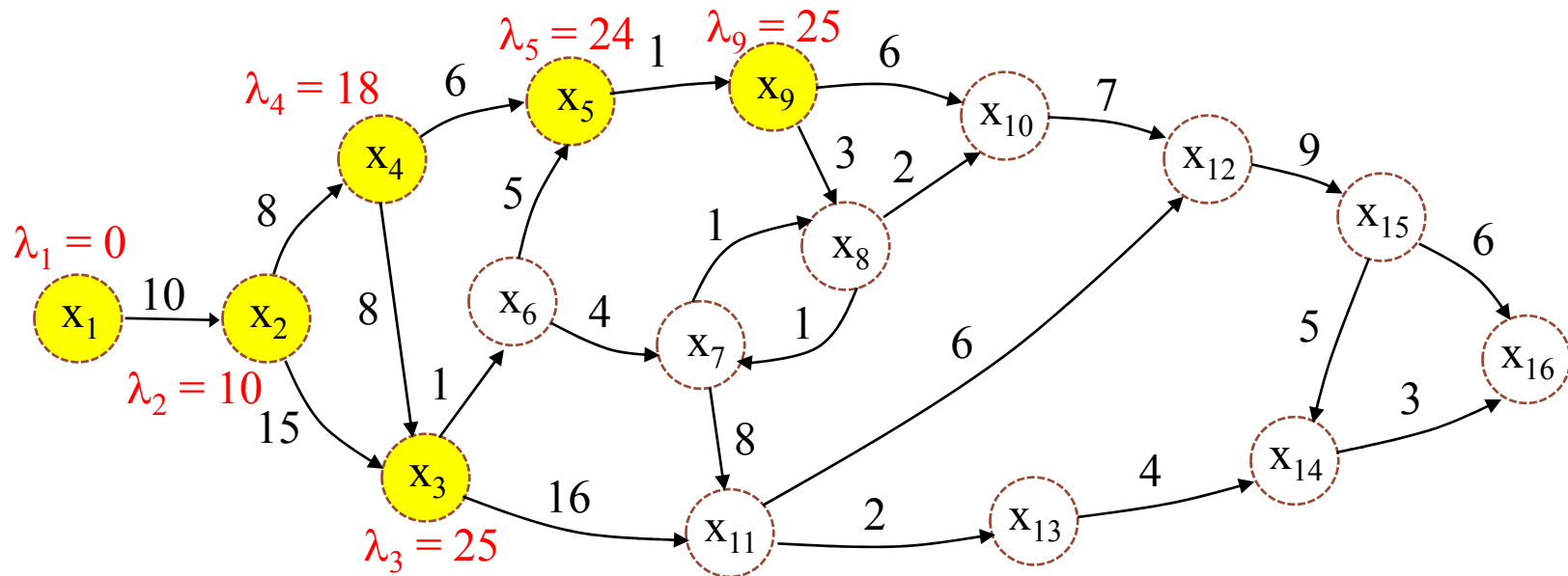
$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

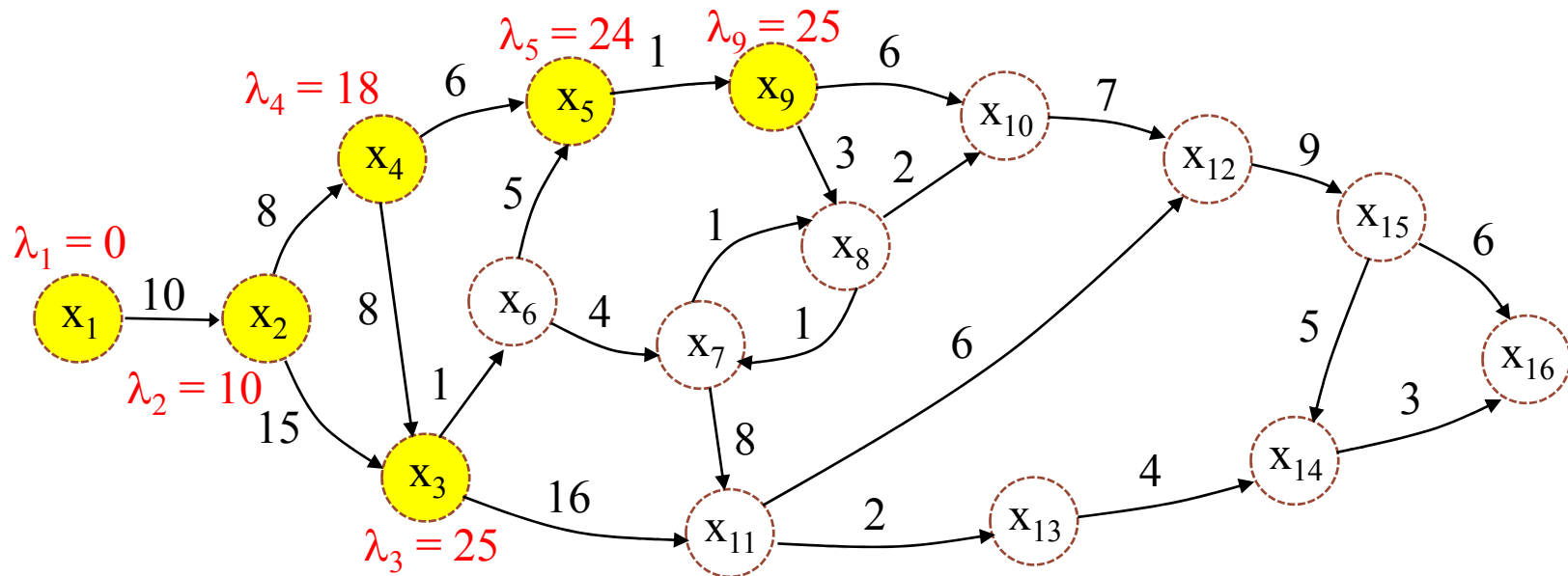
$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

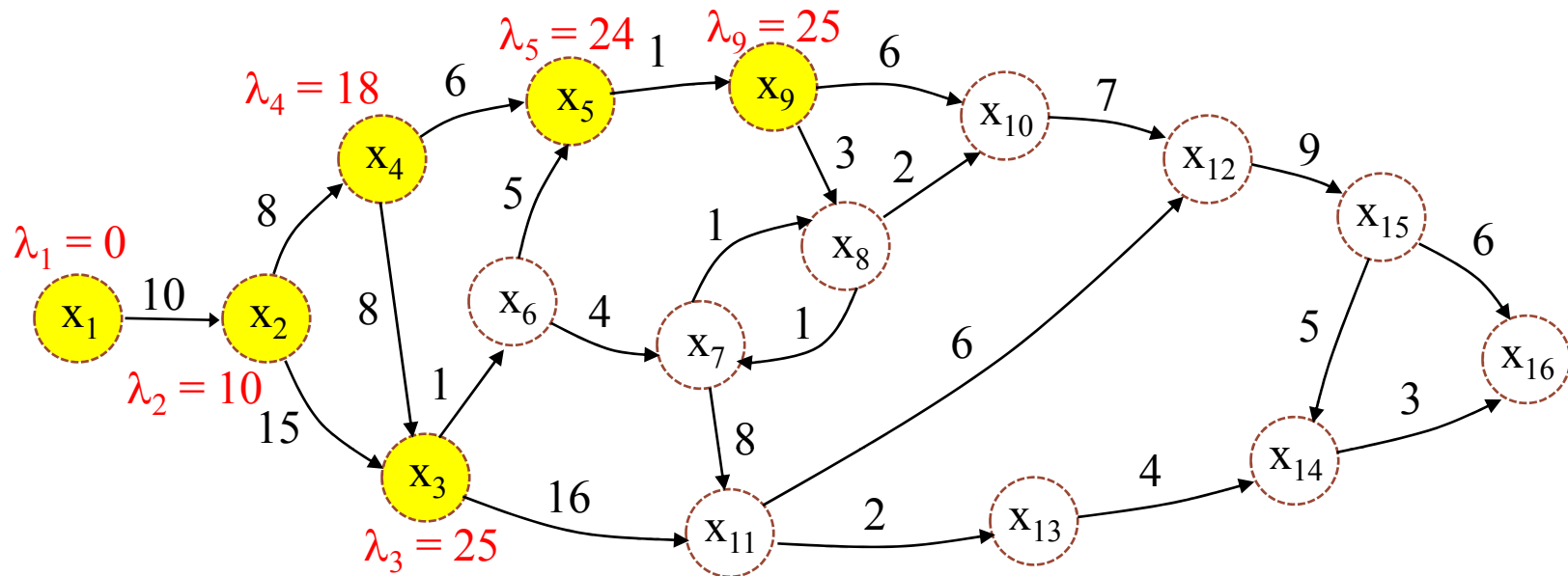
k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

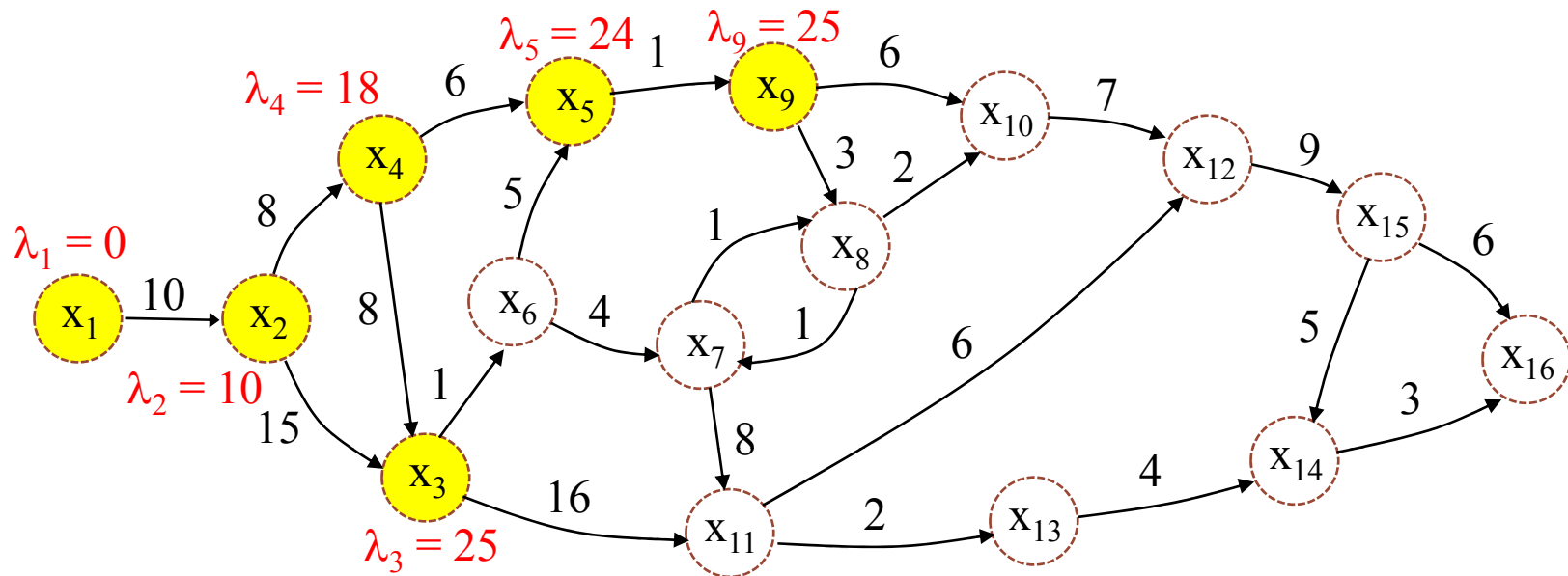
$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

$$v(x_3, x_6)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

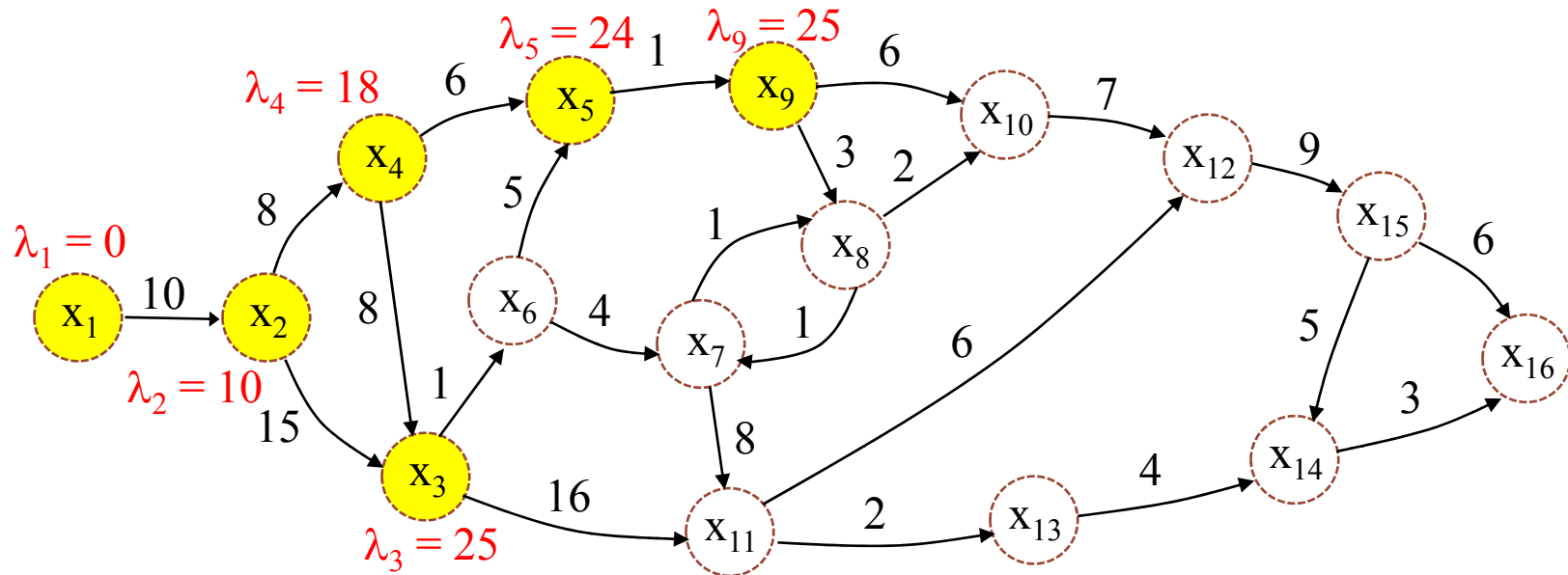
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

$$v(x_3, x_6)$$

$$v(x_9, x_8)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

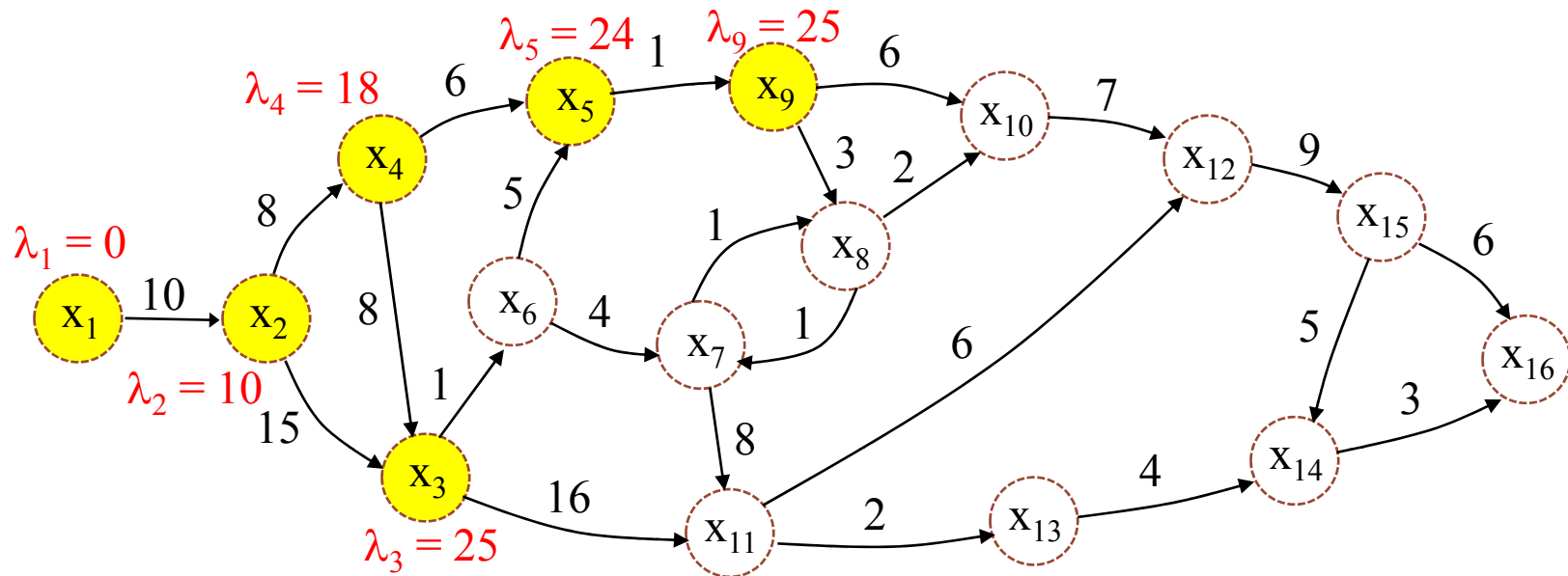
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_9, x_8)$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

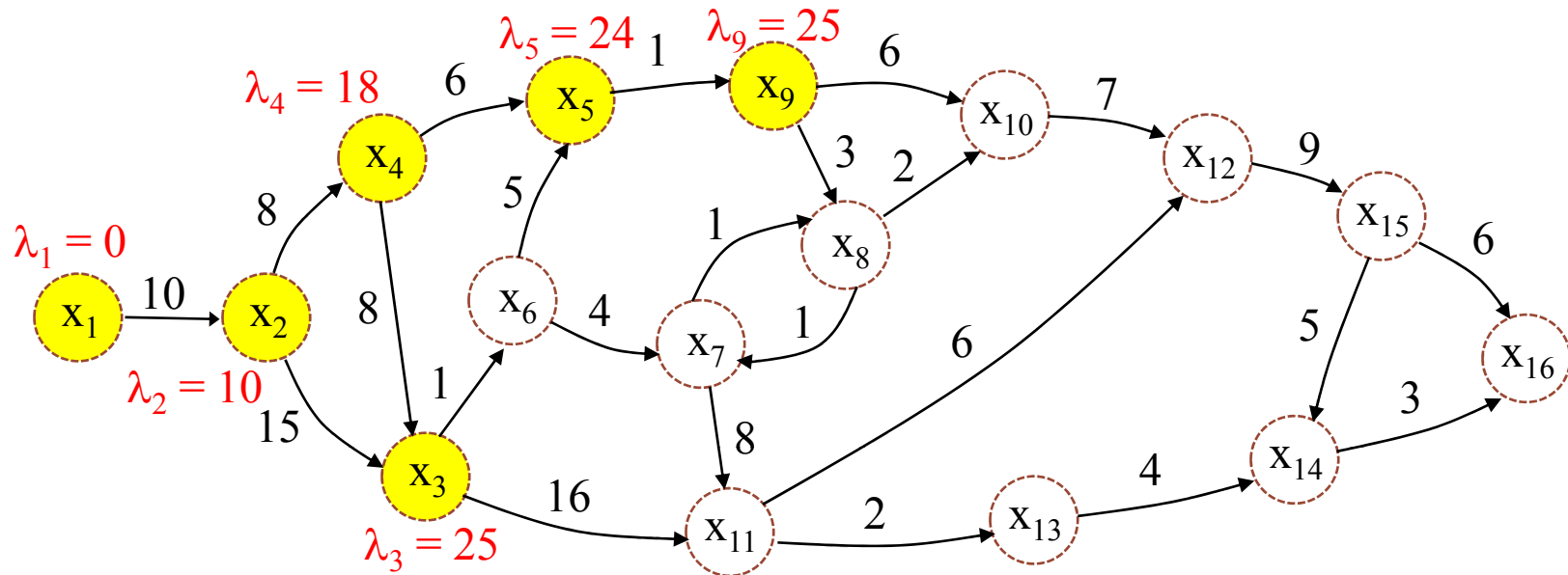
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

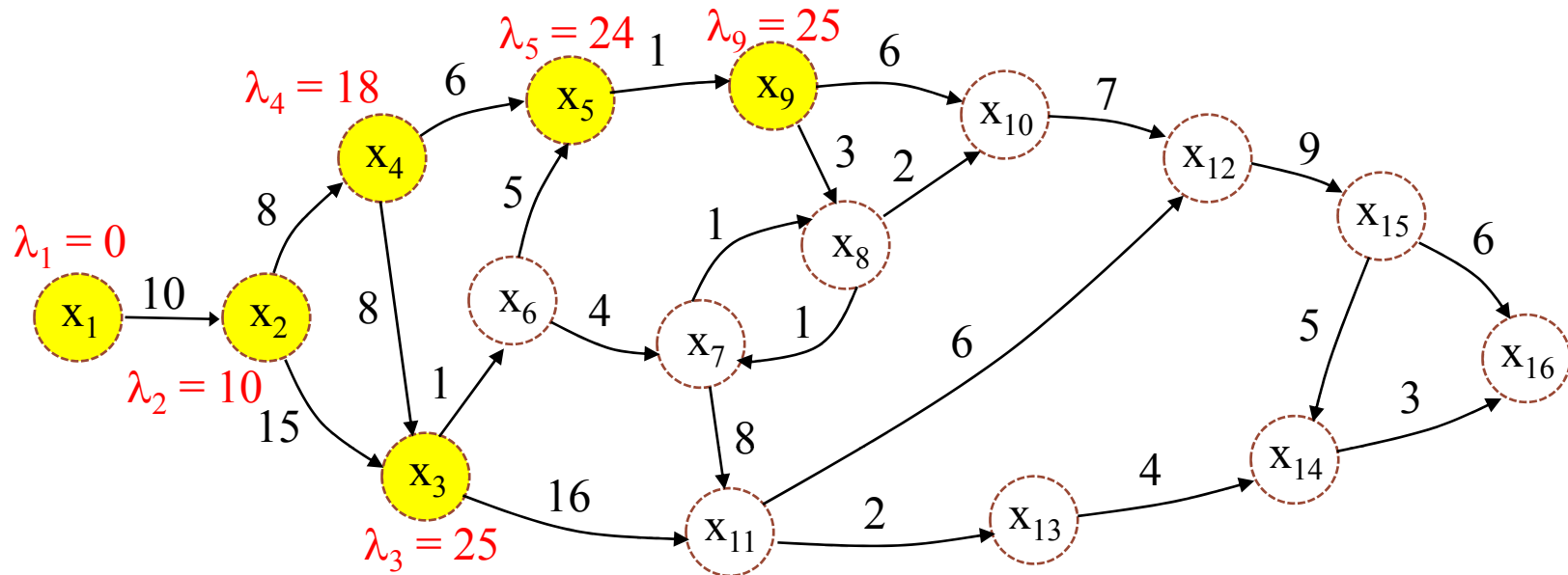
$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

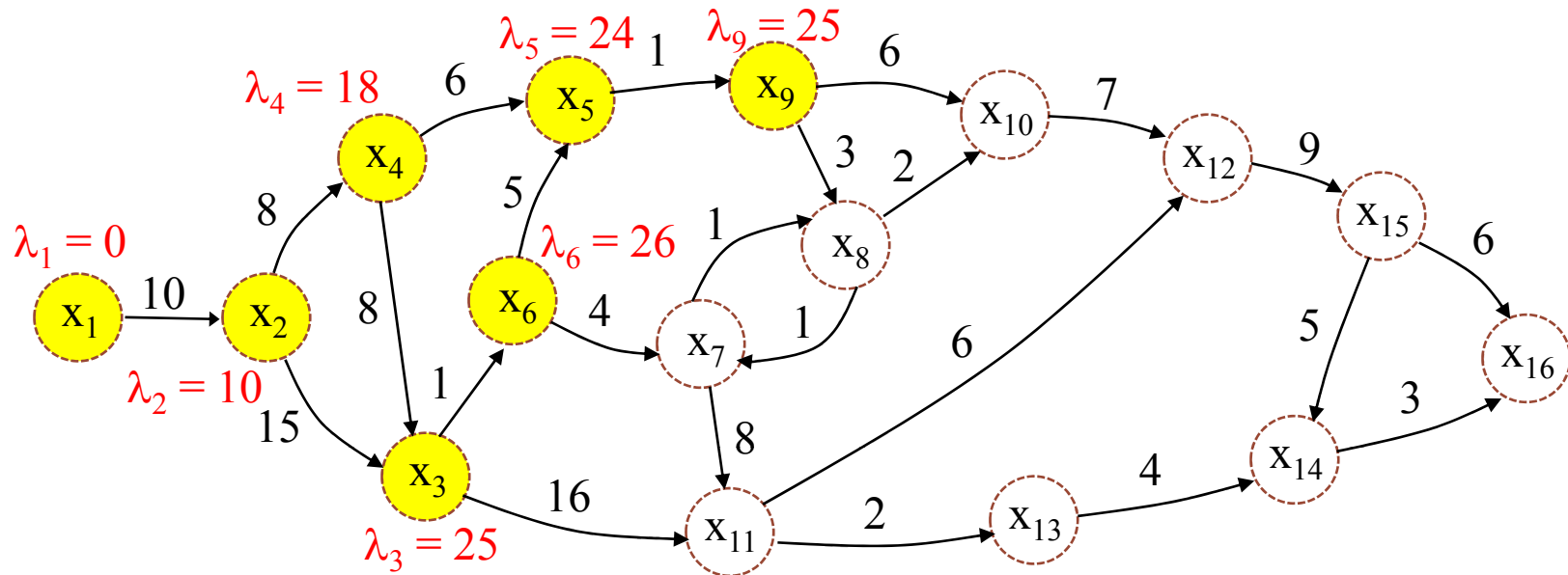
$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

k = 6

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_7 = \{ x_1, x_2, x_4, x_5, x_3, x_9, x_6 \}$$

**k = 4**

$$v(x_2, x_3) \quad \lambda_3 = \lambda_2 + v(x_2, x_3) = 25$$

$$v(x_4, x_3) \quad \lambda_3 = \lambda_4 + v(x_4, x_3) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

$$E_5 = \{ x_1, x_2, x_4, x_5, x_3 \}$$

k = 5

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

$$v(x_5, x_9) \quad \lambda_9 = \lambda_5 + v(x_5, x_9) = 25$$

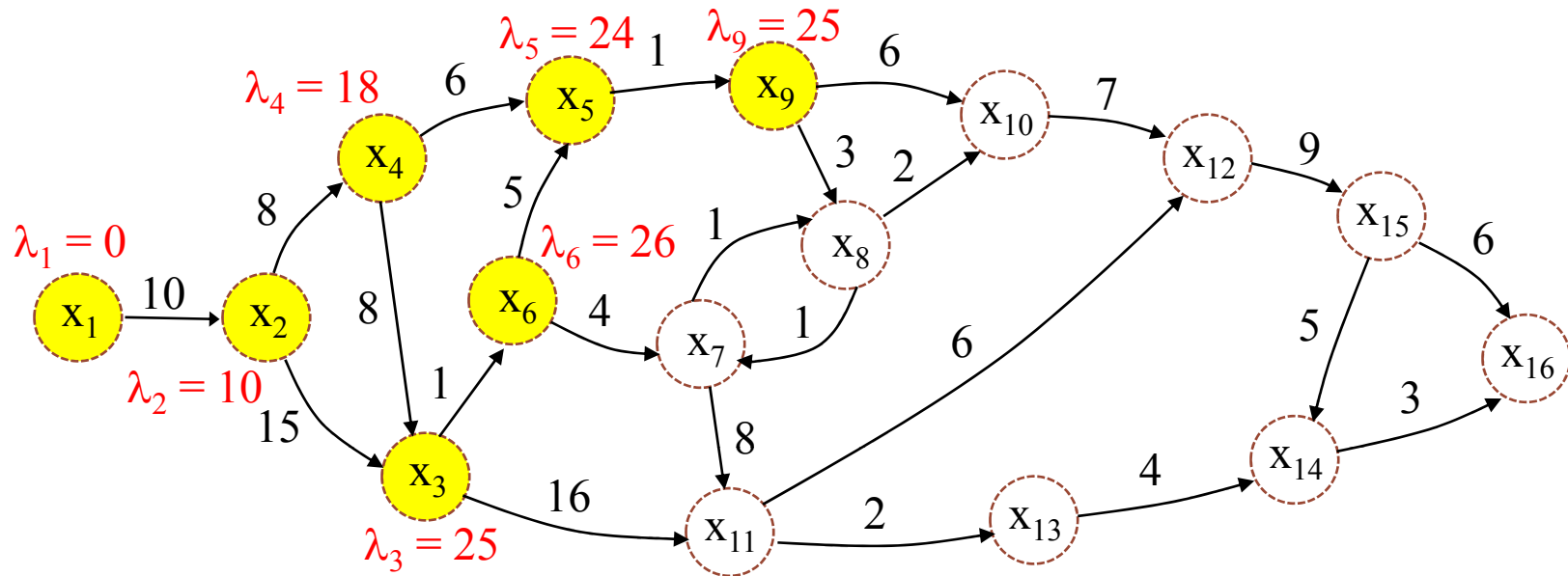
$$E_6 = \{ x_1, x_2, x_4, x_5, x_3, x_9 \}$$

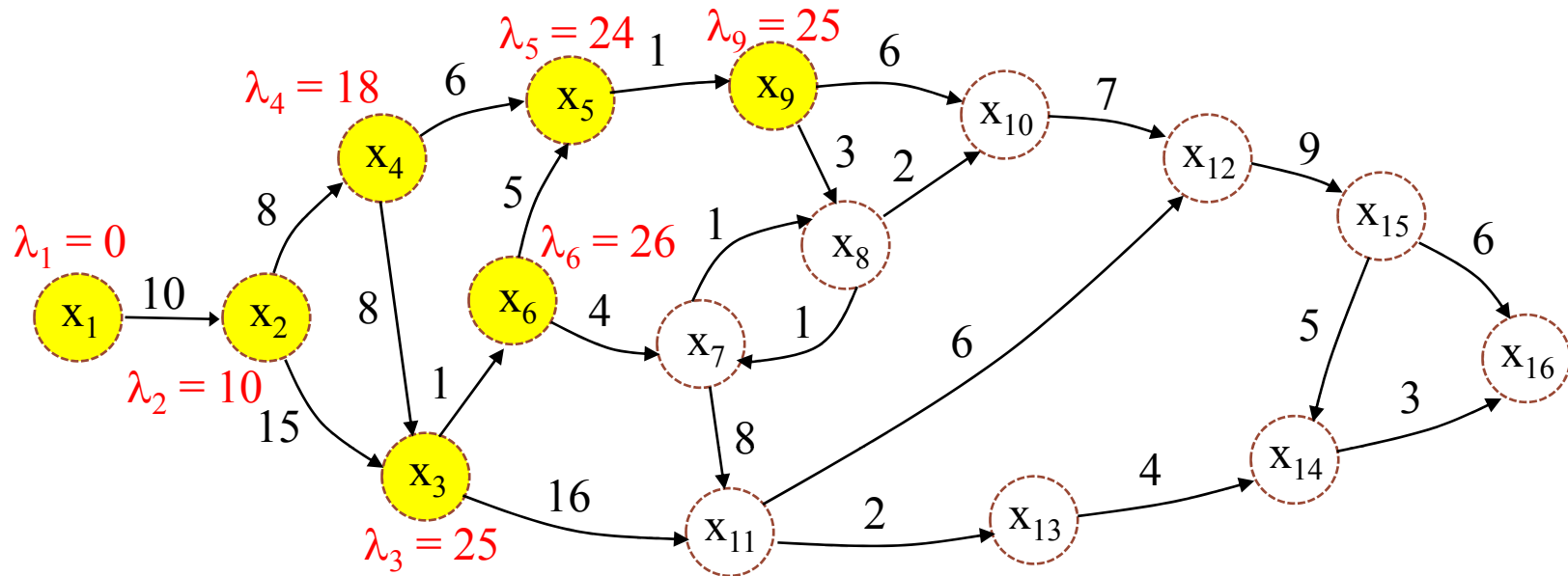
k = 6

$$v(x_3, x_6) \quad \lambda_6 = \lambda_3 + v(x_3, x_6) = 26$$

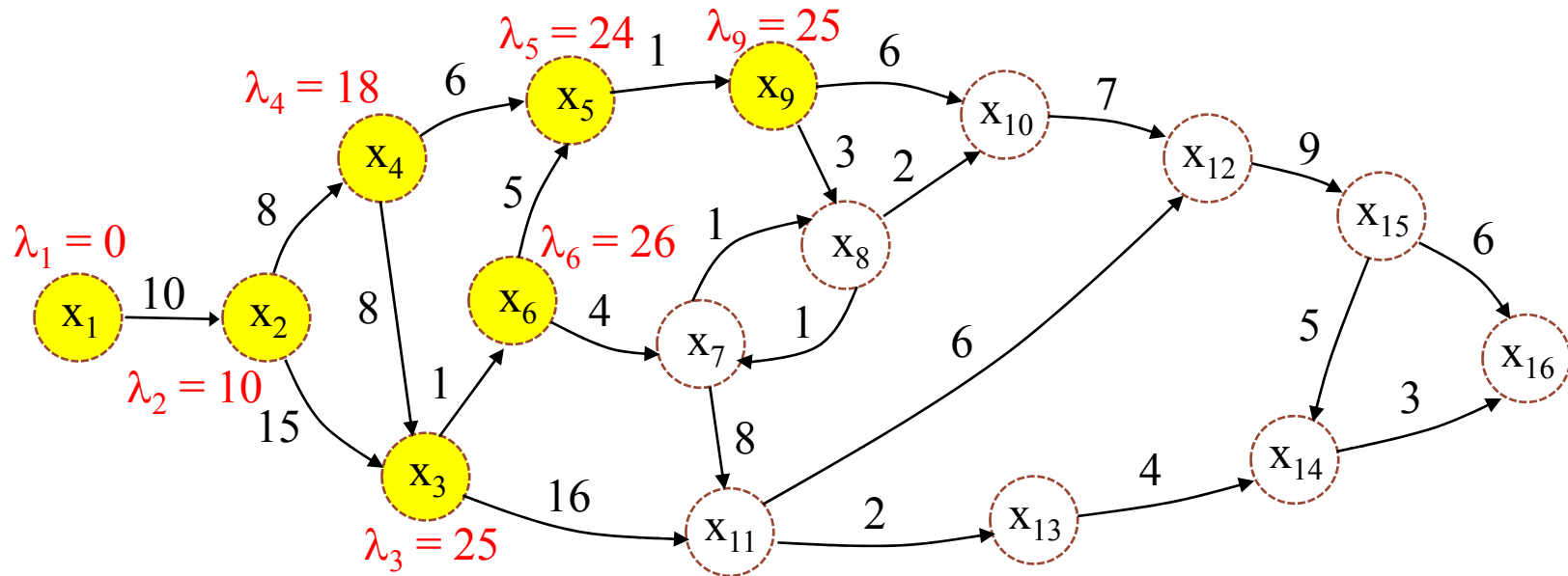
$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

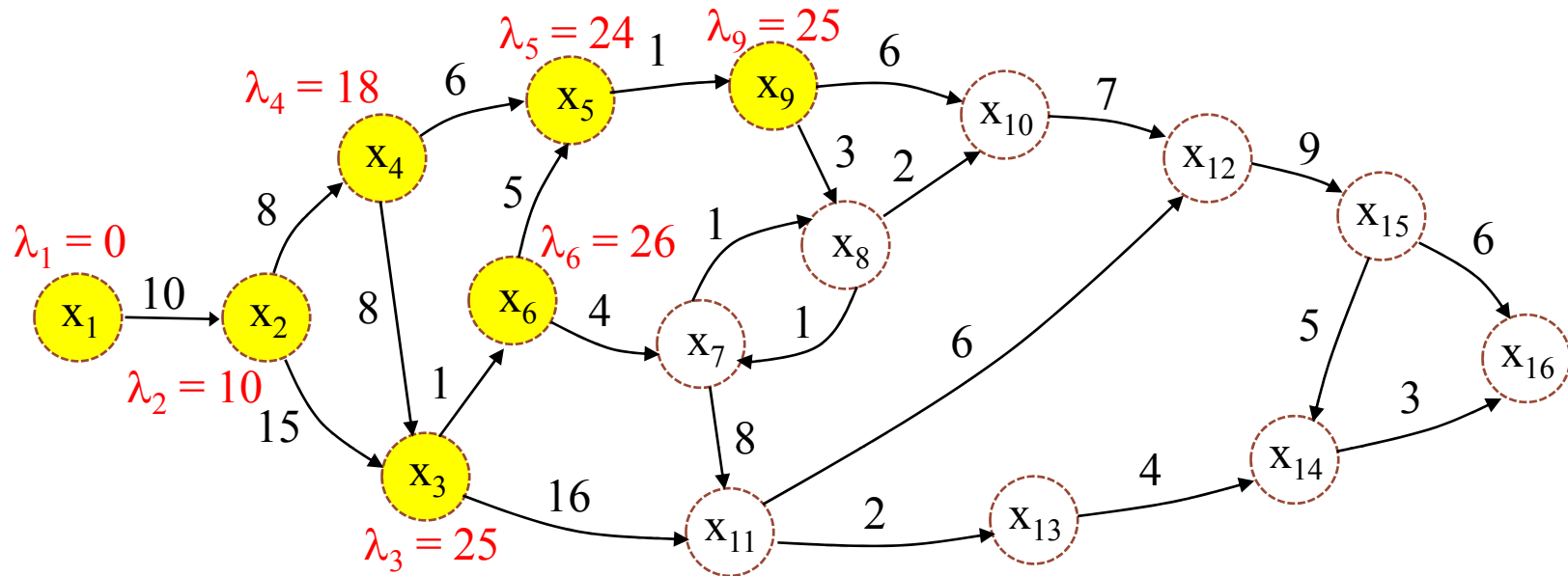
$$E_7 = \{ x_1, x_2, x_4, x_5, x_3, x_9, x_6 \}$$

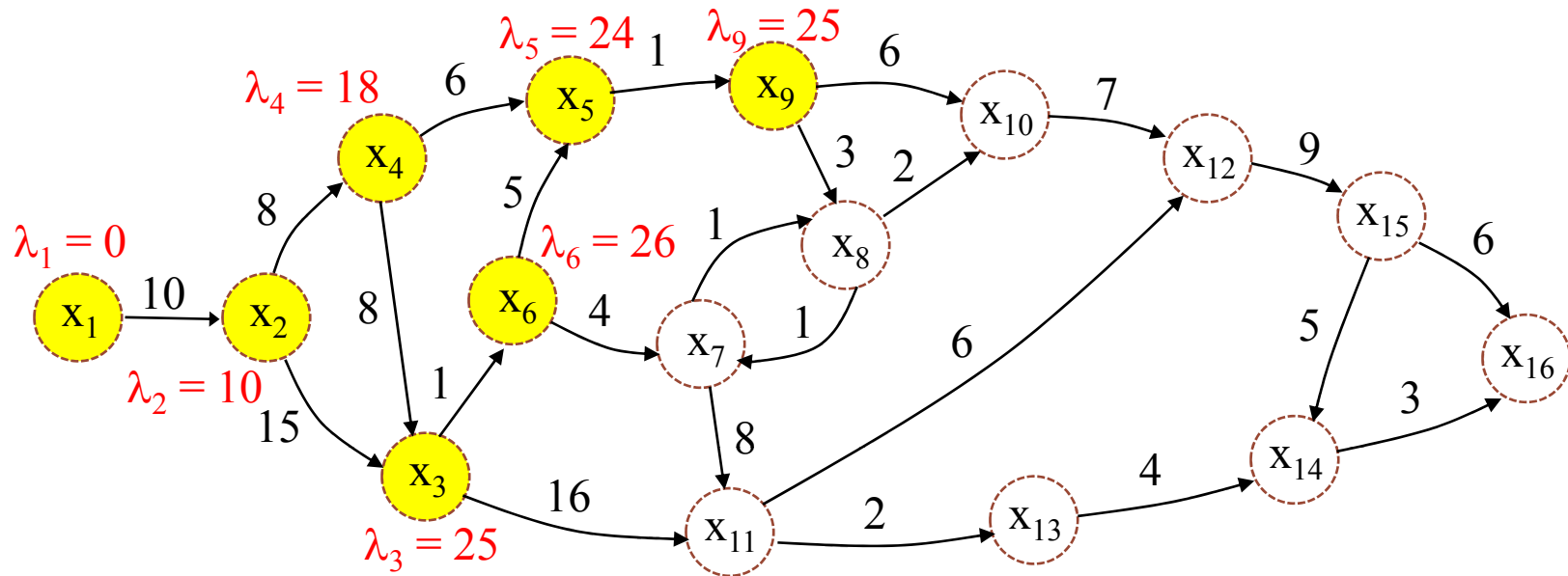
 $k = 7$



$k = 7$ $v(x_3, x_{11})$

 $k = 7$ $v(x_3, x_{11})$ $v(x_6, x_7)$

 $k = 7$ $v(x_3, x_{11})$ $v(x_6, x_7)$ $v(x_9, x_8)$

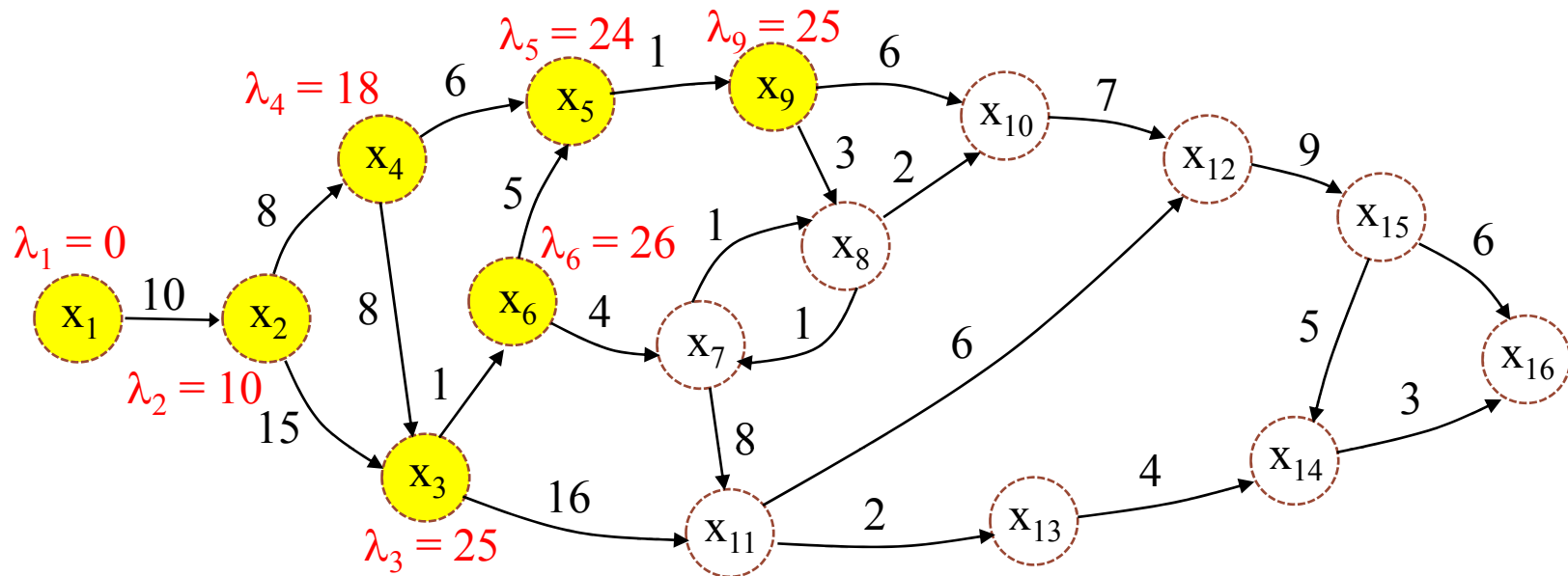


k = 7

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7)$$

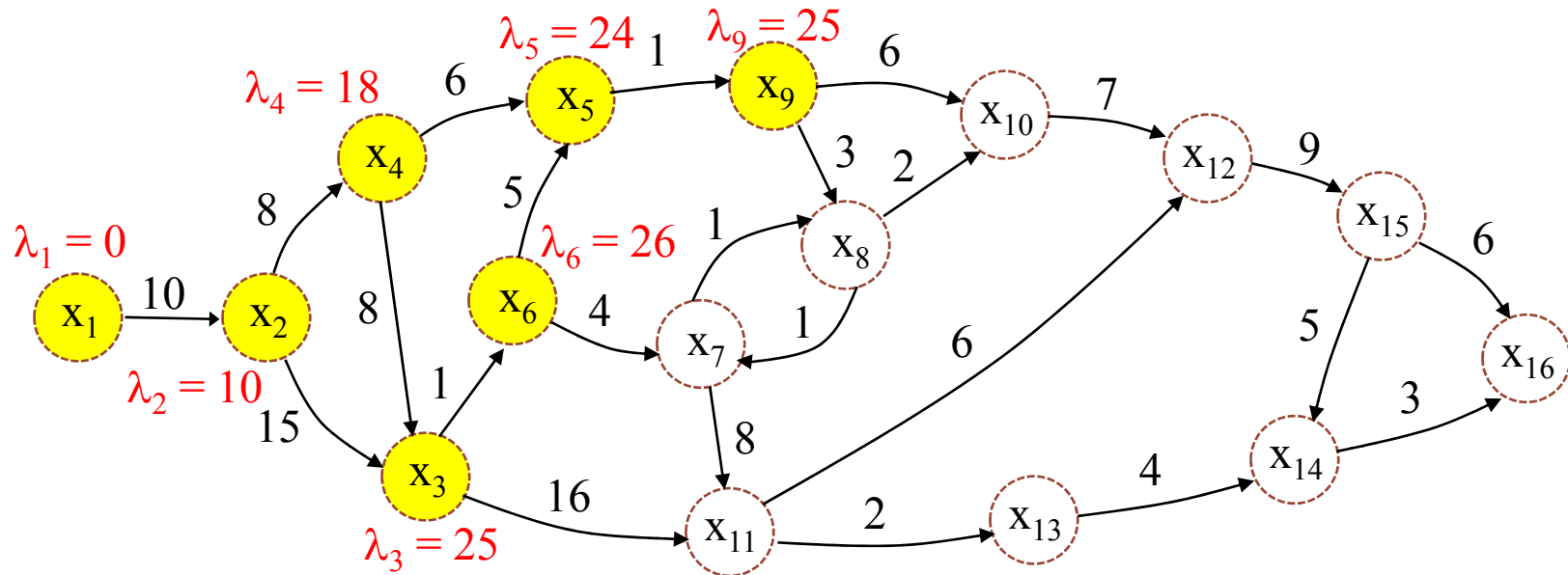
$$v(x_9, x_8)$$

**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

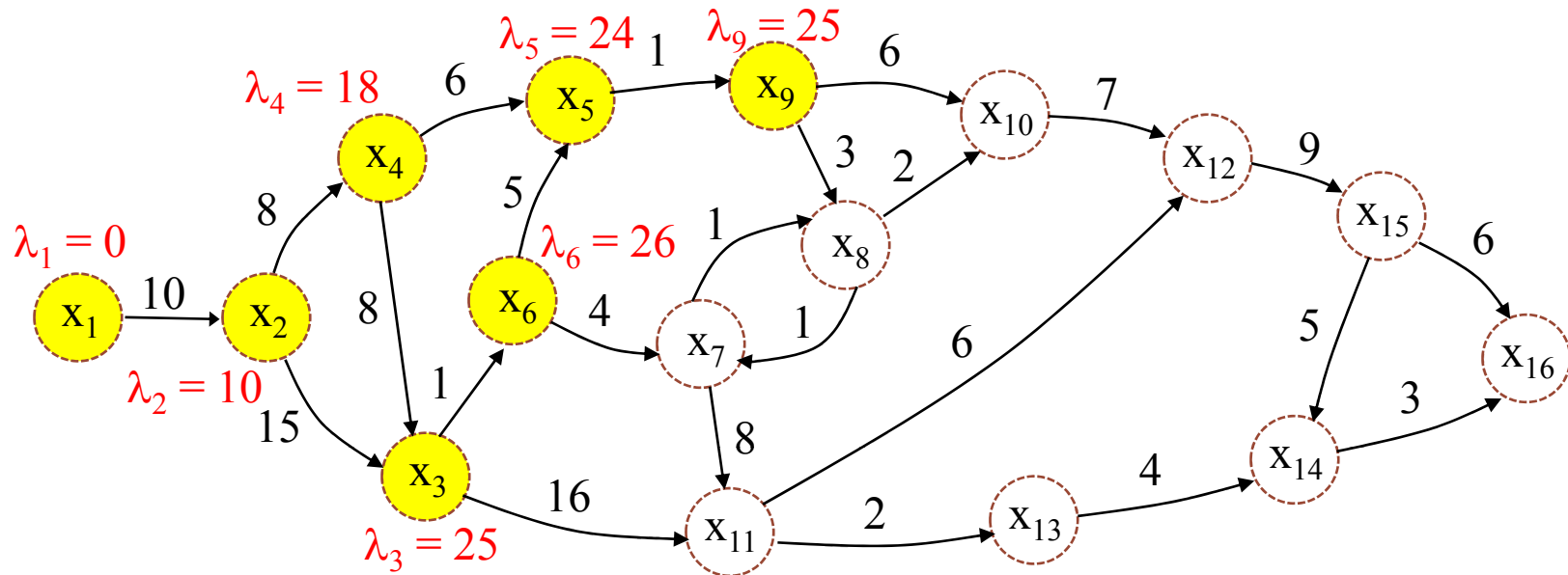
$$v(x_9, x_8)$$

**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

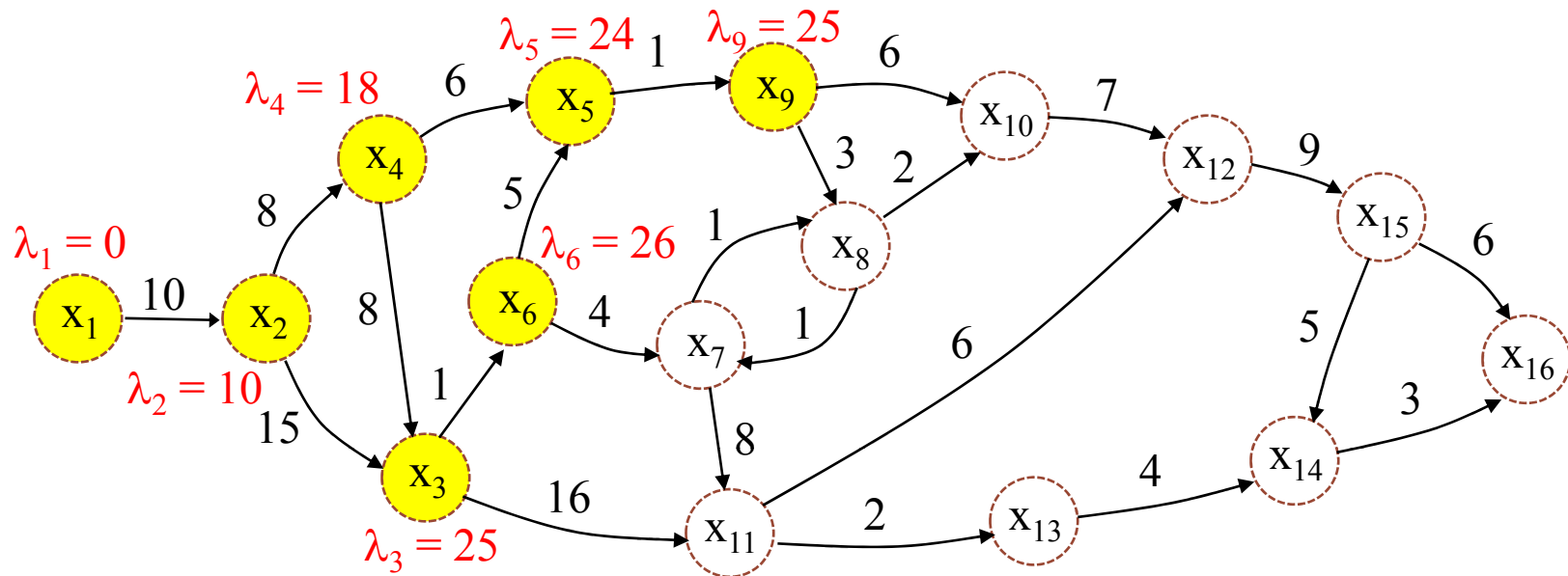


k = 7

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(\mathbf{x}_6, \mathbf{x}_7) \quad \lambda_7 = \lambda_6 + v(\mathbf{x}_6, \mathbf{x}_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

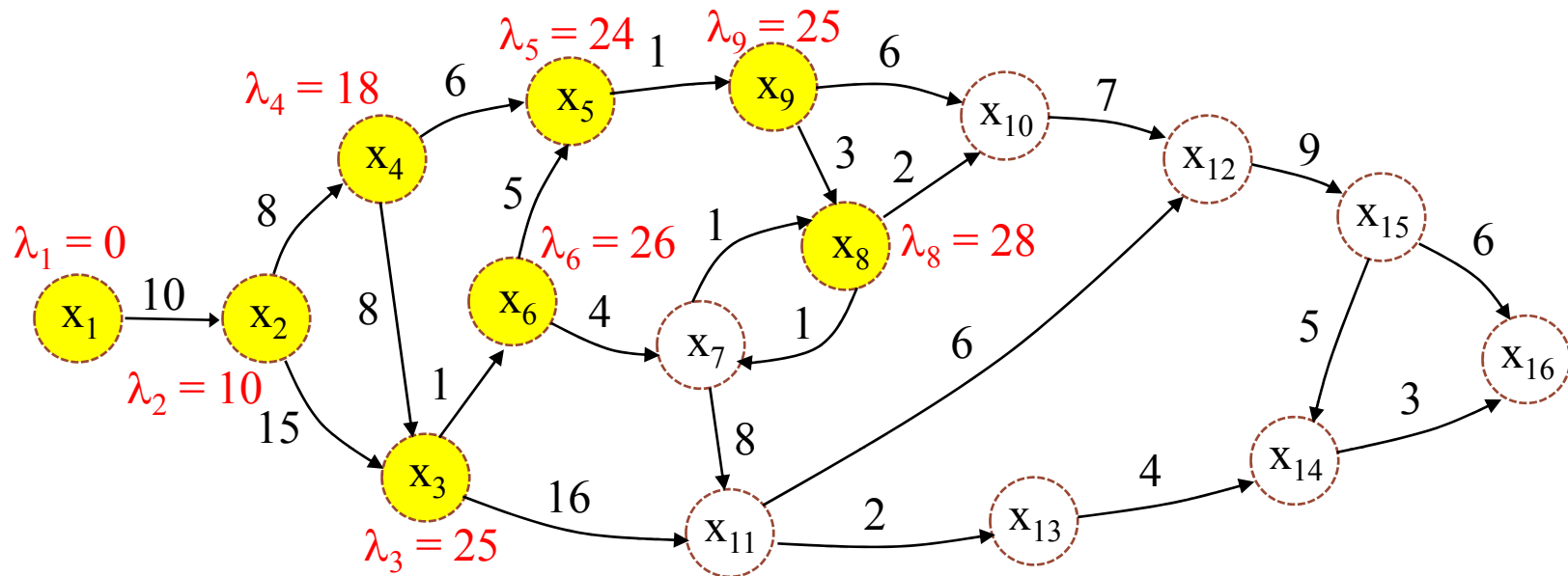
**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

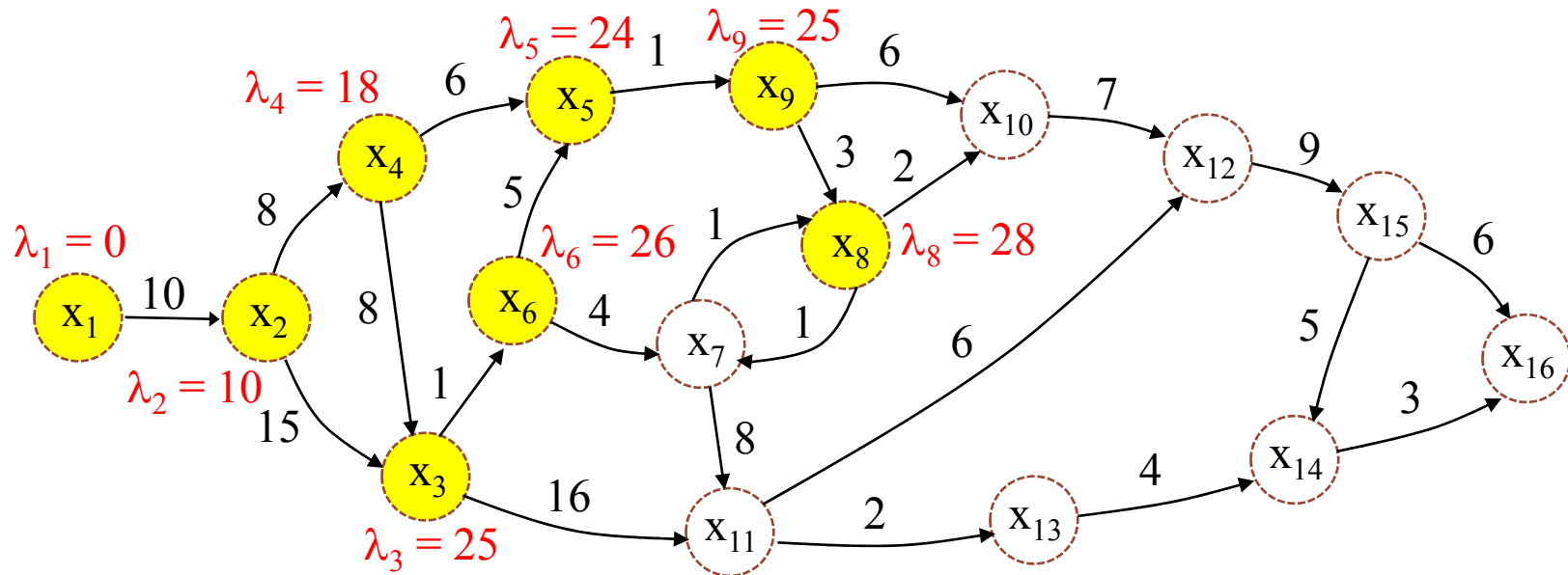
**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

**k = 7**

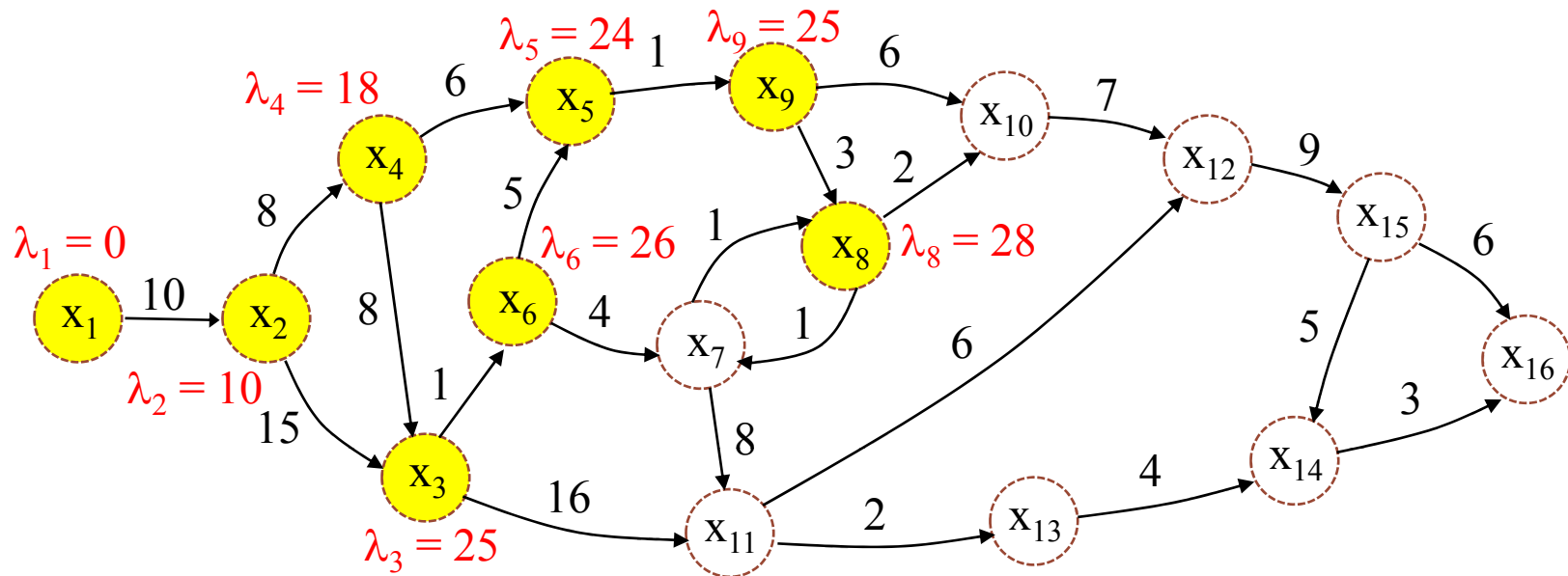
$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

k = 8

**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

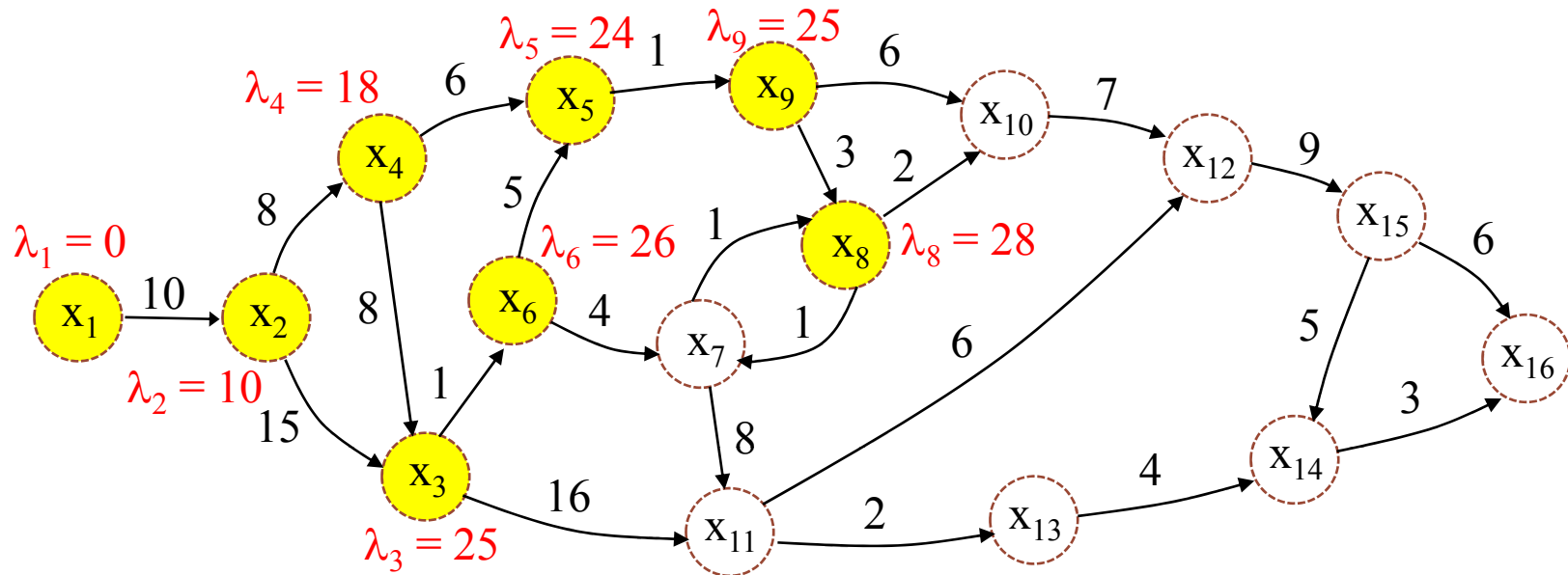
$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_8 = E_7 \cup \{x_8\}$$

k = 8

$$v(x_3, x_{11})$$



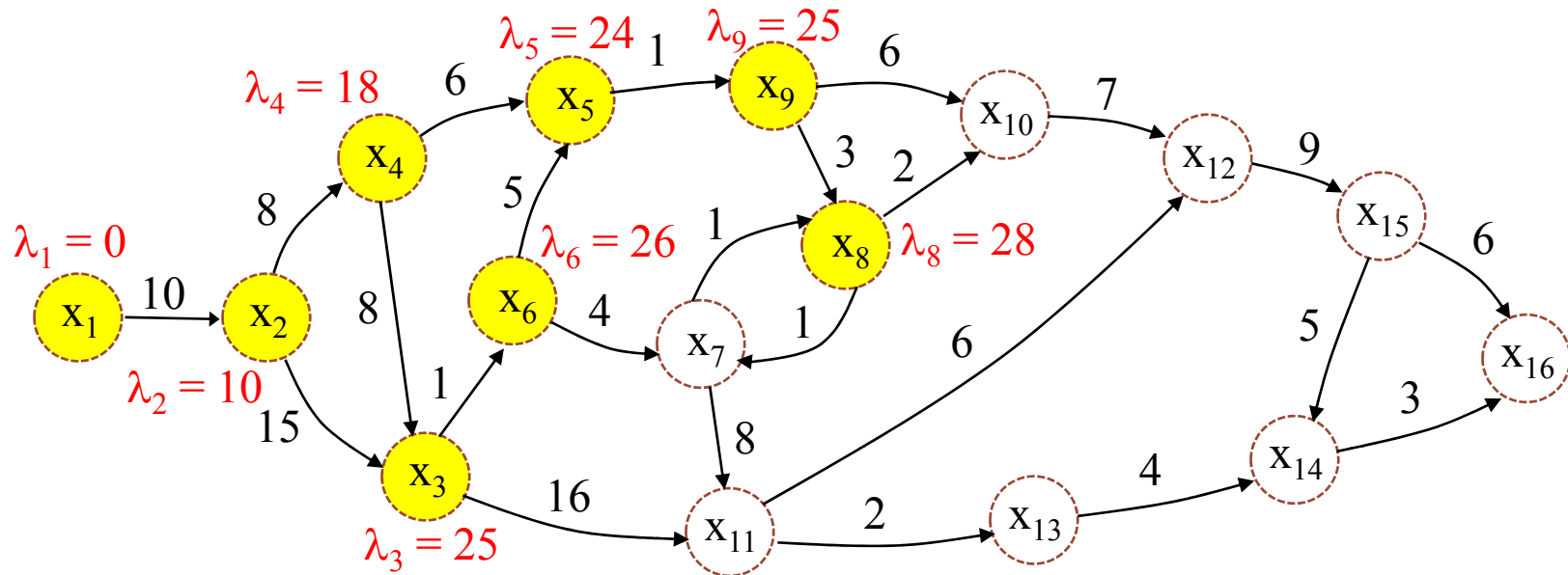
k = 7

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\ v(x_9, x_8) \quad \lambda_8 &= \lambda_9 + v(x_9, x_8) = 28 \end{aligned}$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

k = 8

$$\begin{aligned} v(x_3, x_{11}) \\ v(x_6, x_7) \end{aligned}$$

**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

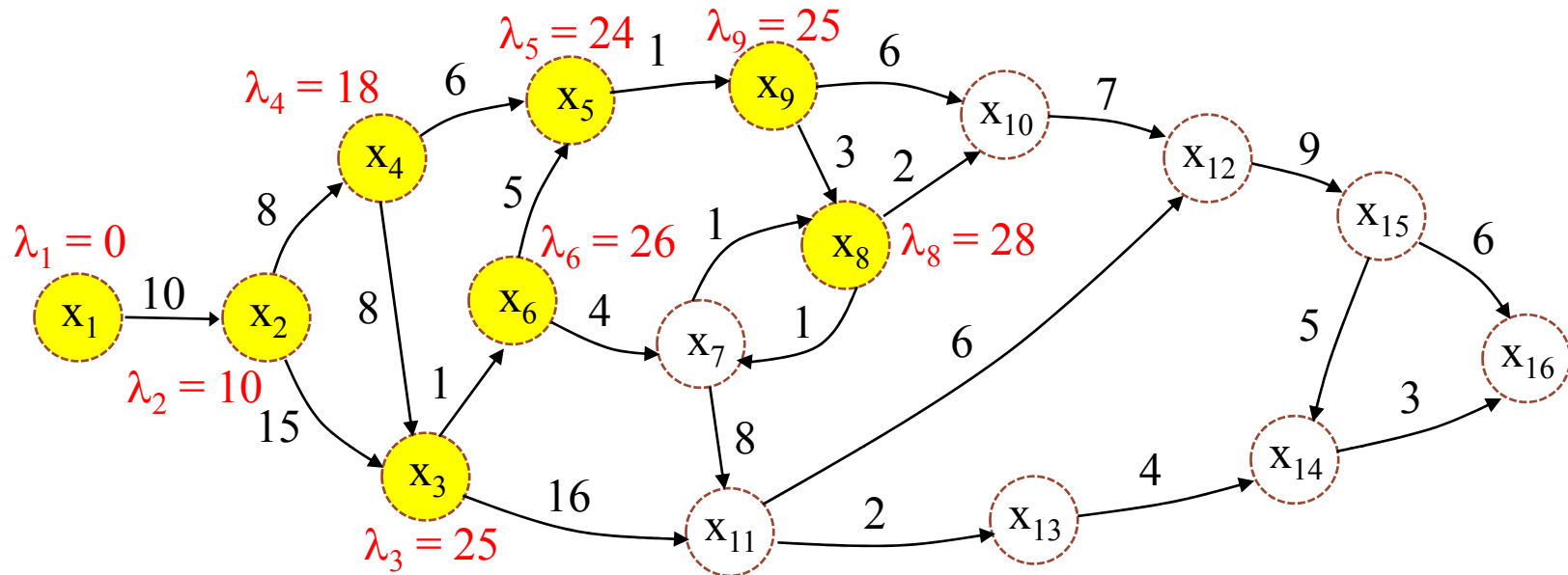
$$E_8 = E_7 \cup \{x_8\}$$

k = 8

$$v(x_3, x_{11})$$

$$v(x_6, x_7)$$

$$v(x_8, x_7)$$

**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_8 = E_7 \cup \{x_8\}$$

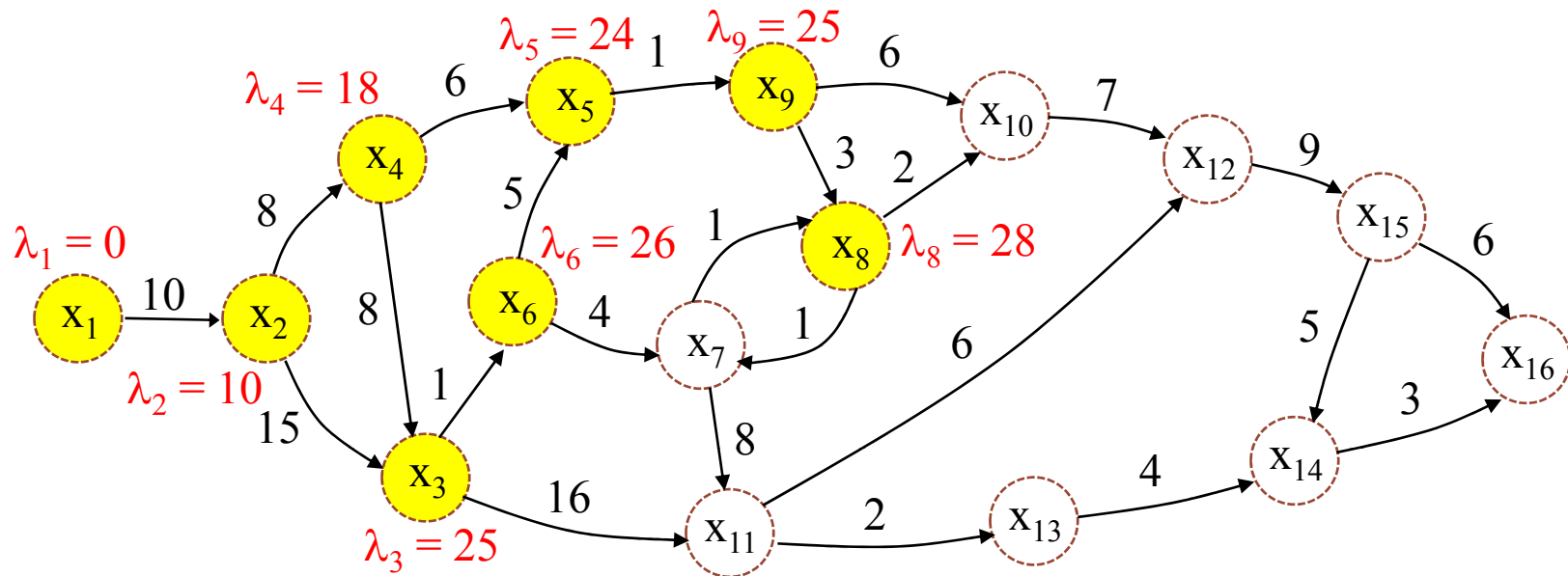
k = 8

$$v(x_3, x_{11})$$

$$v(x_6, x_7)$$

$$v(x_8, x_7)$$

$$v(x_9, x_{10})$$

**k = 7**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7) \quad \lambda_7 = \lambda_6 + v(x_6, x_7) = 30$$

$$v(x_9, x_8) \quad \lambda_8 = \lambda_9 + v(x_9, x_8) = 28$$

$$E_8 = E_7 \cup \{x_8\}$$

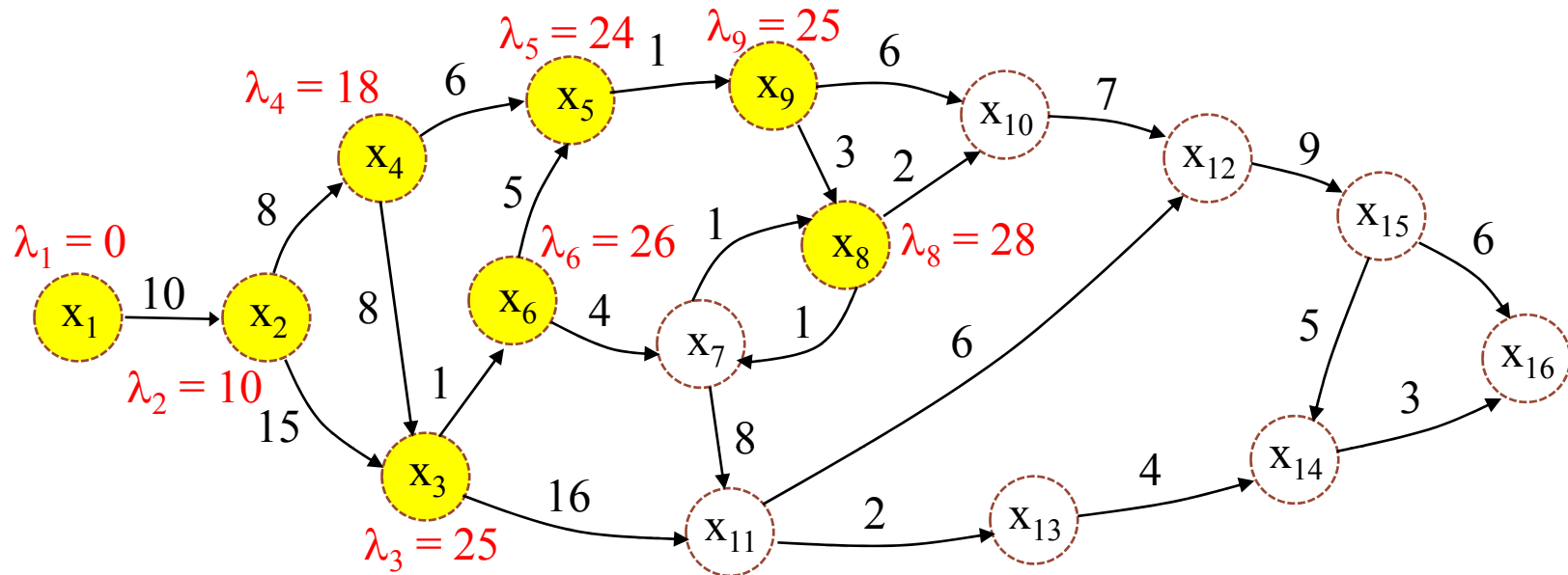
k = 8

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_6, x_7)$$

$$v(x_8, x_7)$$

$$v(x_9, x_{10})$$



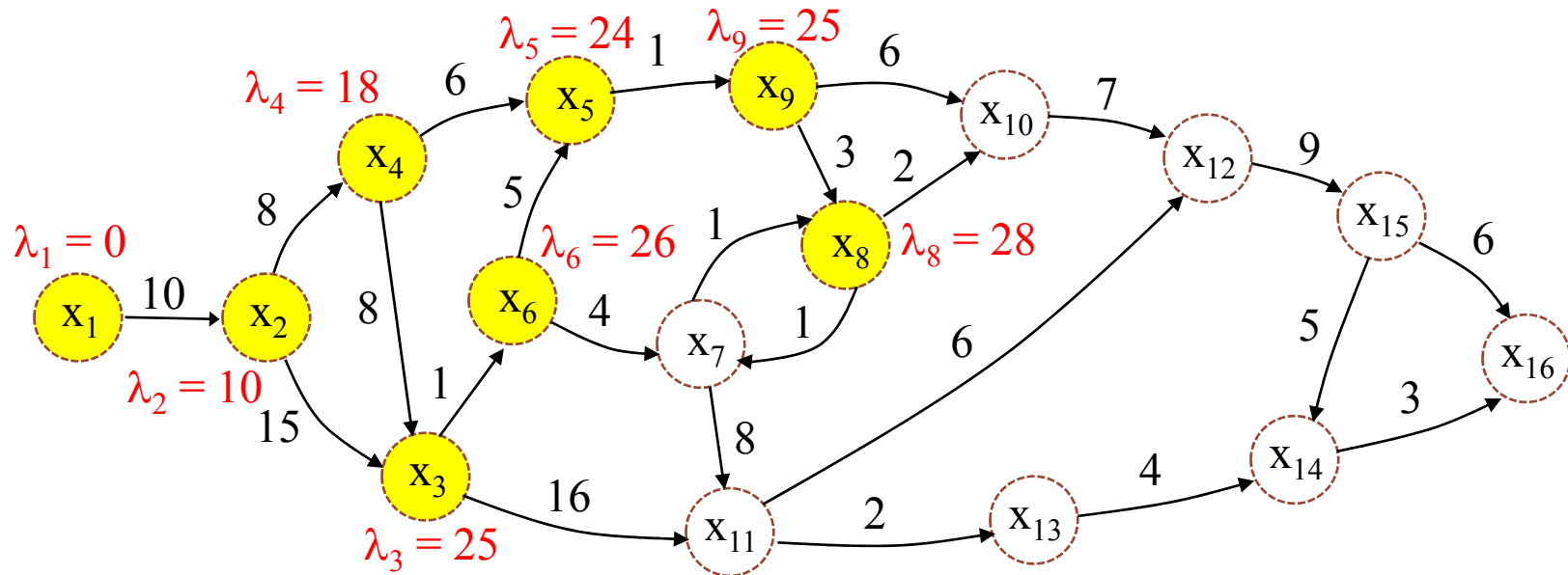
k = 7

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\ v(x_9, x_8) \quad \lambda_8 &= \lambda_9 + v(x_9, x_8) = 28 \end{aligned}$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

k = 8

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\ v(x_8, x_7) \quad \lambda_7 &= \lambda_8 + v(x_8, x_7) = 29 \\ v(x_9, x_{10}) \end{aligned}$$



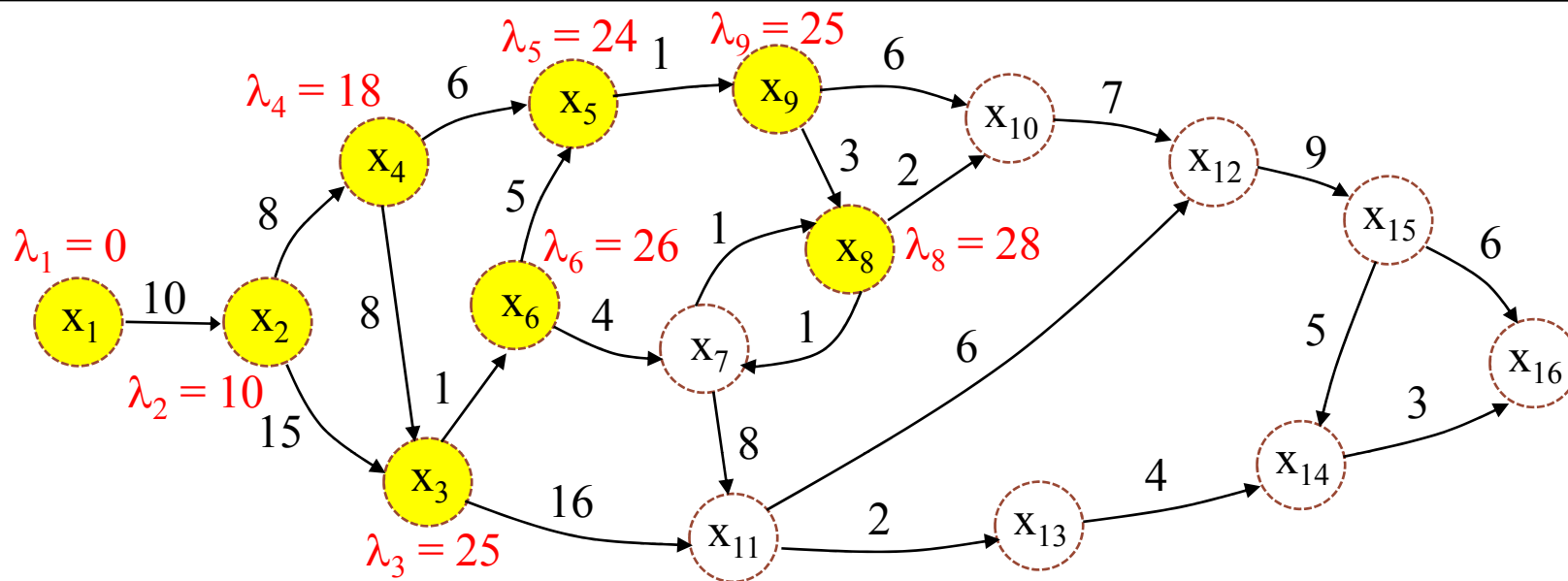
k = 7

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\ v(x_9, x_8) \quad \lambda_8 &= \lambda_9 + v(x_9, x_8) = 28 \end{aligned}$$

$$E_8 = E_7 \cup \{ \mathbf{x_8} \}$$

k = 8

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\ v(x_8, x_7) \quad \lambda_7 &= \lambda_8 + v(x_8, x_7) = 29 \\ v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31 \end{aligned}$$

**k = 7**

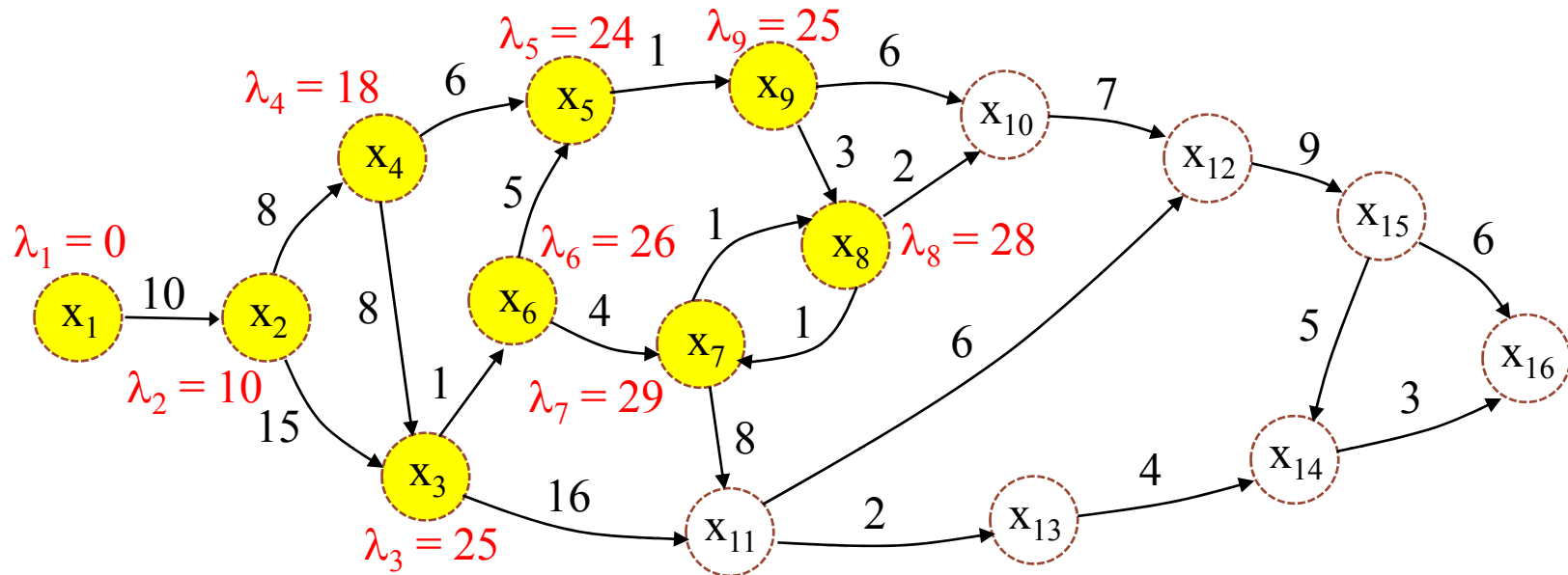
$$\begin{aligned}
 v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\
 v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\
 v(x_9, x_8) \quad \lambda_8 &= \lambda_9 + v(x_9, x_8) = 28
 \end{aligned}$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

k = 8

$$\begin{aligned}
 v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\
 v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\
 v(x_8, x_7) \quad \lambda_7 &= \lambda_8 + v(x_8, x_7) = 29 \\
 v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31
 \end{aligned}$$

$$E_9 = E_8 \cup \{ \mathbf{x}_7 \}$$

**k = 7**

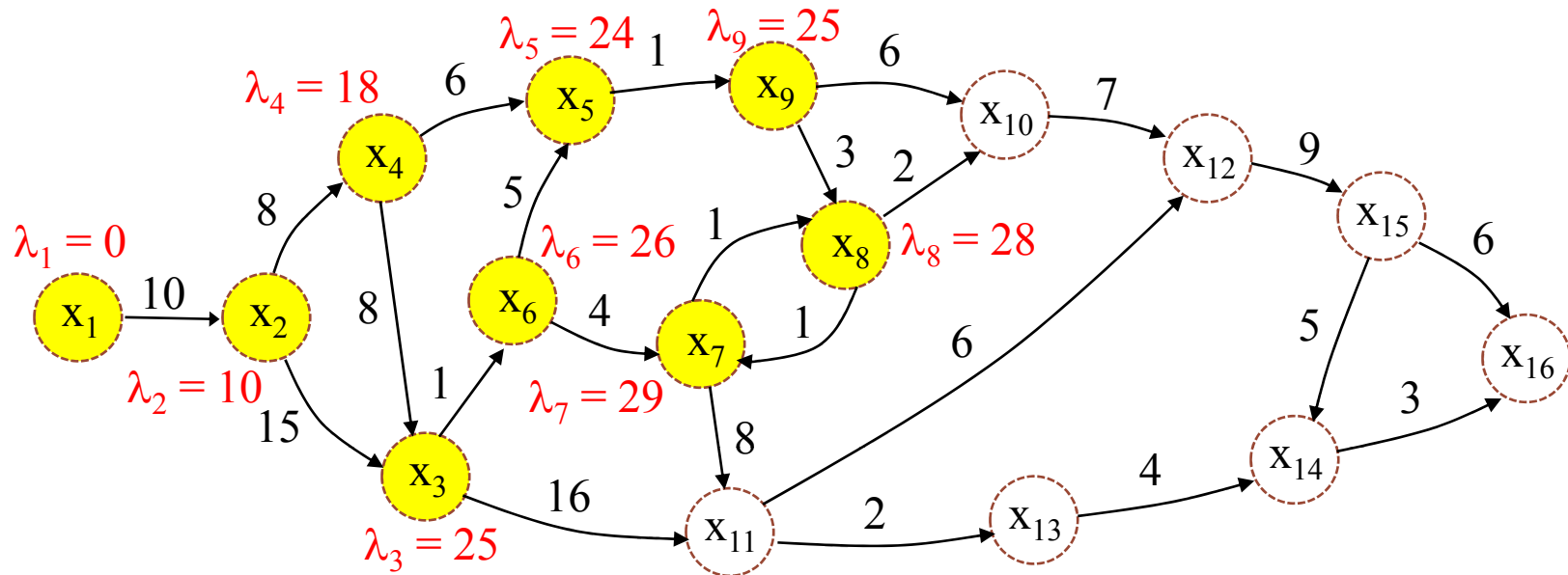
$$\begin{aligned}
 v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\
 v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\
 v(x_9, x_8) \quad \lambda_8 &= \lambda_9 + v(x_9, x_8) = 28
 \end{aligned}$$

$$E_8 = E_7 \cup \{ \mathbf{x}_8 \}$$

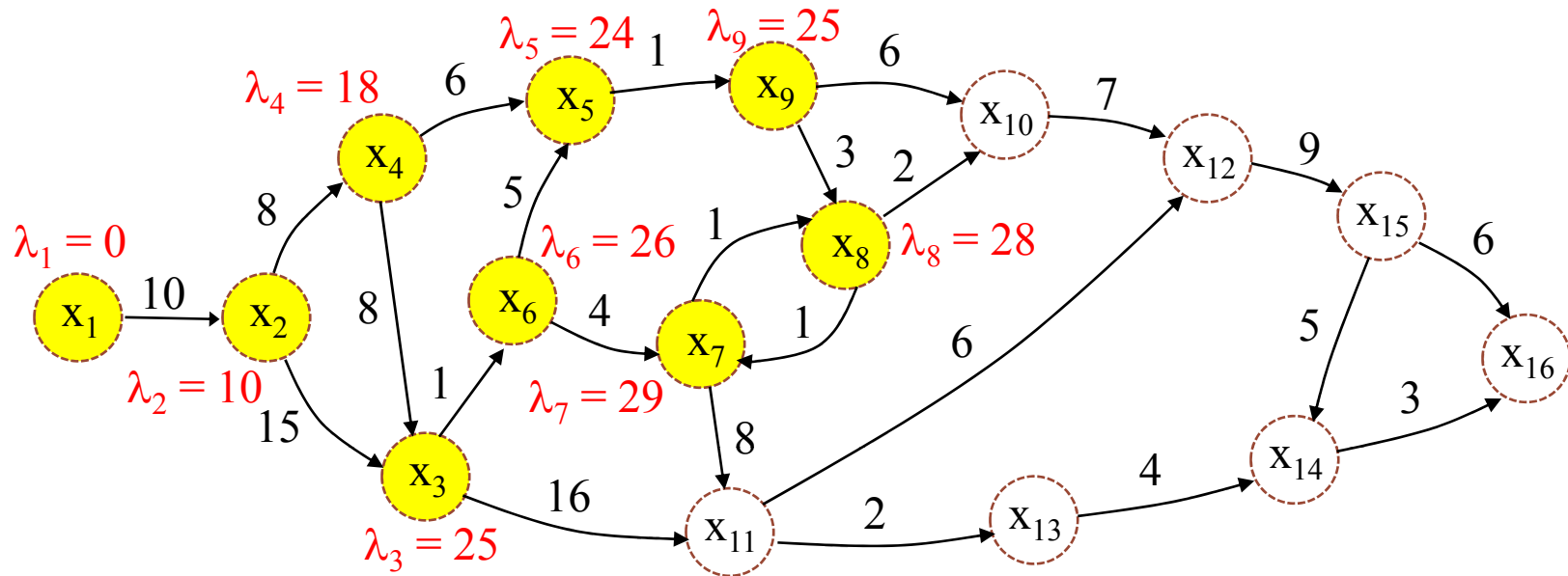
k = 8

$$\begin{aligned}
 v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\
 v(x_6, x_7) \quad \lambda_7 &= \lambda_6 + v(x_6, x_7) = 30 \\
 v(x_8, x_7) \quad \lambda_7 &= \lambda_8 + v(x_8, x_7) = 29 \\
 v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31
 \end{aligned}$$

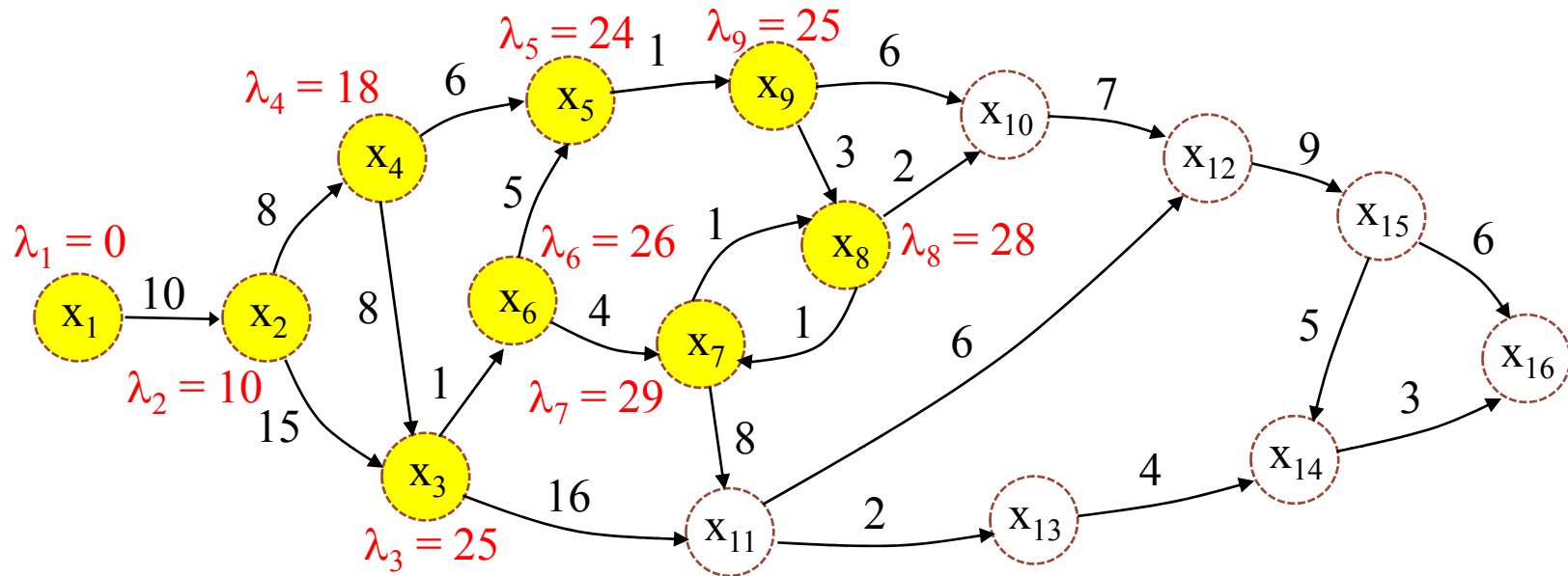
$$E_9 = E_8 \cup \{ \mathbf{x}_7 \}$$



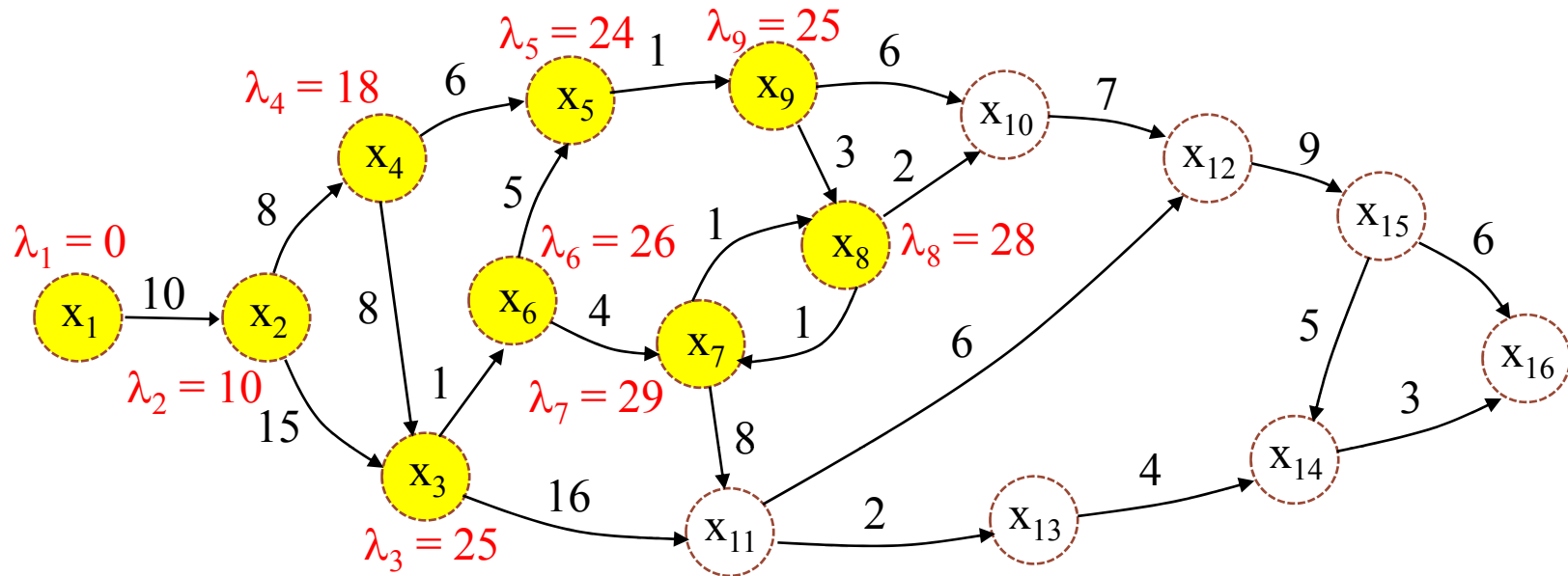
$k = 9$



$k=9$ $v(x_3, x_{11})$

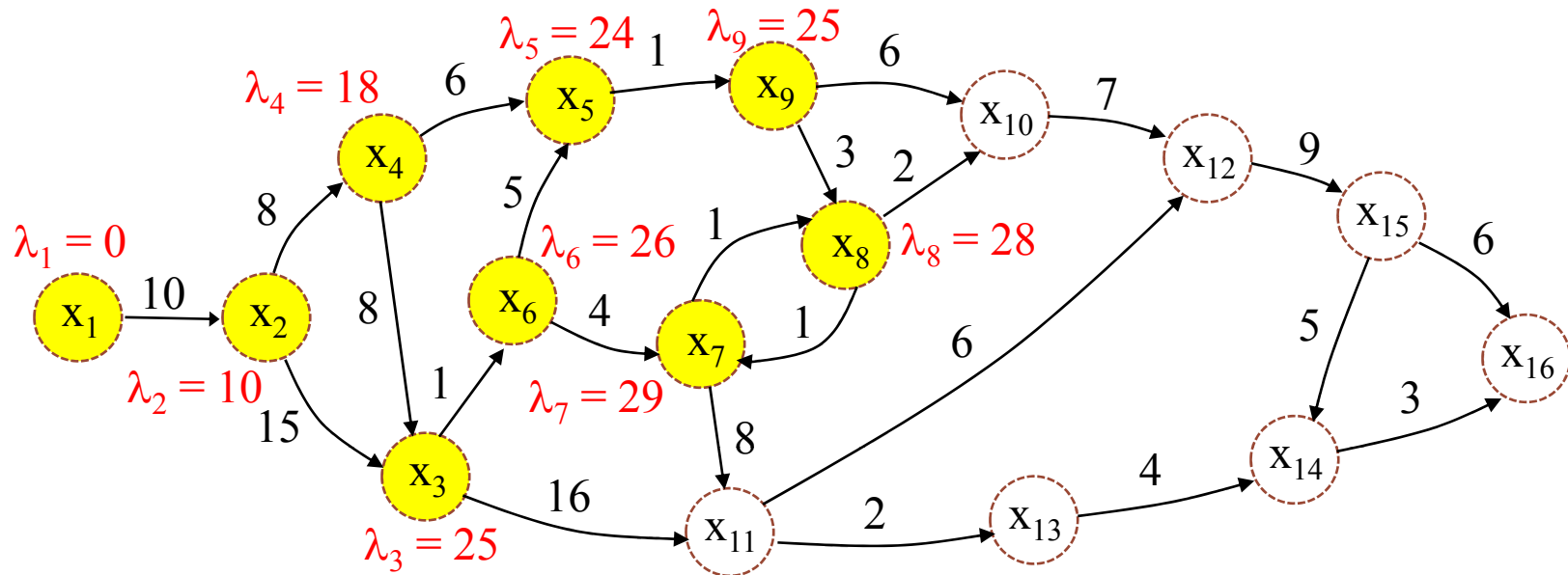


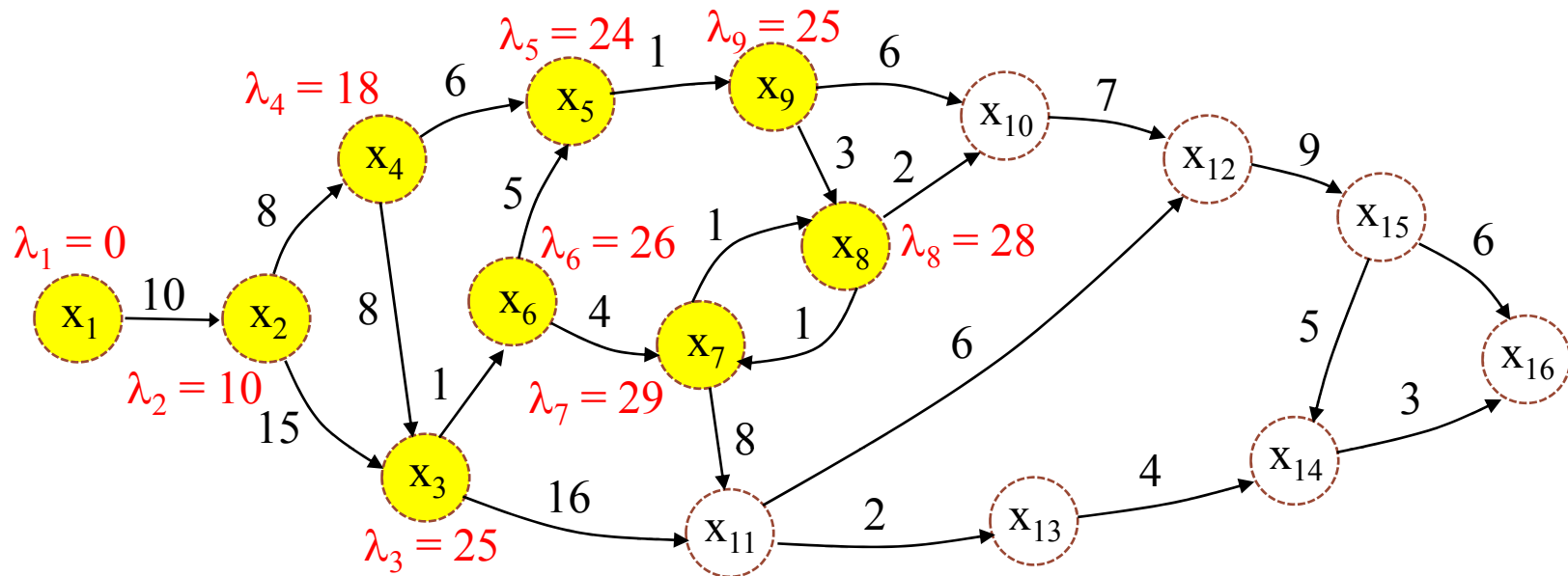
$k=9$ $v(x_3, x_{11})$
 $v(x_7, x_{11})$



k = 9

- $v(x_3, x_{11})$
- $v(x_7, x_{11})$
- $v(x_8, x_{10})$

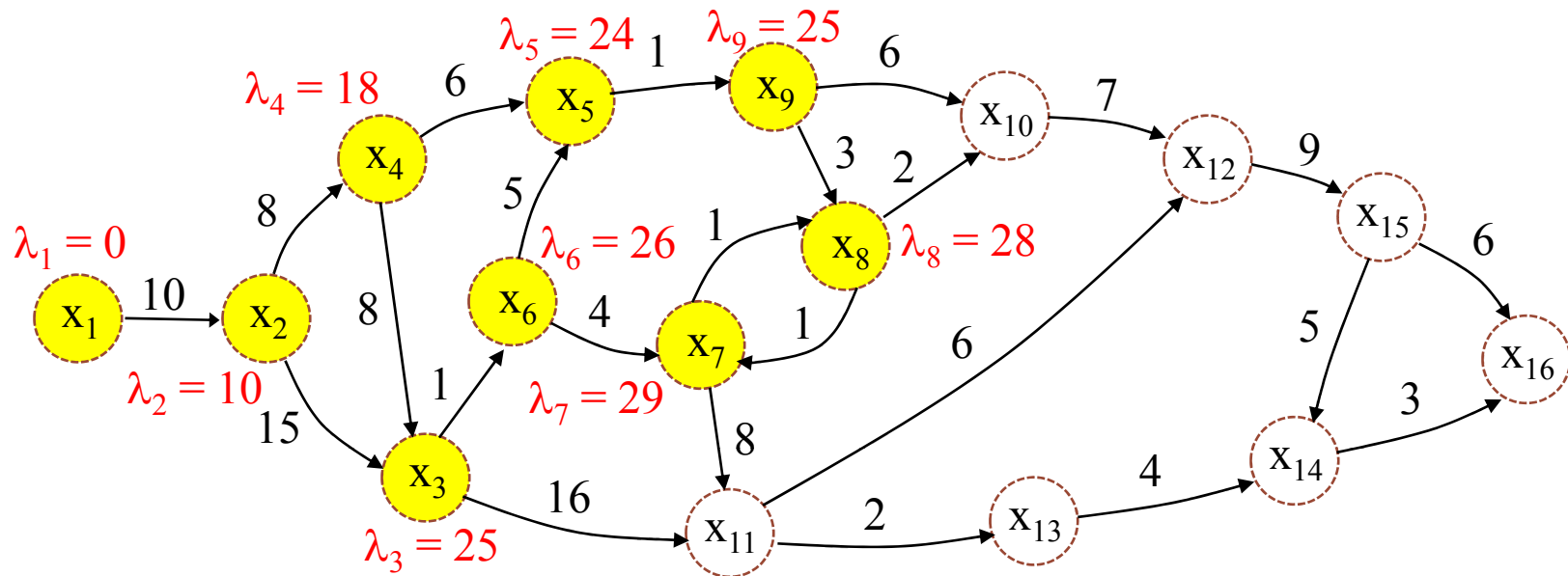
**k = 9** $v(x_3, x_{11})$ $v(x_7, x_{11})$ $v(x_8, x_{10})$ $v(x_9, x_{10})$



k=9

$$\lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$v(x_3, x_{11})$
 $v(x_7, x_{11})$
 $v(x_8, x_{10})$
 $v(x_9, x_{10})$

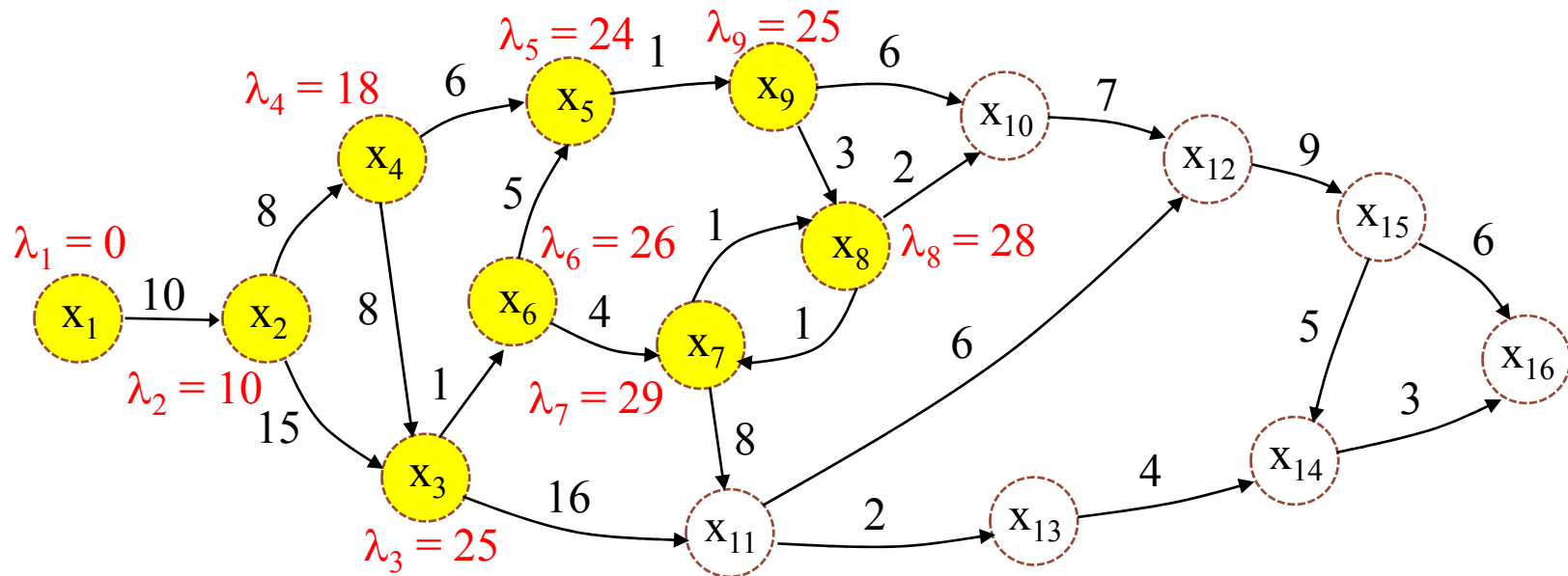
**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10})$$

$$v(x_9, x_{10})$$

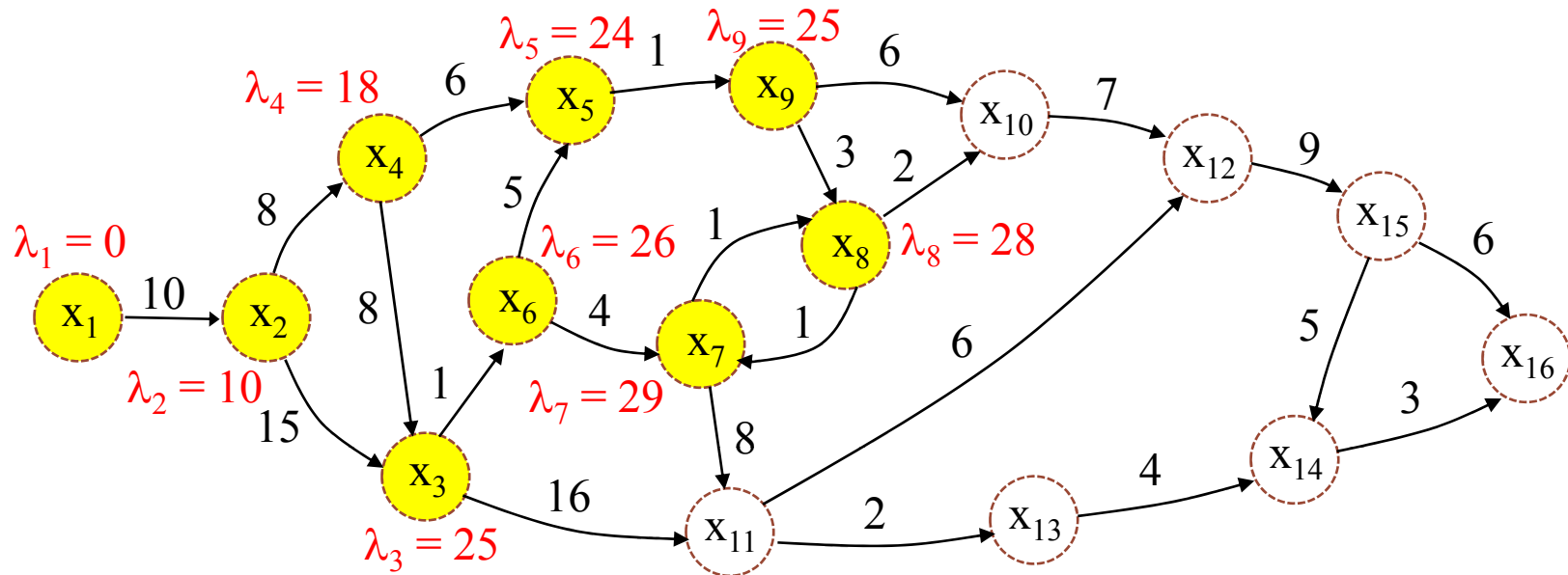
**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10})$$

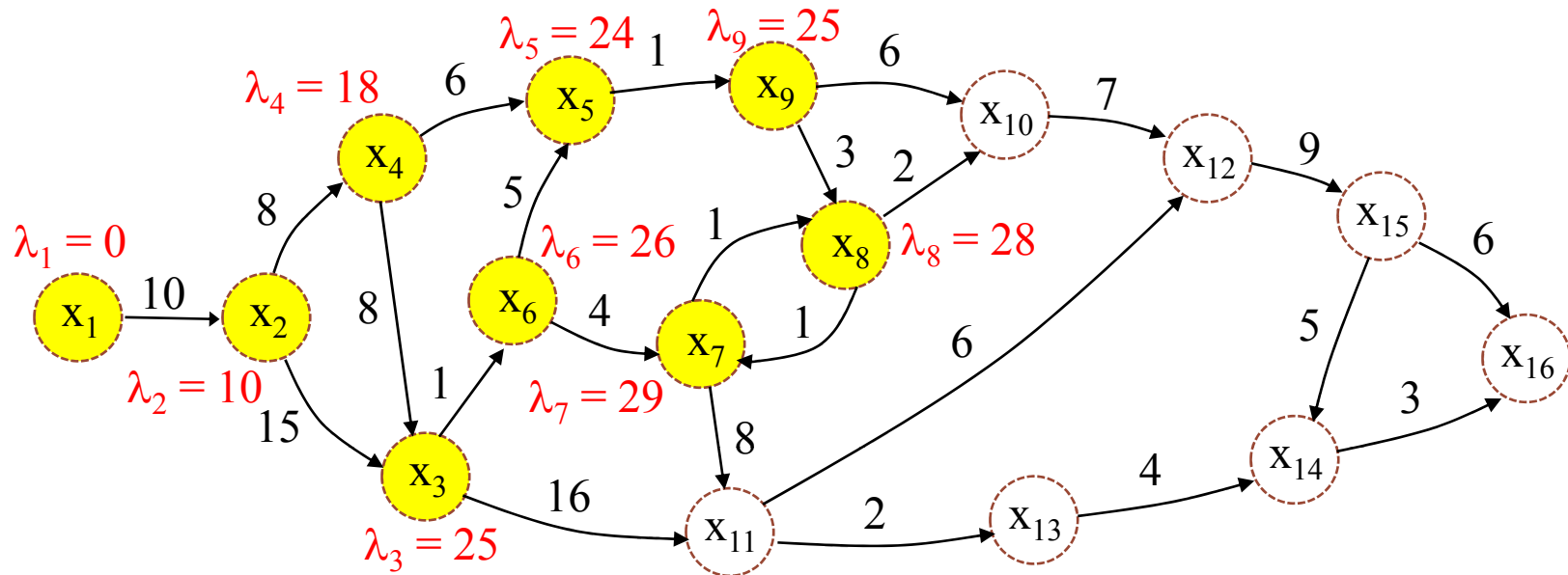
**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

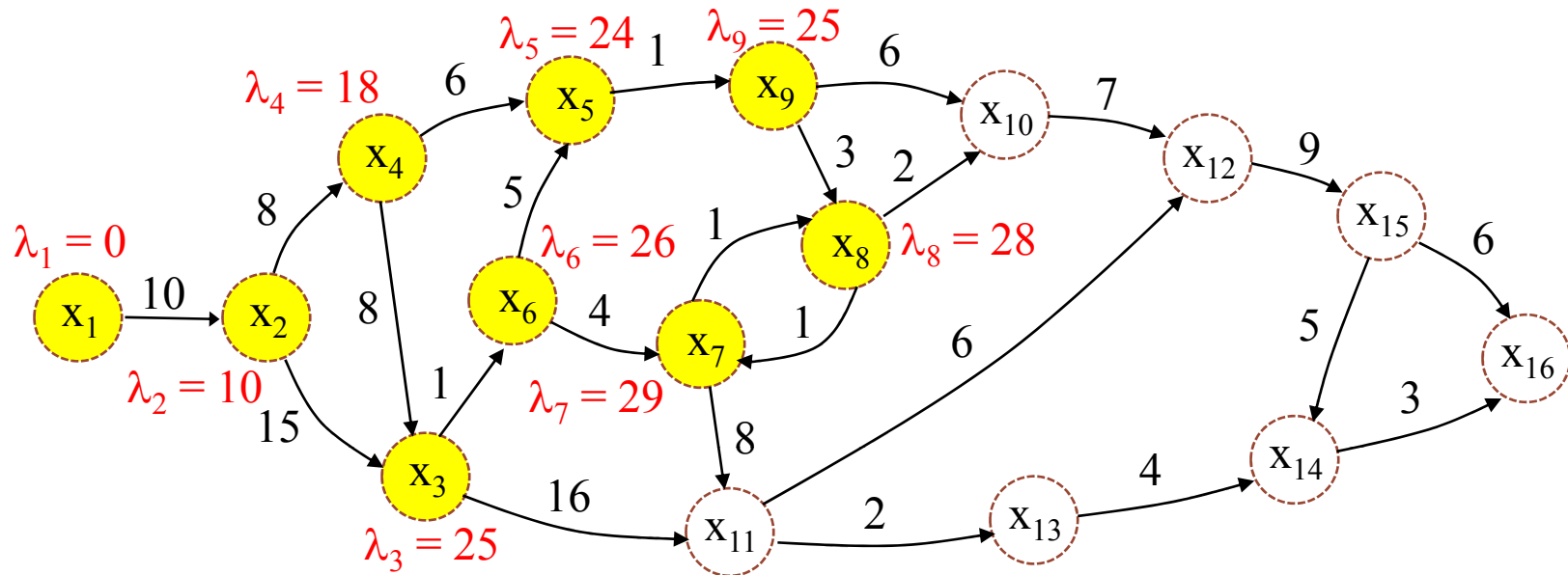
$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

**k = 9**

$$\begin{array}{ll}
 v(x_3, x_{11}) & \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41 \\
 v(x_7, x_{11}) & \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37 \\
 v(x_8, x_{10}) & \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30 \\
 v(x_9, x_{10}) & \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31
 \end{array}$$

**k=9**

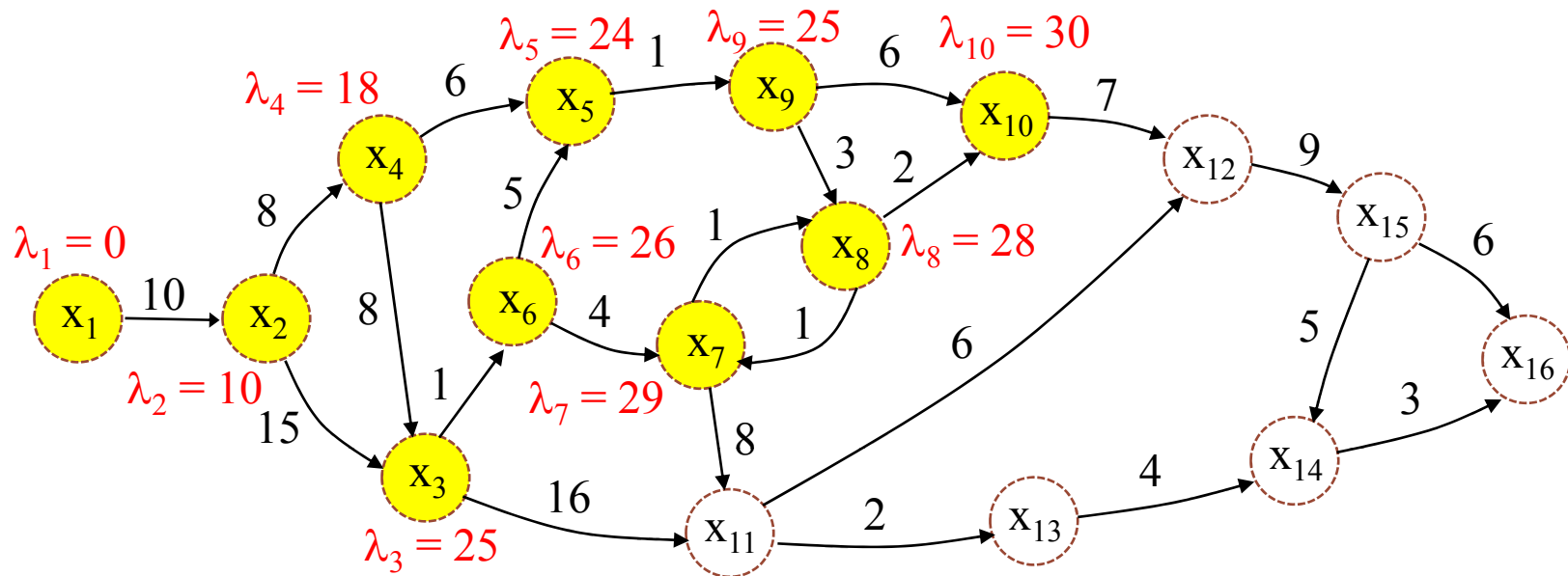
$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

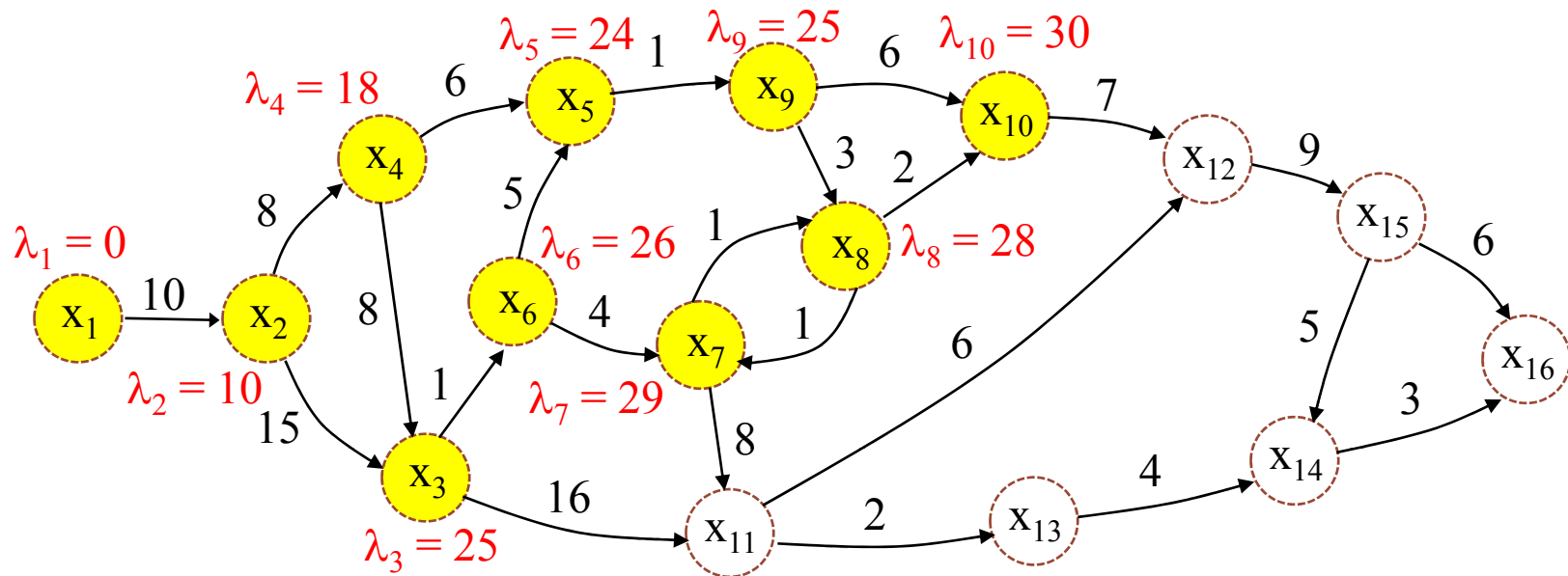
$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

$$E_{10} = E_9 \cup \{ \mathbf{x}_{10} \}$$

**k=9**

$$\begin{array}{ll}
 v(x_3, x_{11}) & \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41 \\
 v(x_7, x_{11}) & \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37 \\
 v(x_8, x_{10}) & \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30 \\
 v(x_9, x_{10}) & \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31
 \end{array}$$

$$E_{10} = E_9 \cup \{ \mathbf{x}_{10} \}$$

**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

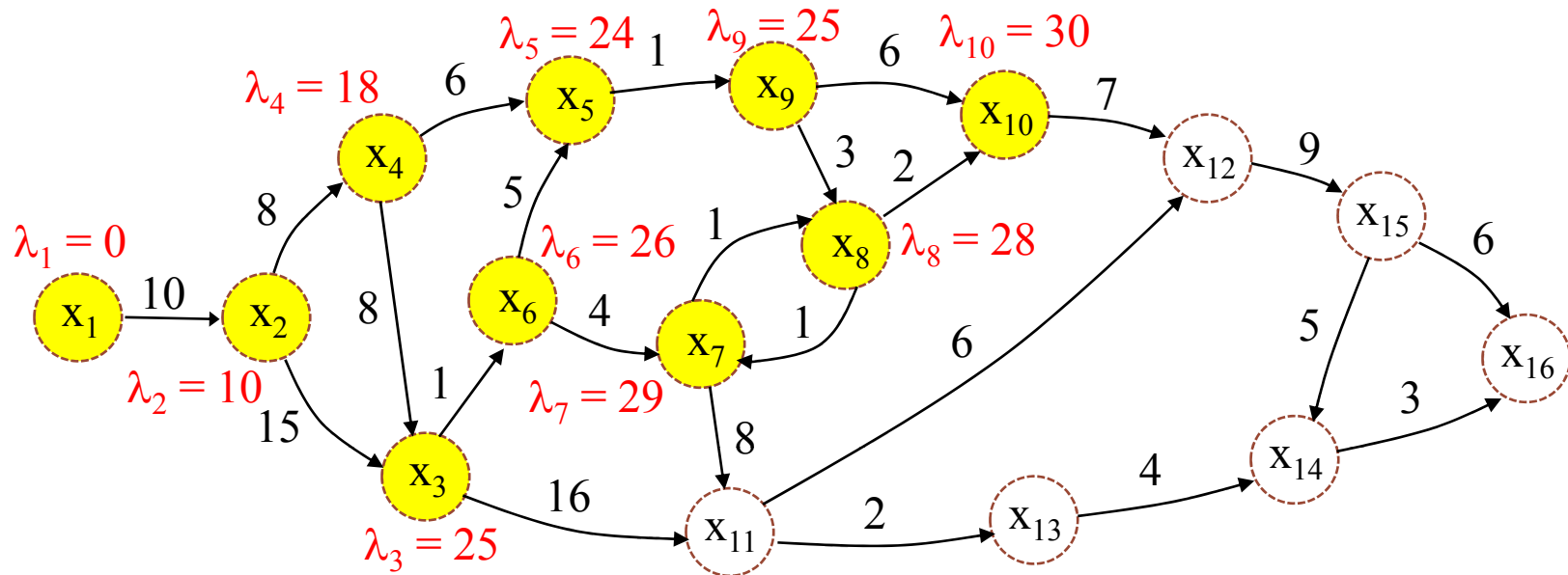
$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

$$E_{10} = E_9 \cup \{ \mathbf{x}_{10} \}$$

k = 10

**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

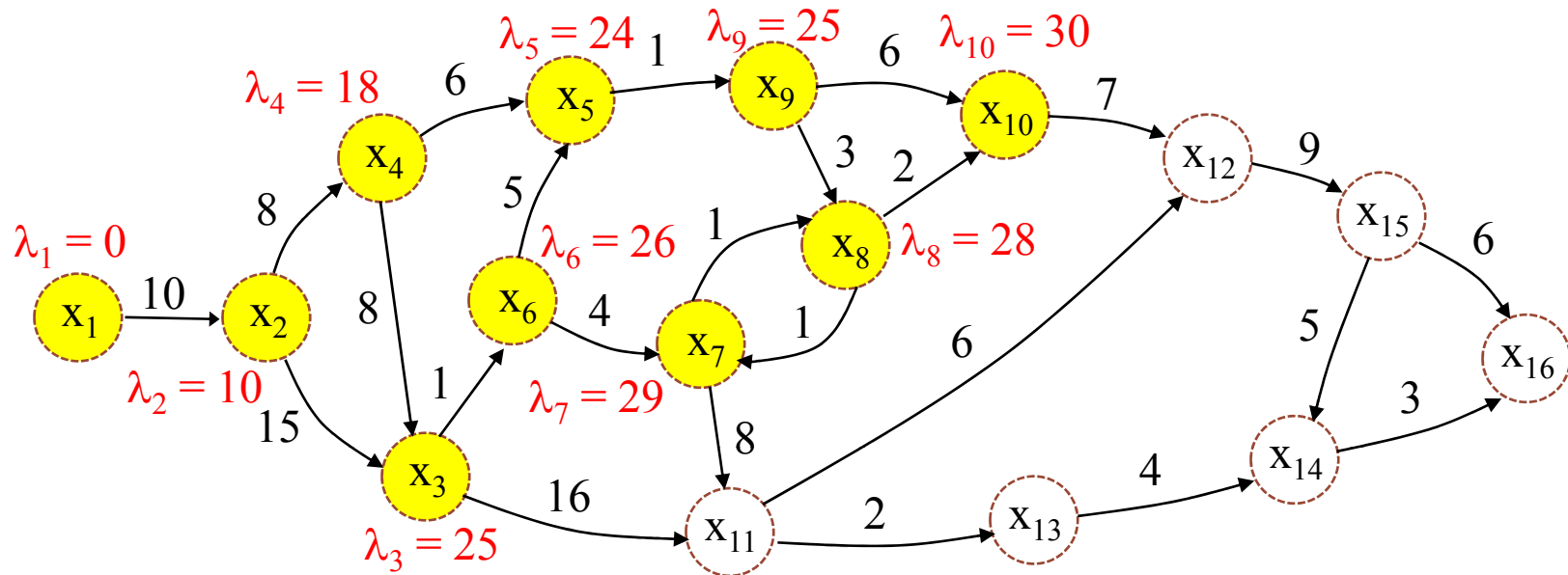
$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

$$E_{10} = E_9 \cup \{x_{10}\}$$

k = 10

$$v(x_3, x_{11})$$



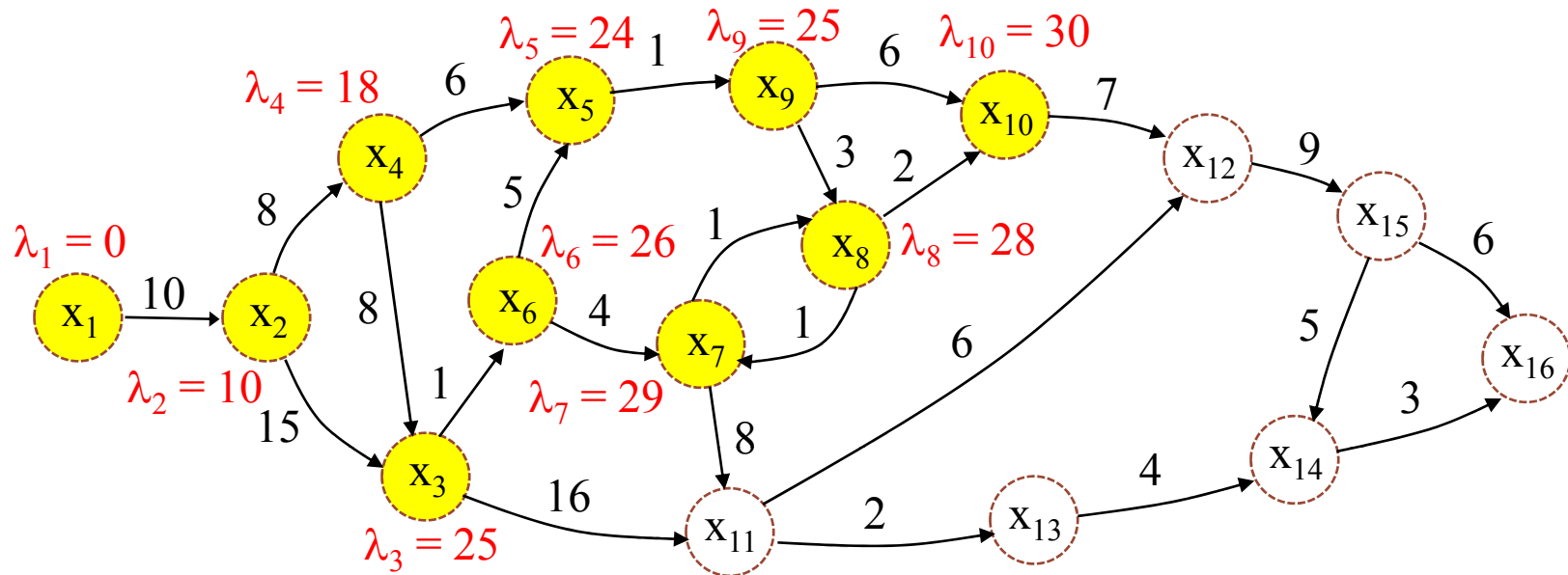
k = 9

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_7, x_{11}) \quad \lambda_{11} &= \lambda_7 + v(x_7, x_{11}) = 37 \\ v(x_8, x_{10}) \quad \lambda_{10} &= \lambda_8 + v(x_8, x_{10}) = 30 \\ v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31 \end{aligned}$$

$$E_{10} = E_9 \cup \{x_{10}\}$$

k = 10

$$\begin{aligned} v(x_3, x_{11}) \\ v(x_7, x_{11}) \end{aligned}$$



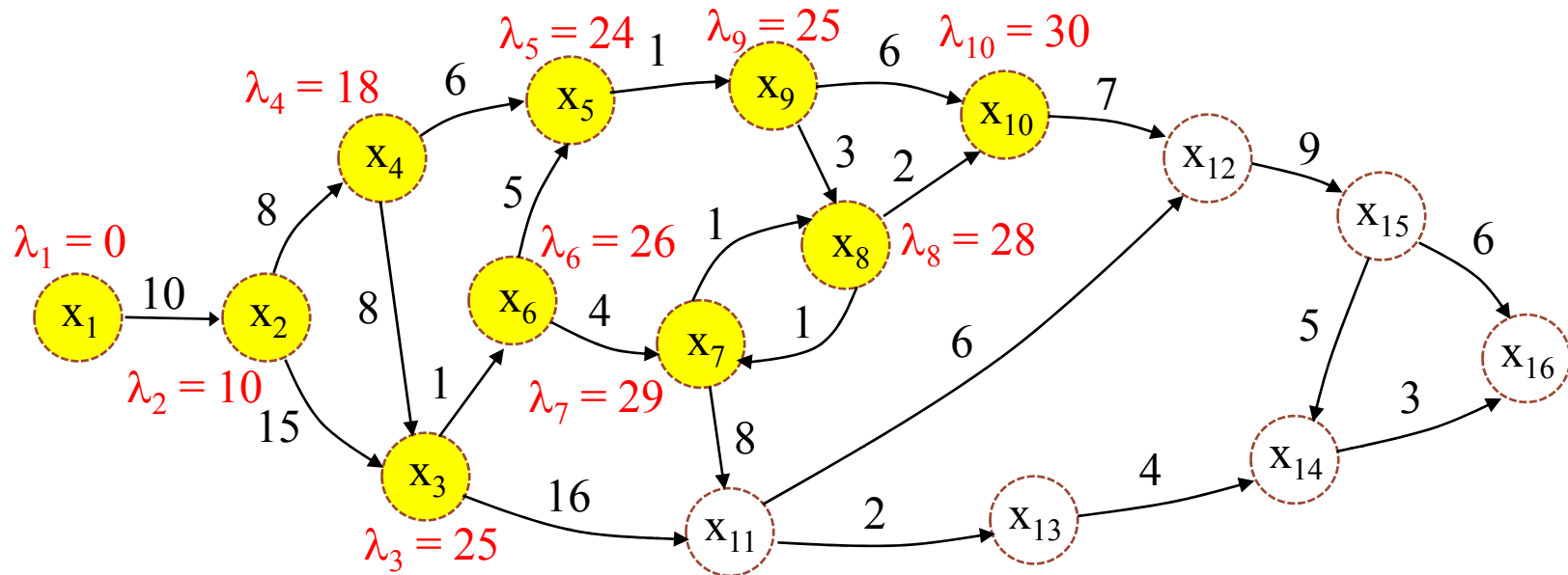
k = 9

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_7, x_{11}) \quad \lambda_{11} &= \lambda_7 + v(x_7, x_{11}) = 37 \\ v(x_8, x_{10}) \quad \lambda_{10} &= \lambda_8 + v(x_8, x_{10}) = 30 \\ v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31 \end{aligned}$$

$$E_{10} = E_9 \cup \{ \mathbf{x}_{10} \}$$

k = 10

$$\begin{aligned} v(x_3, x_{11}) \\ v(x_7, x_{11}) \\ v(x_{10}, x_{12}) \end{aligned}$$

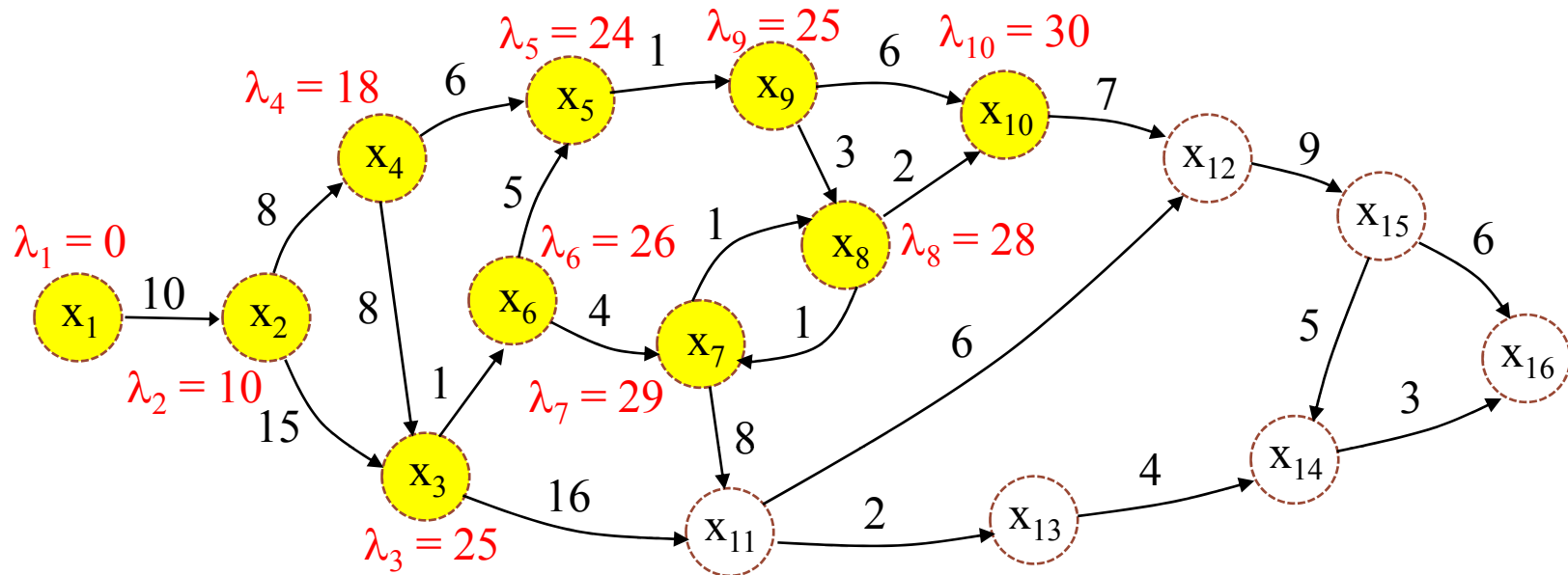
**k = 9**

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_7, x_{11}) \quad \lambda_{11} &= \lambda_7 + v(x_7, x_{11}) = 37 \\ v(x_8, x_{10}) \quad \lambda_{10} &= \lambda_8 + v(x_8, x_{10}) = 30 \\ v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31 \end{aligned}$$

$$E_{10} = E_9 \cup \{ \mathbf{x}_{10} \}$$

k = 10

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_7, x_{11}) \\ v(x_{10}, x_{12}) \end{aligned}$$

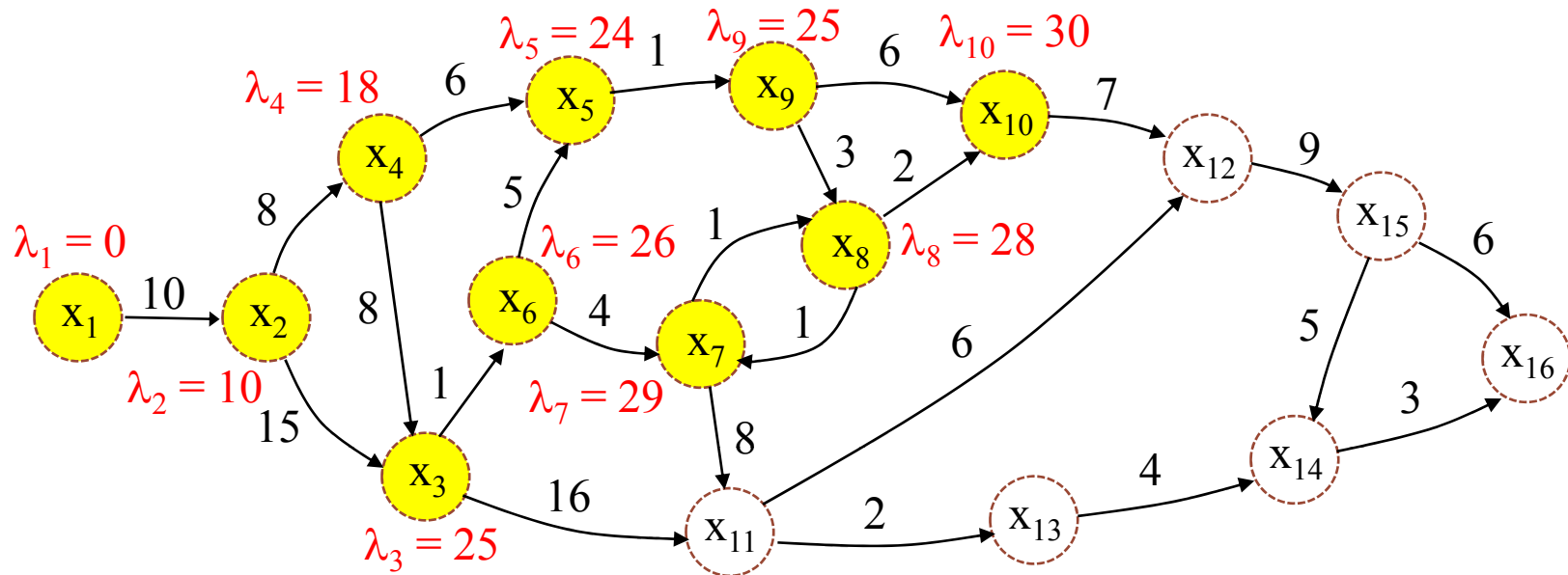
**k = 9**

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_7, x_{11}) \quad \lambda_{11} &= \lambda_7 + v(x_7, x_{11}) = 37 \\ v(x_8, x_{10}) \quad \lambda_{10} &= \lambda_8 + v(x_8, x_{10}) = 30 \\ v(x_9, x_{10}) \quad \lambda_{10} &= \lambda_9 + v(x_9, x_{10}) = 31 \end{aligned}$$

$$E_{10} = E_9 \cup \{x_{10}\}$$

k = 10

$$\begin{aligned} v(x_3, x_{11}) \quad \lambda_{11} &= \lambda_3 + v(x_3, x_{11}) = 41 \\ v(x_7, x_{11}) \quad \lambda_{11} &= \lambda_7 + v(x_7, x_{11}) = 37 \\ v(x_{10}, x_{12}) \end{aligned}$$

**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

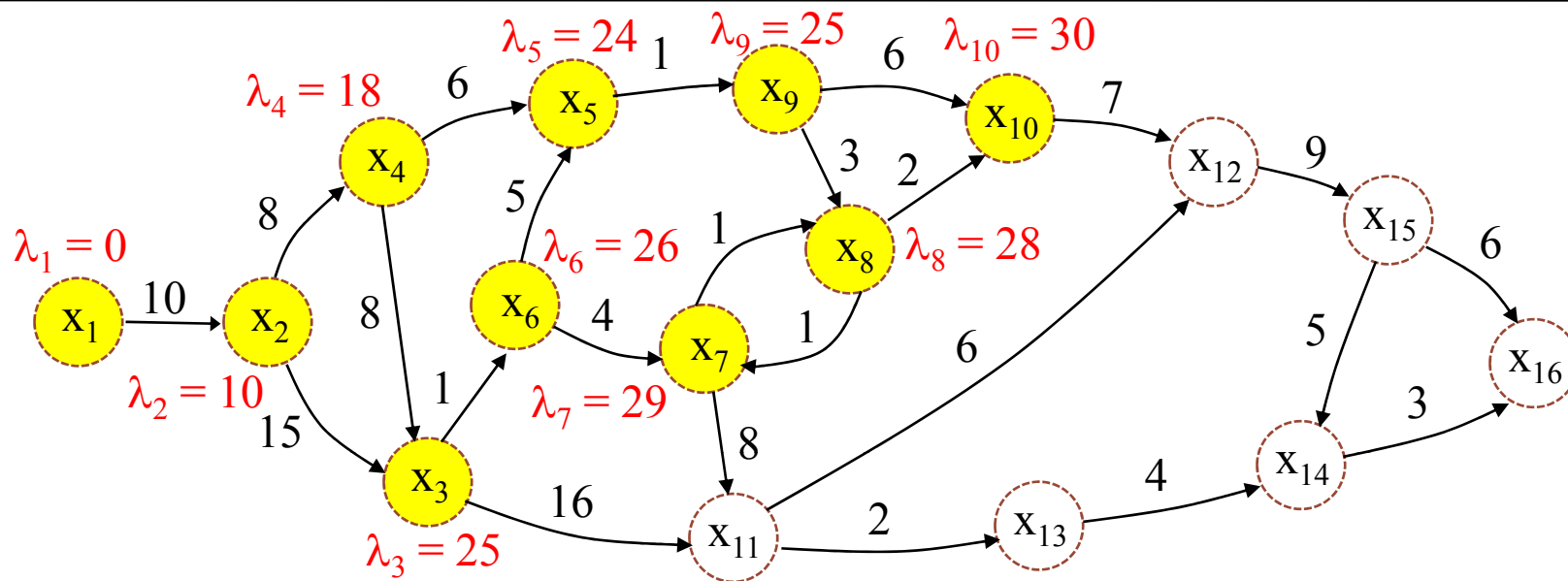
$$E_{10} = E_9 \cup \{x_{10}\}$$

k = 10

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

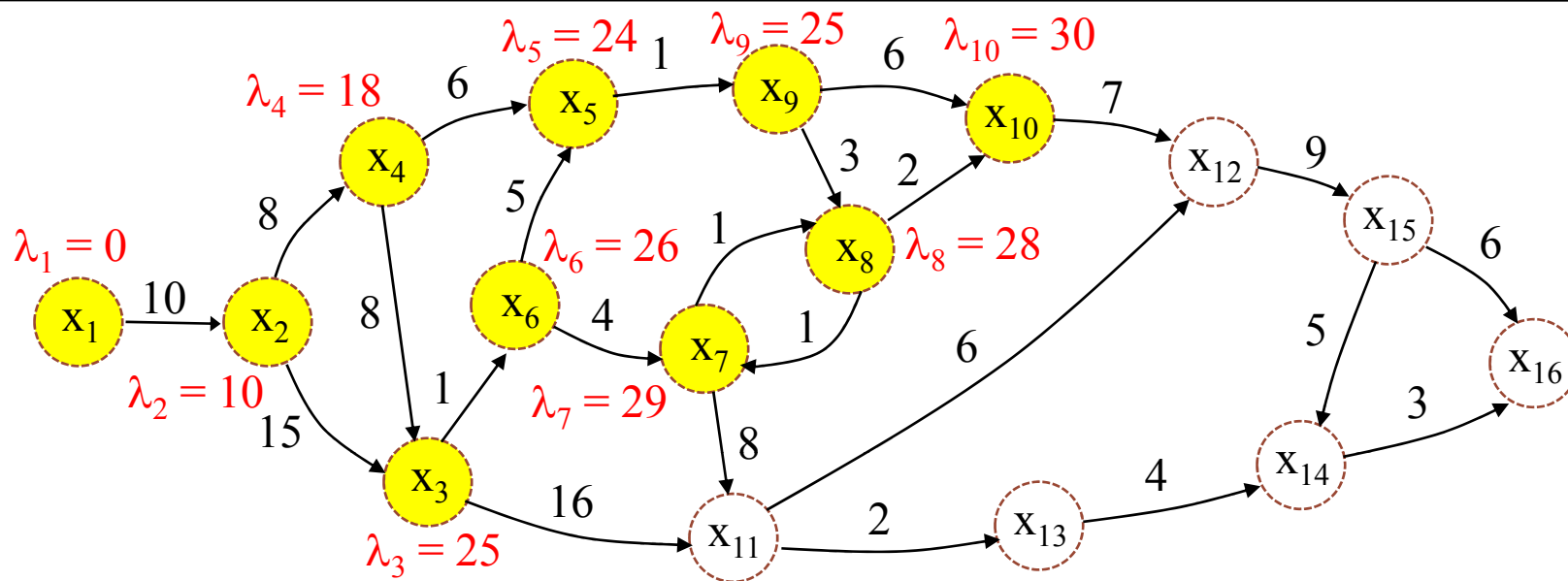
$$E_{10} = E_9 \cup \{x_{10}\}$$

k = 10

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

**k = 9**

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_8, x_{10}) \quad \lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$$

$$v(x_9, x_{10}) \quad \lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$$

$$E_{10} = E_9 \cup \{x_{10}\}$$

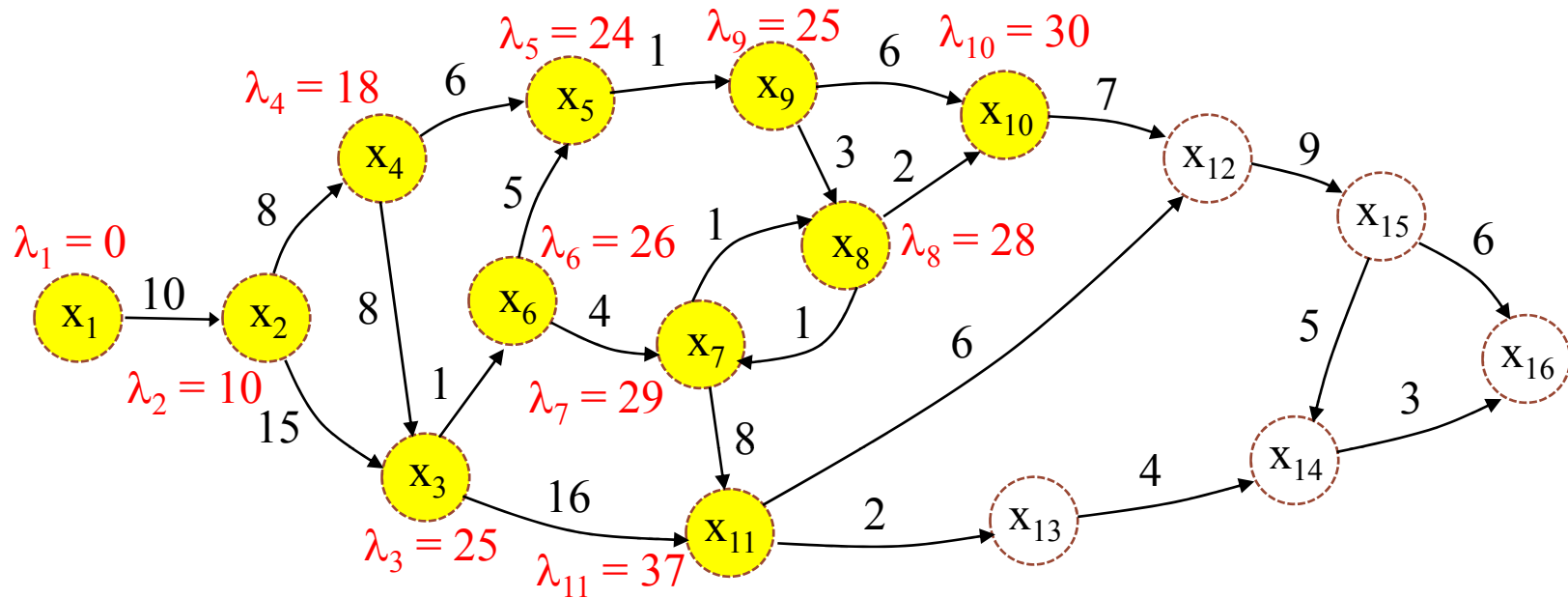
k = 10

$$v(x_3, x_{11}) \quad \lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$$

$$v(x_7, x_{11}) \quad \lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$$

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

$$E_{11} = E_{10} \cup \{x_{11}\}$$

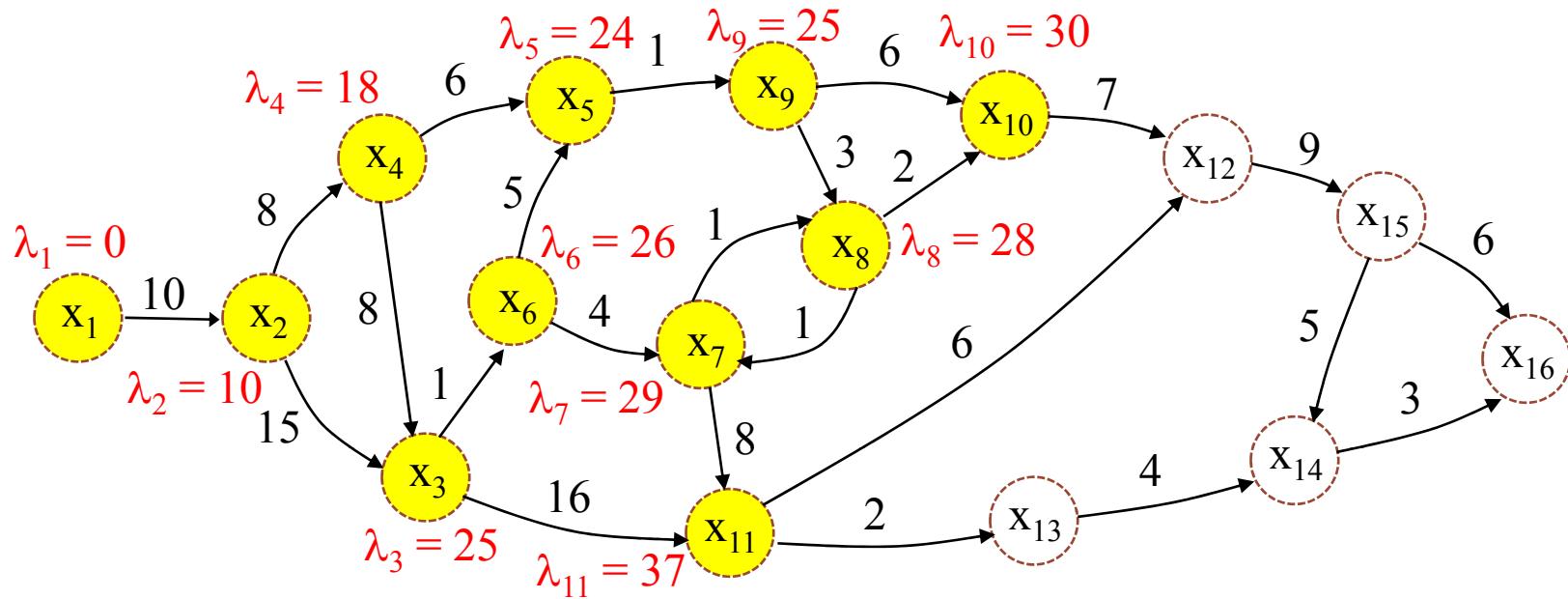


<u>k = 9</u>	$v(x_3, x_{11})$	$\lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$
	$v(x_7, x_{11})$	$\lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$
	$v(x_8, x_{10})$	$\lambda_{10} = \lambda_8 + v(x_8, x_{10}) = 30$
	$v(x_9, x_{10})$	$\lambda_{10} = \lambda_9 + v(x_9, x_{10}) = 31$

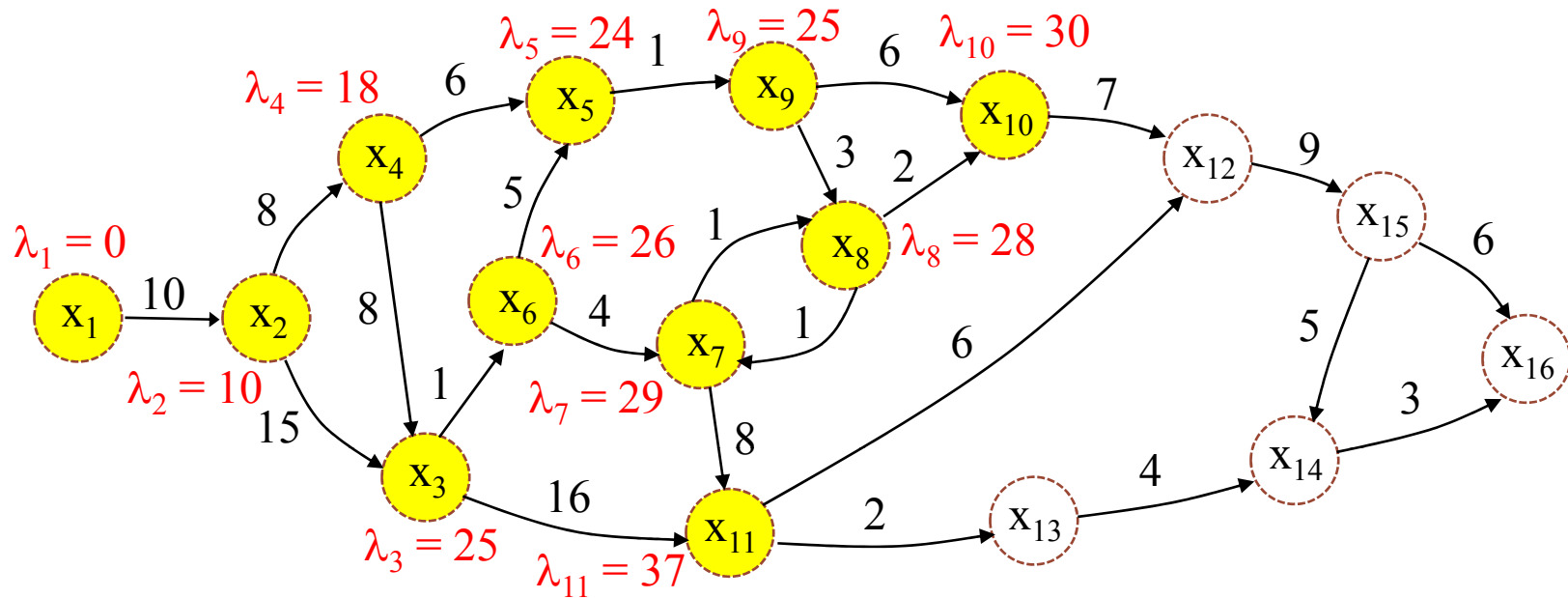
$$E_{10} = E_9 \cup \{ \mathbf{x}_{10} \}$$

<u>k = 10</u>	$v(x_3, x_{11})$	$\lambda_{11} = \lambda_3 + v(x_3, x_{11}) = 41$
	$v(x_7, x_{11})$	$\lambda_{11} = \lambda_7 + v(x_7, x_{11}) = 37$
	$v(x_{10}, x_{12})$	$\lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$

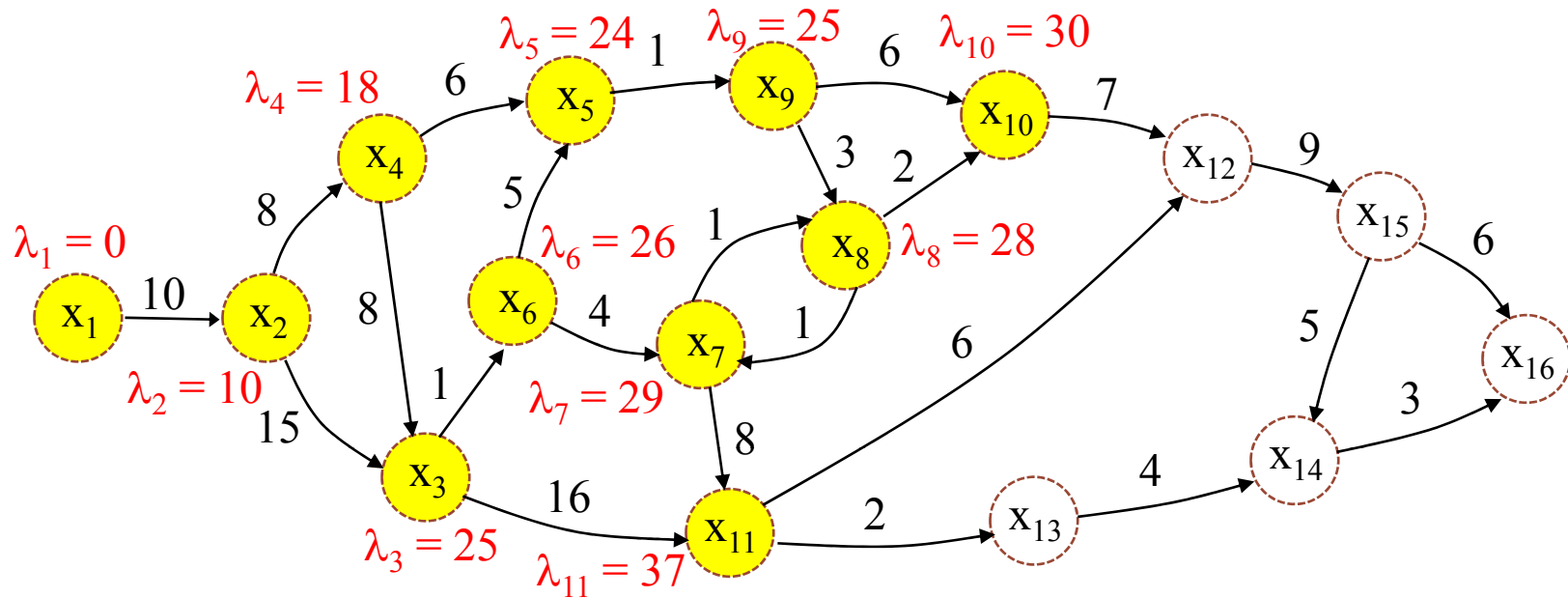
$$\mathbf{E}_{11} = \mathbf{E}_{10} \cup \{ \mathbf{x}_{11} \}$$



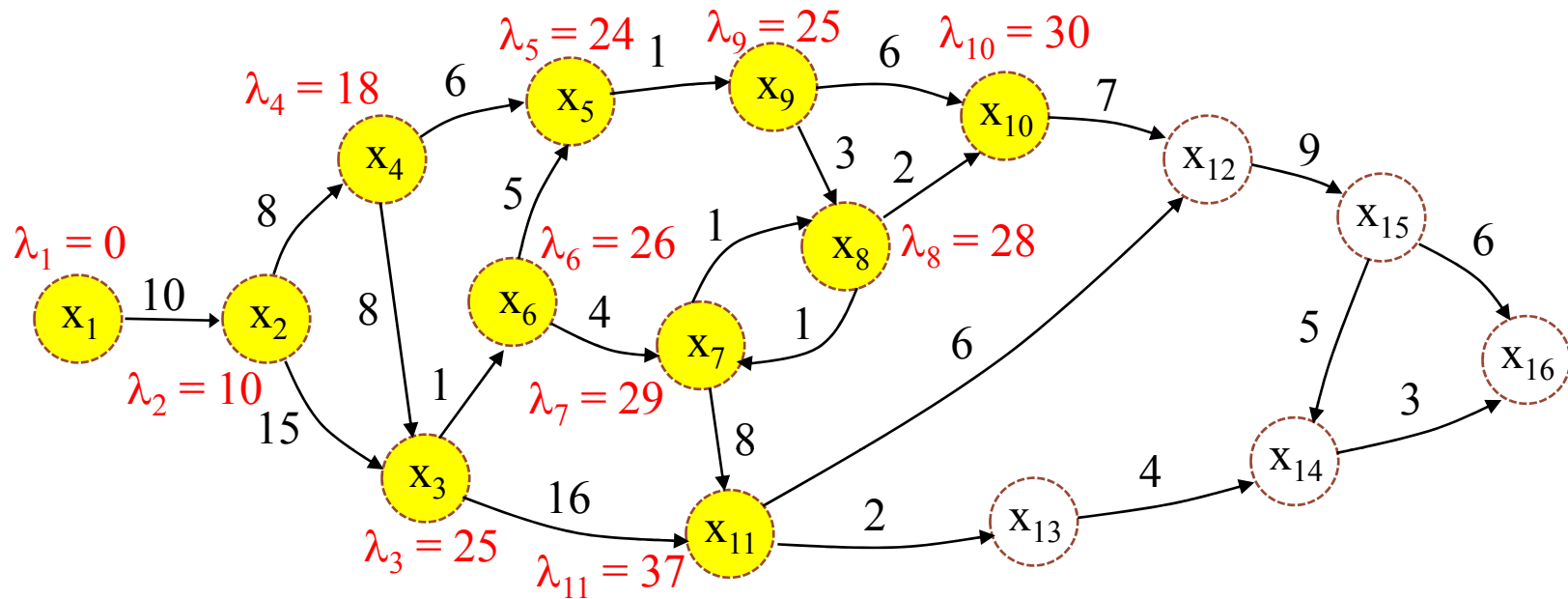
k = 11



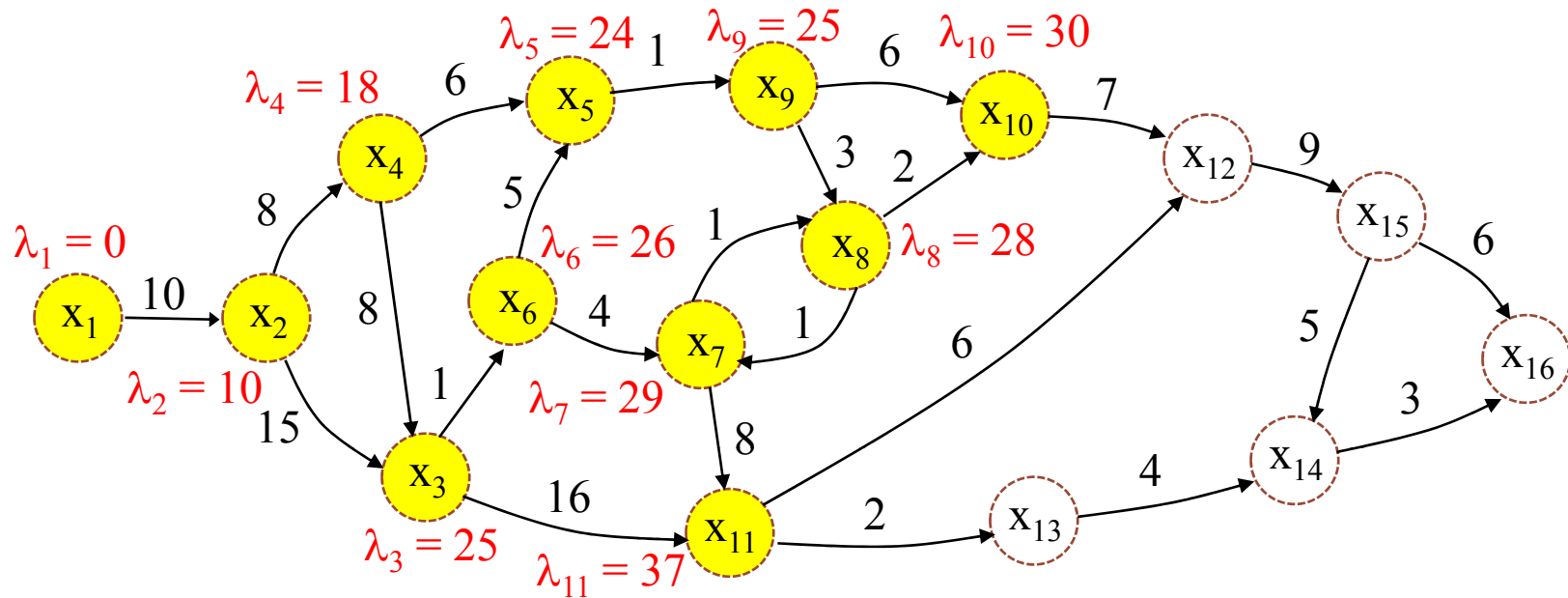
$k = 11$ $v(x_{10}, x_{12})$



k = 11 $v(x_{10}, x_{12})$
 $v(x_{11}, x_{13})$



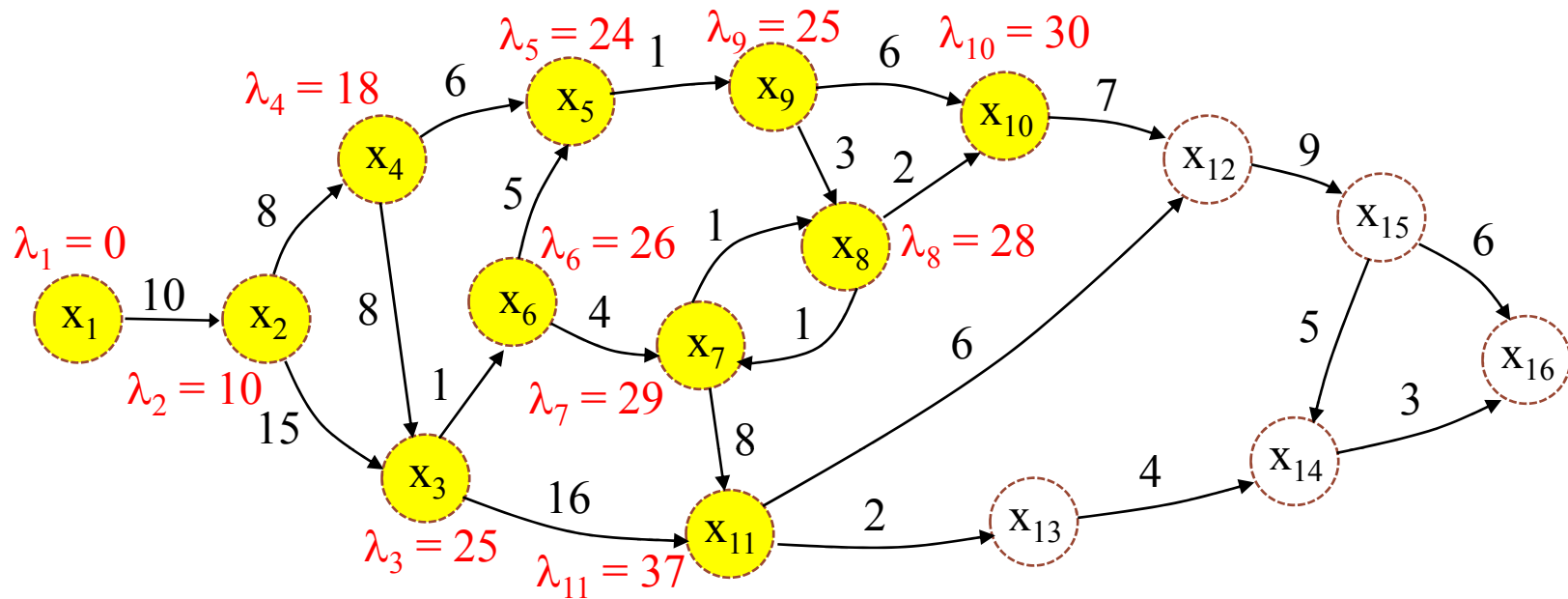
k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13})$



k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

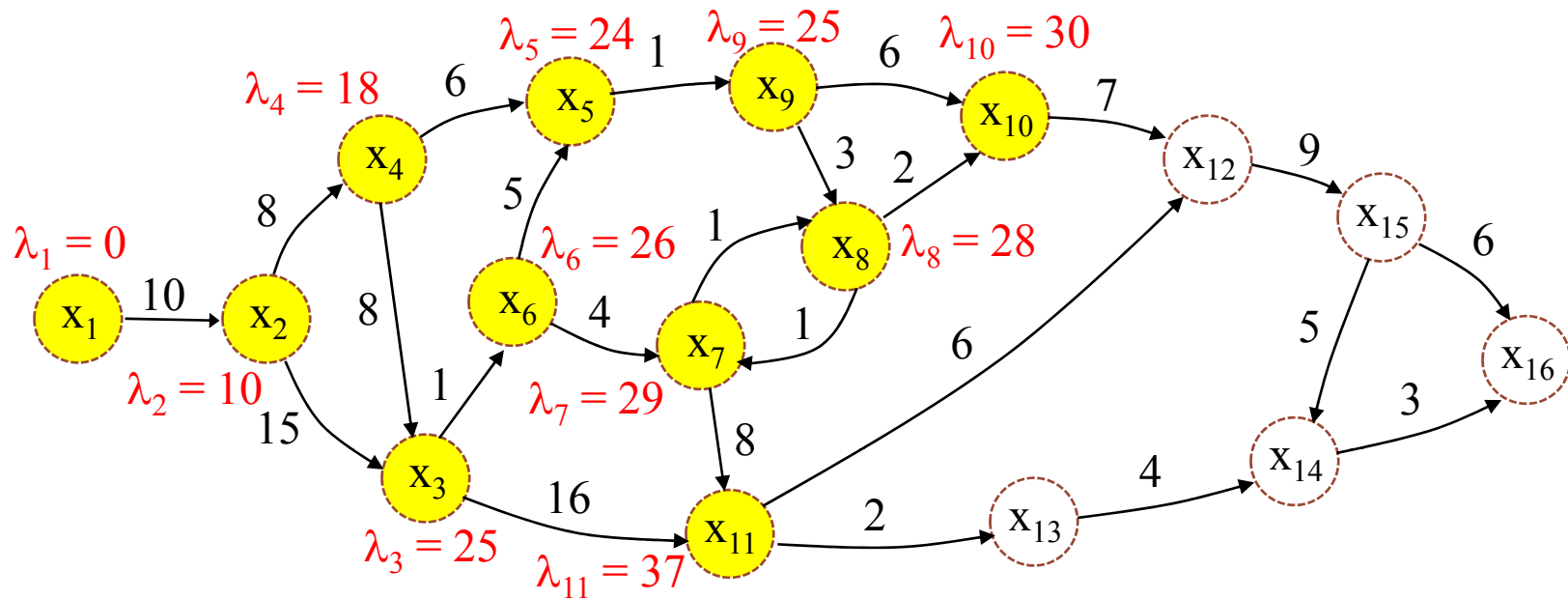
$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$



k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

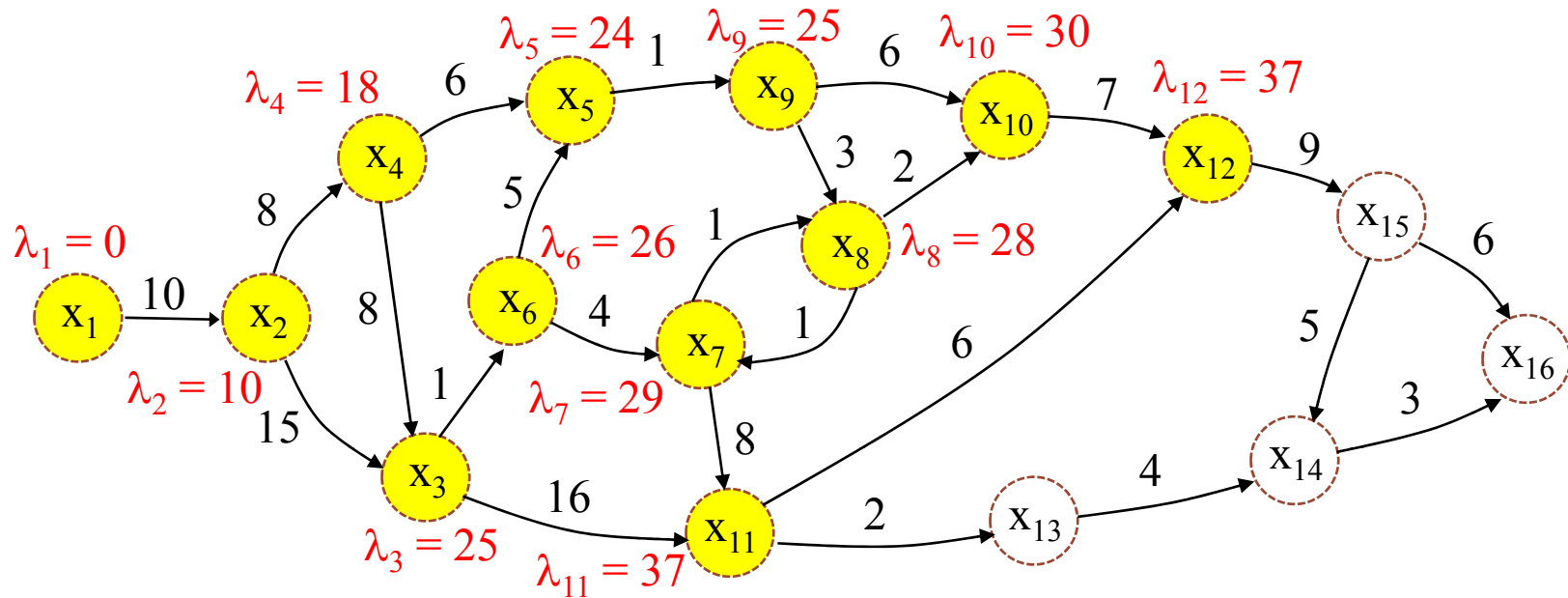


k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

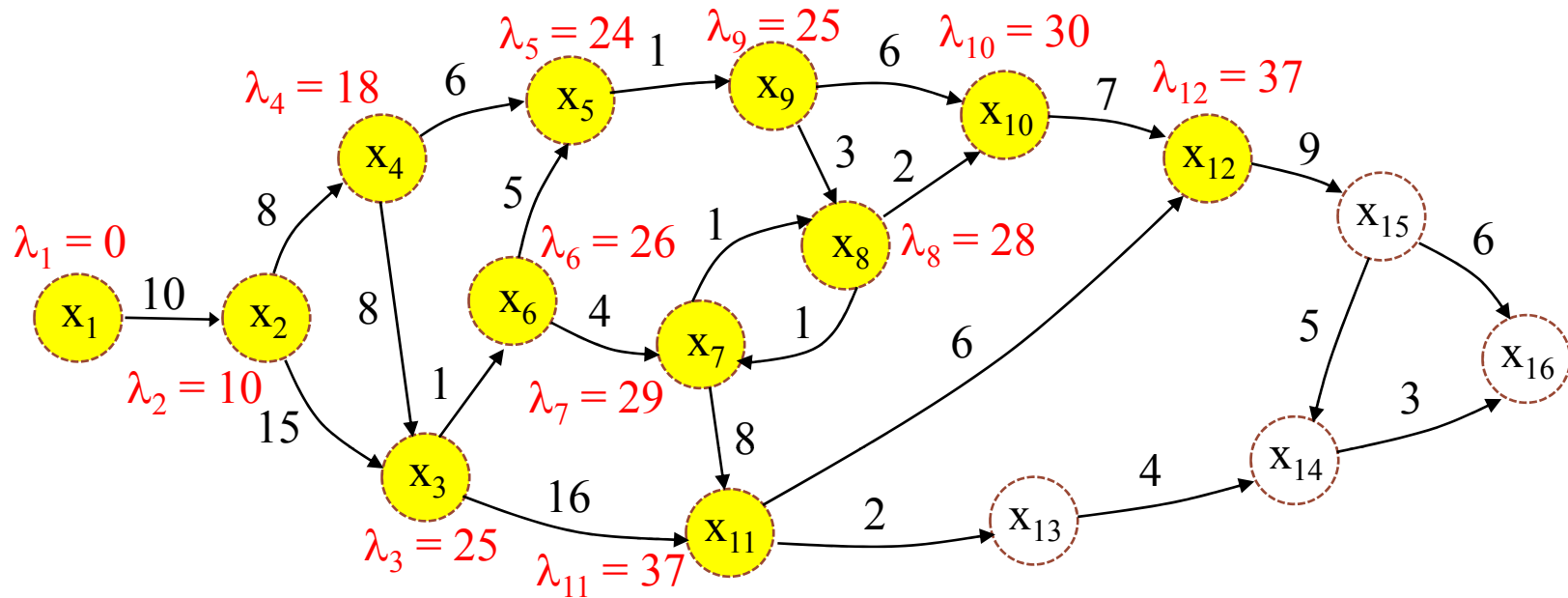
$$E_{12} = E_{11} \cup \{x_{12}\}$$



k = 11

$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$E_{12} = E_{11} \cup \{x_{12}\}$



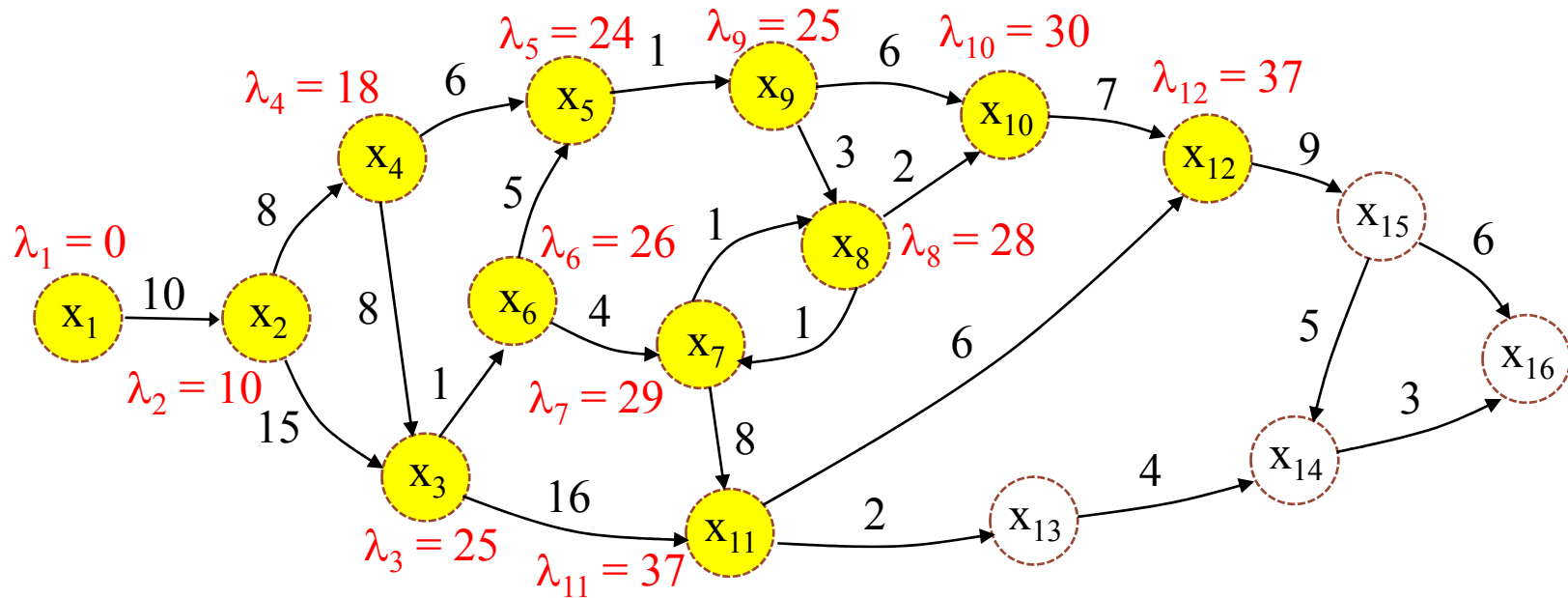
k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

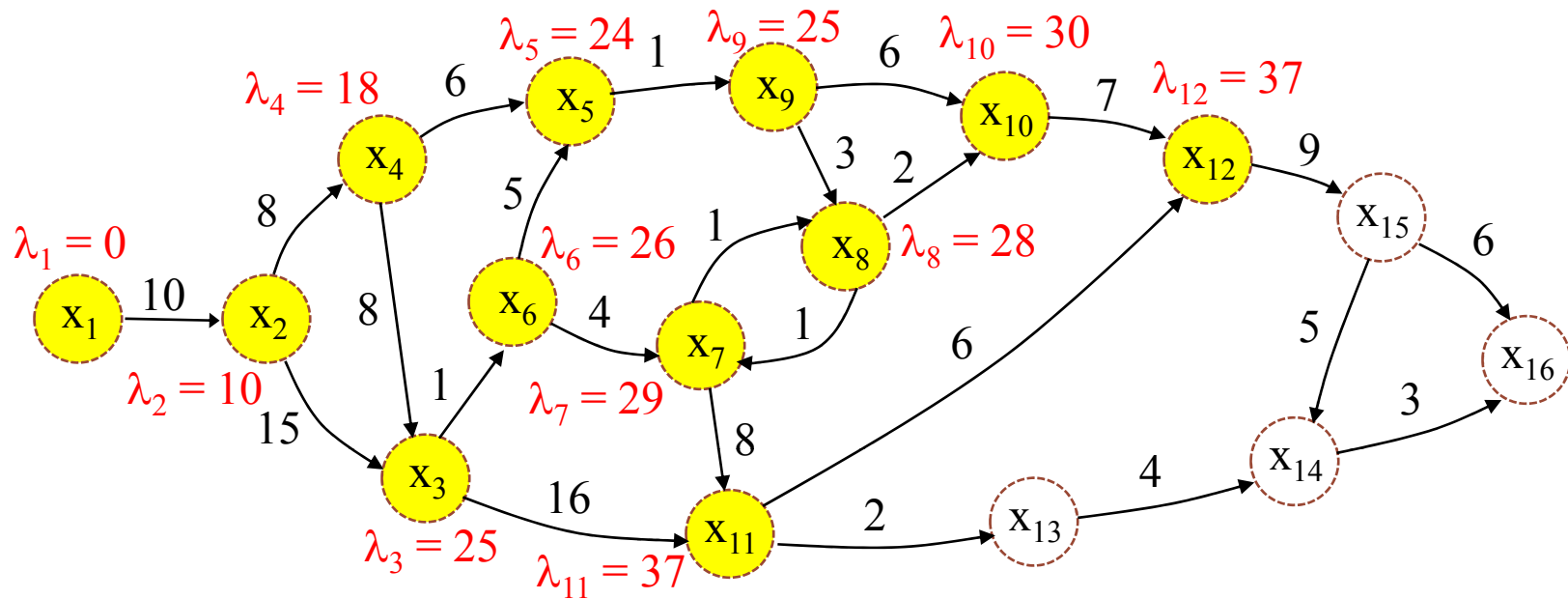
k = 12



k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12 $v(x_{11}, x_{13})$



k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

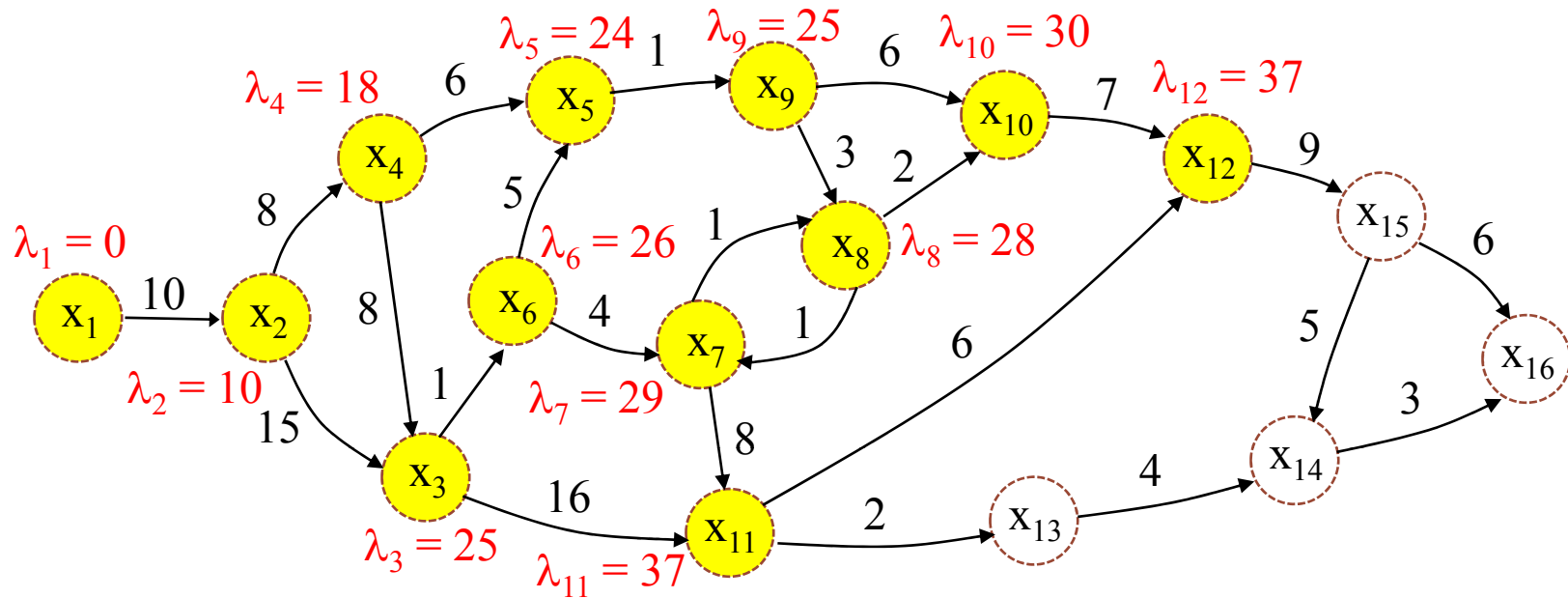
$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12

$$v(x_{11}, x_{13})$$

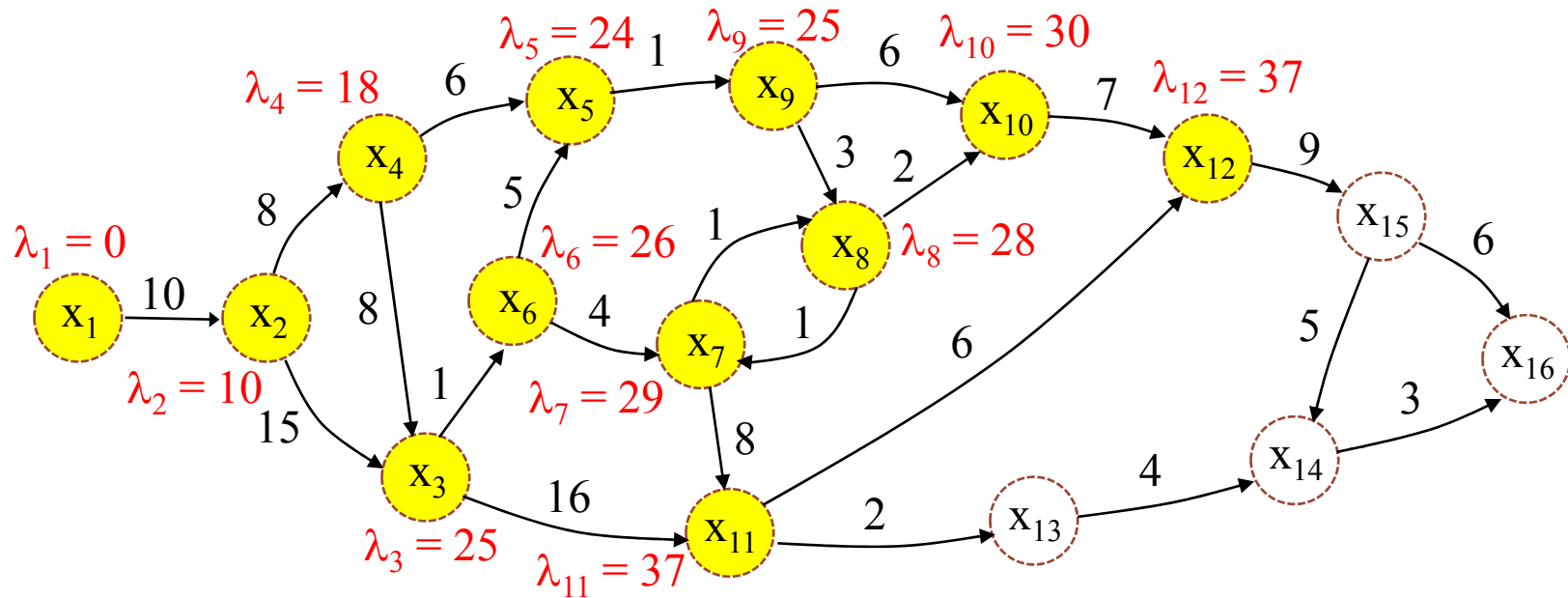
$$v(x_{12}, x_{15})$$



k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

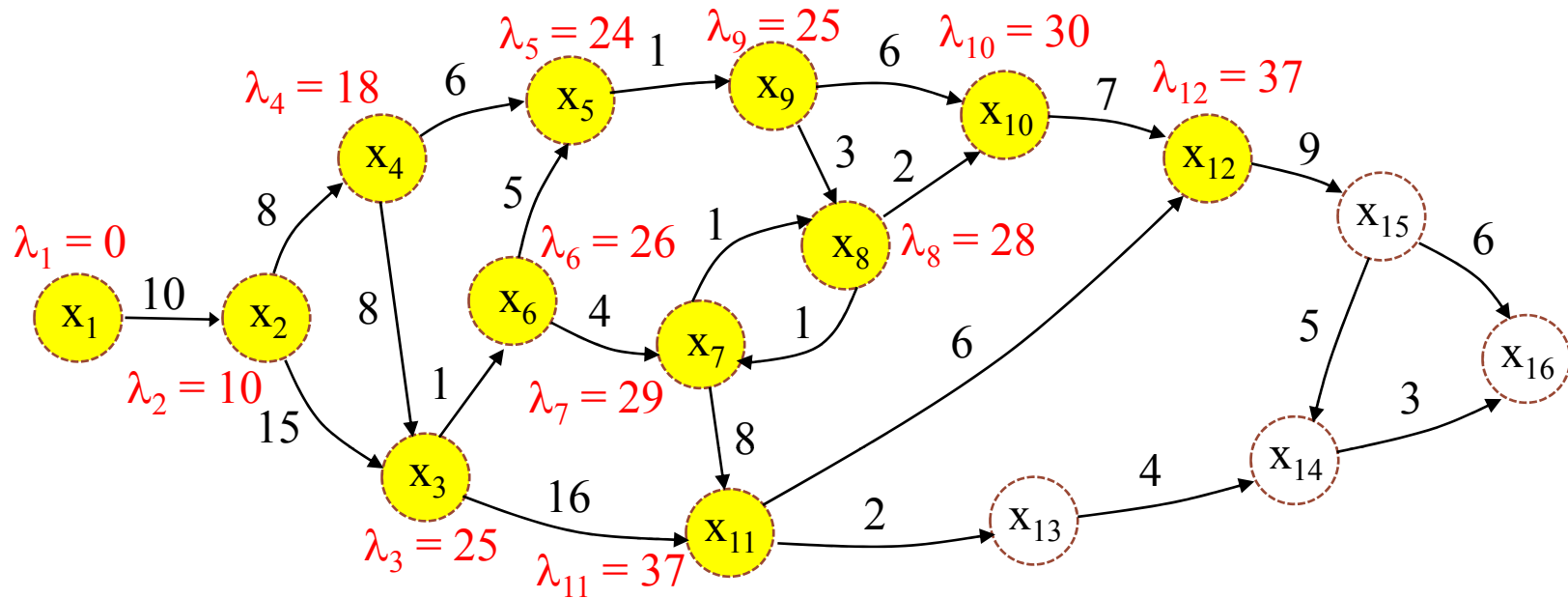
k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15})$



k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$



k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

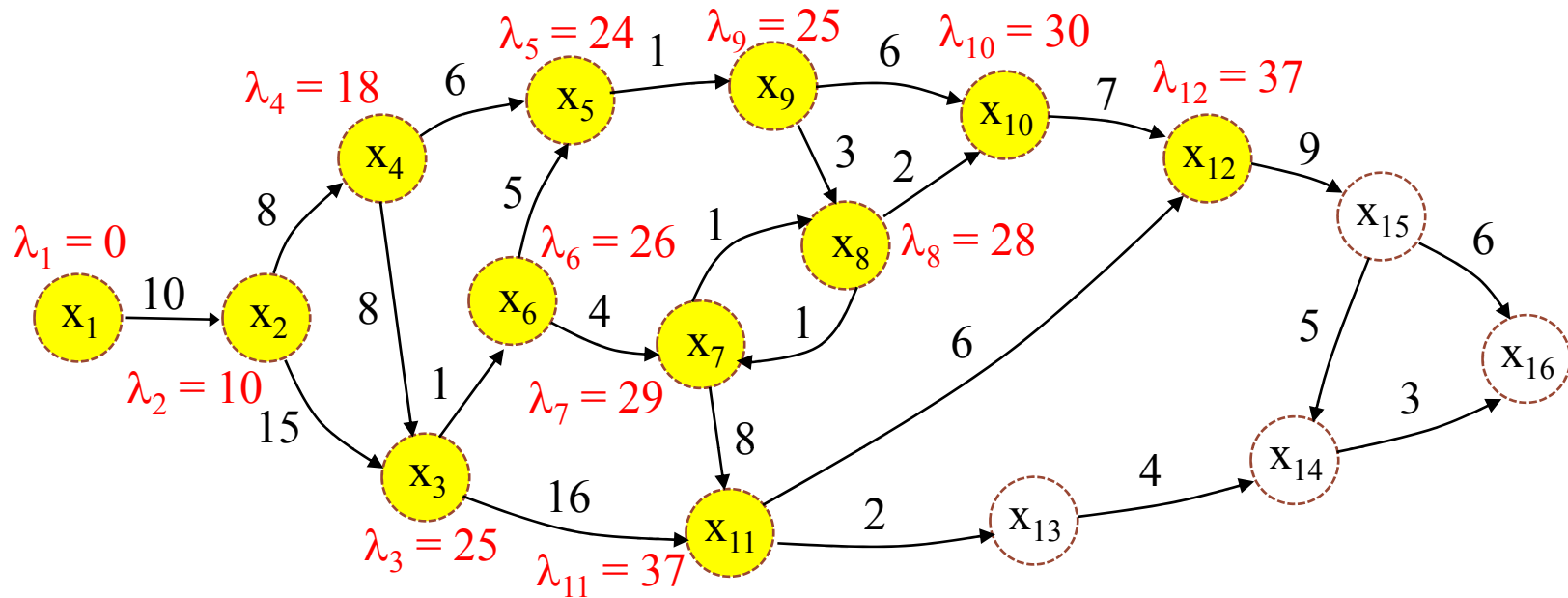
$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$



k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

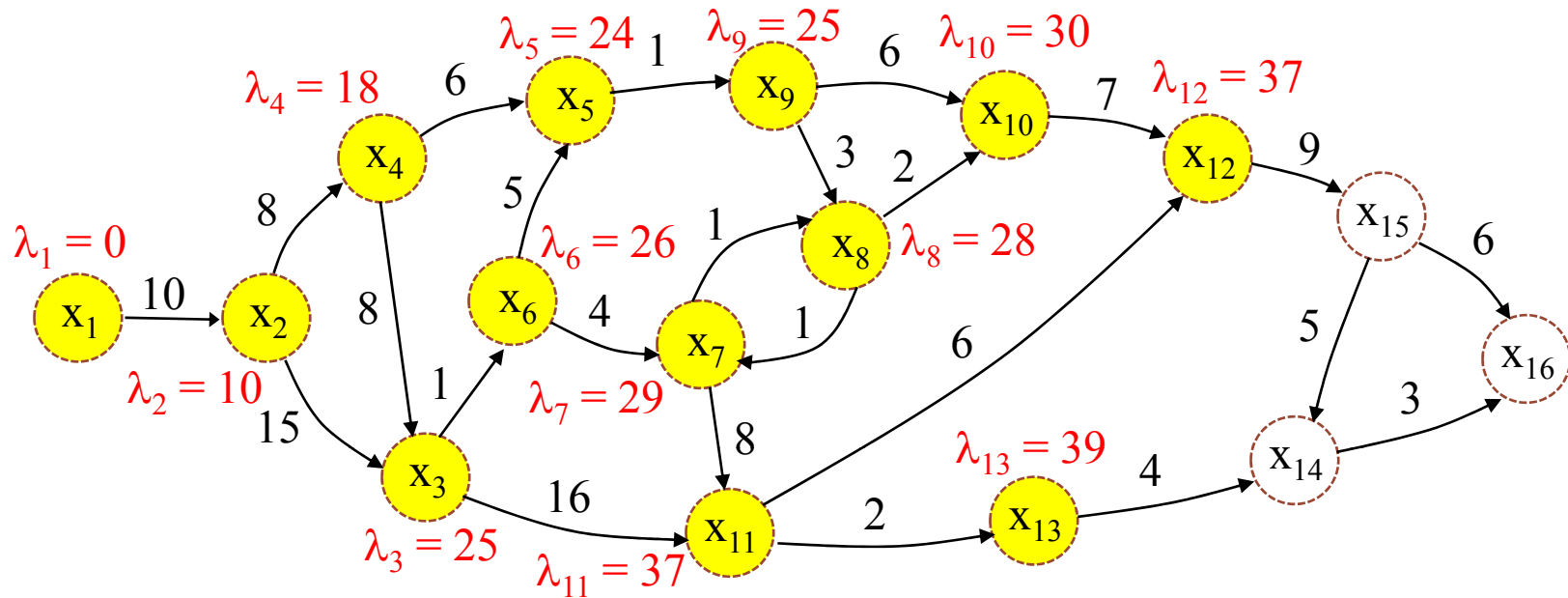
$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$E_{13} = E_{12} \cup \{x_{13}\}$$



k = 11

$$v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$$

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

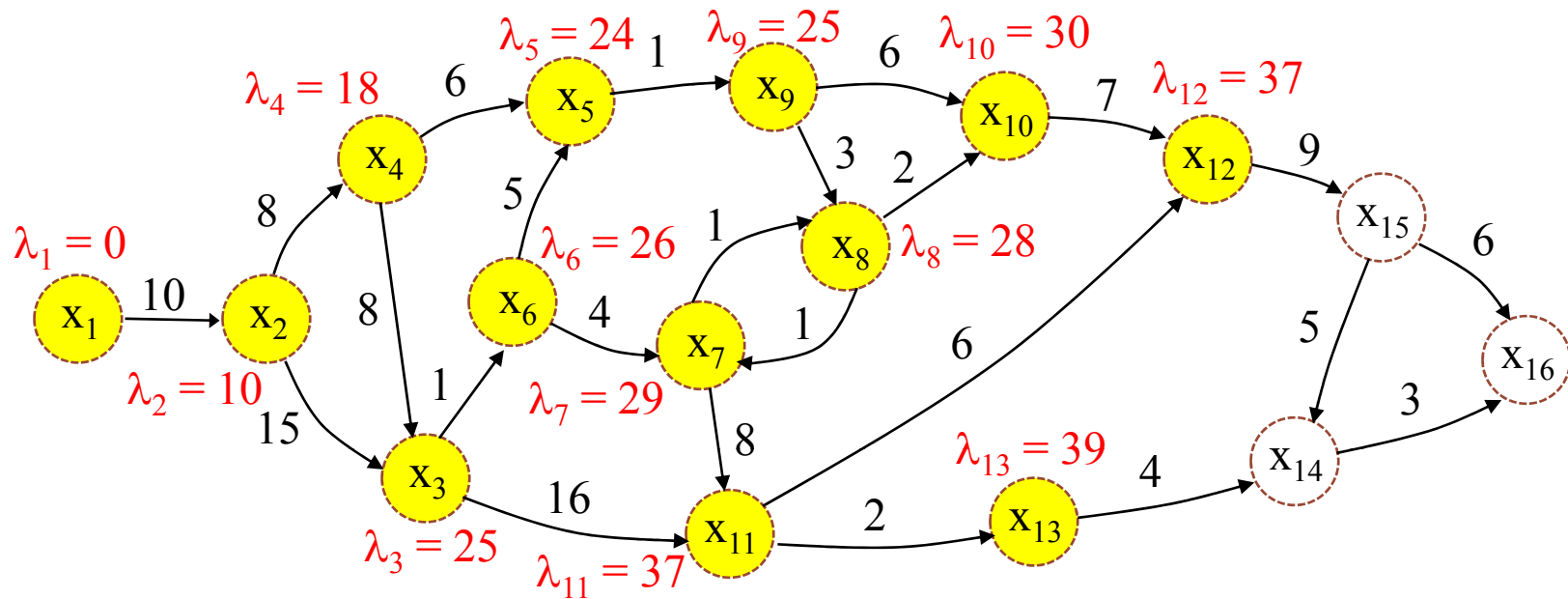
$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12

$$v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$$

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$E_{13} = E_{12} \cup \{x_{13}\}$$



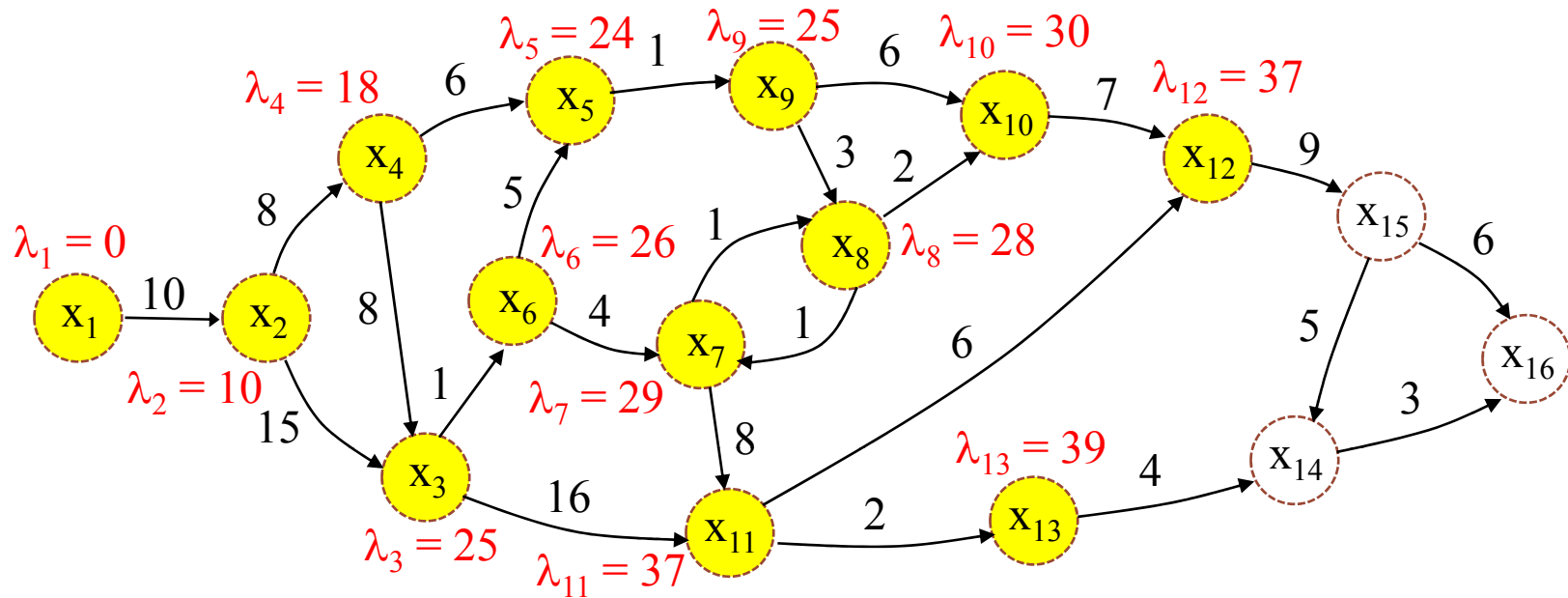
k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{ x_{12} \}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$

$$E_{13} = E_{12} \cup \{ x_{13} \}$$

k = 13



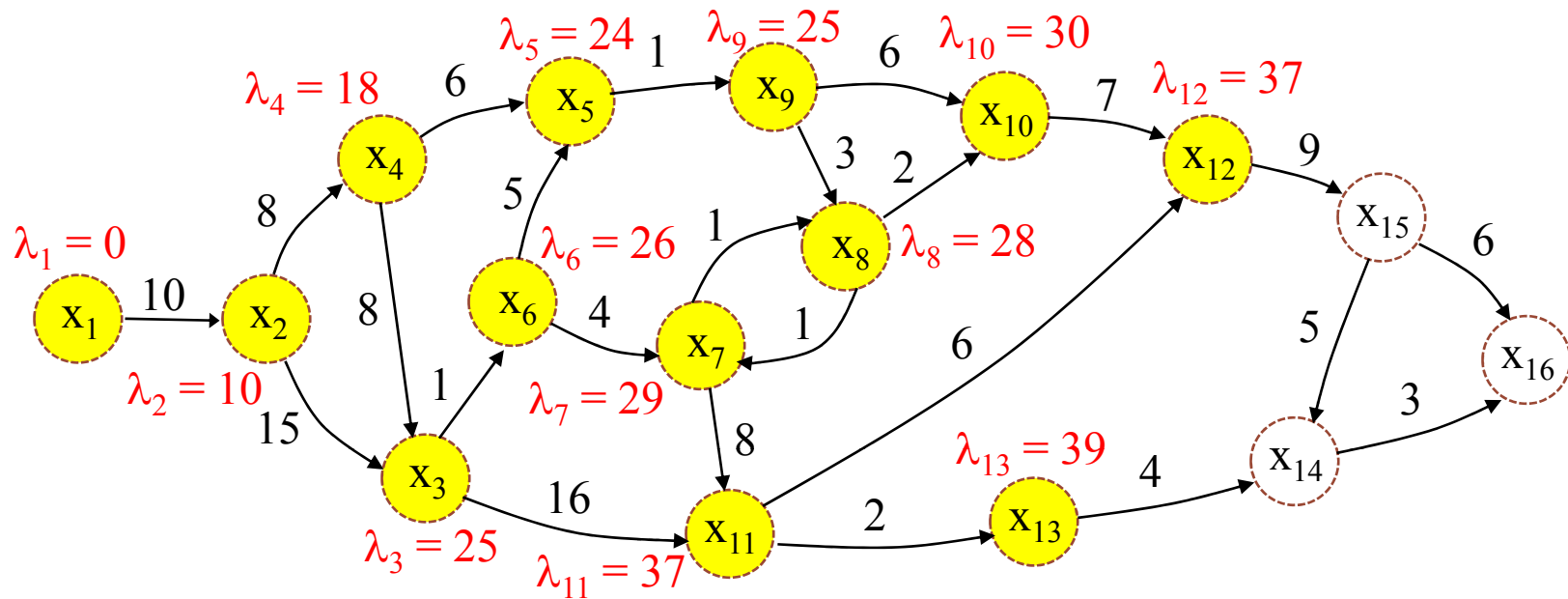
k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{ x_{12} \}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$

$$E_{13} = E_{12} \cup \{ x_{13} \}$$

k = 13 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{13}, x_{14})$



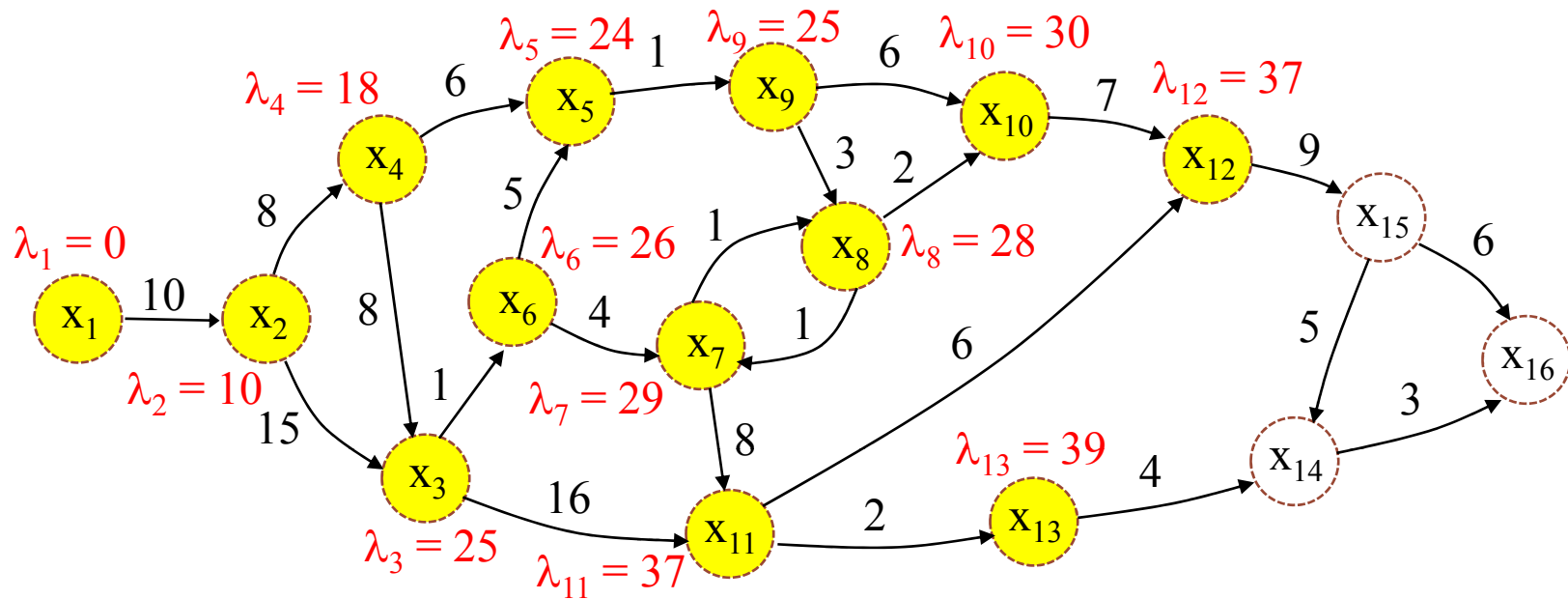
k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$

$$E_{13} = E_{12} \cup \{x_{13}\}$$

k = 13 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{13}, x_{14}) \quad \lambda_{14} = \lambda_{13} + v(x_{13}, x_{14}) = 43$



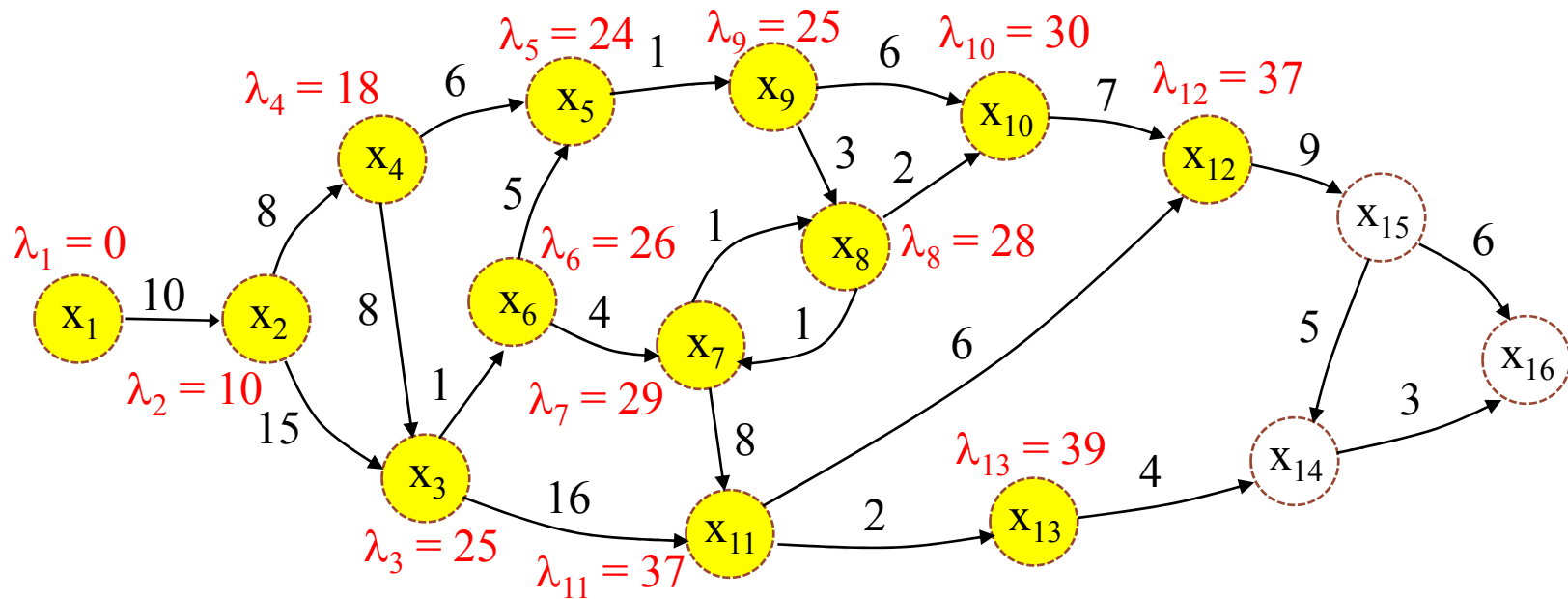
k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$

$$E_{13} = E_{12} \cup \{x_{13}\}$$

k = 13 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{13}, x_{14}) \quad \lambda_{14} = \lambda_{13} + v(x_{13}, x_{14}) = 43$



k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

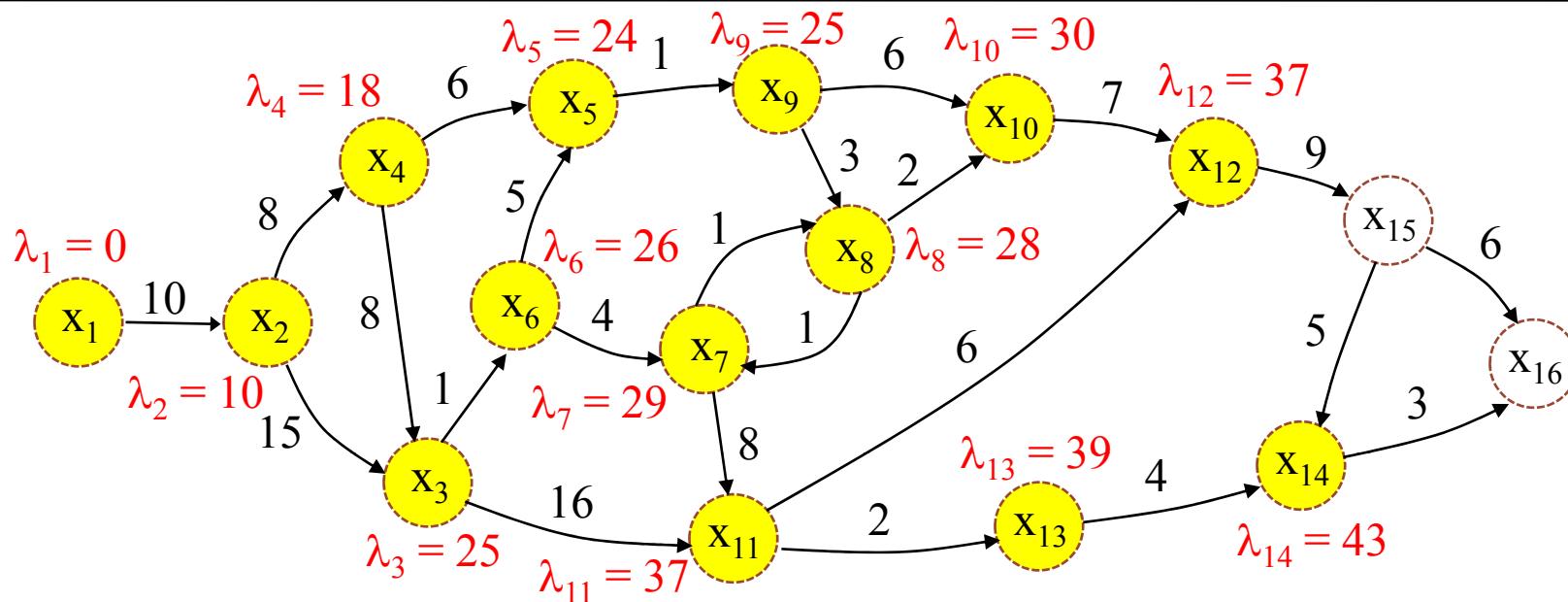
$$E_{12} = E_{11} \cup \{x_{12}\}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$

$$E_{13} = E_{12} \cup \{x_{13}\}$$

k = 13 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{13}, x_{14}) \quad \lambda_{14} = \lambda_{13} + v(x_{13}, x_{14}) = 43$

$$E_{14} = E_{13} \cup \{x_{14}\}$$



k = 11 $v(x_{10}, x_{12}) \quad \lambda_{12} = \lambda_{10} + v(x_{10}, x_{12}) = 37$
 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$

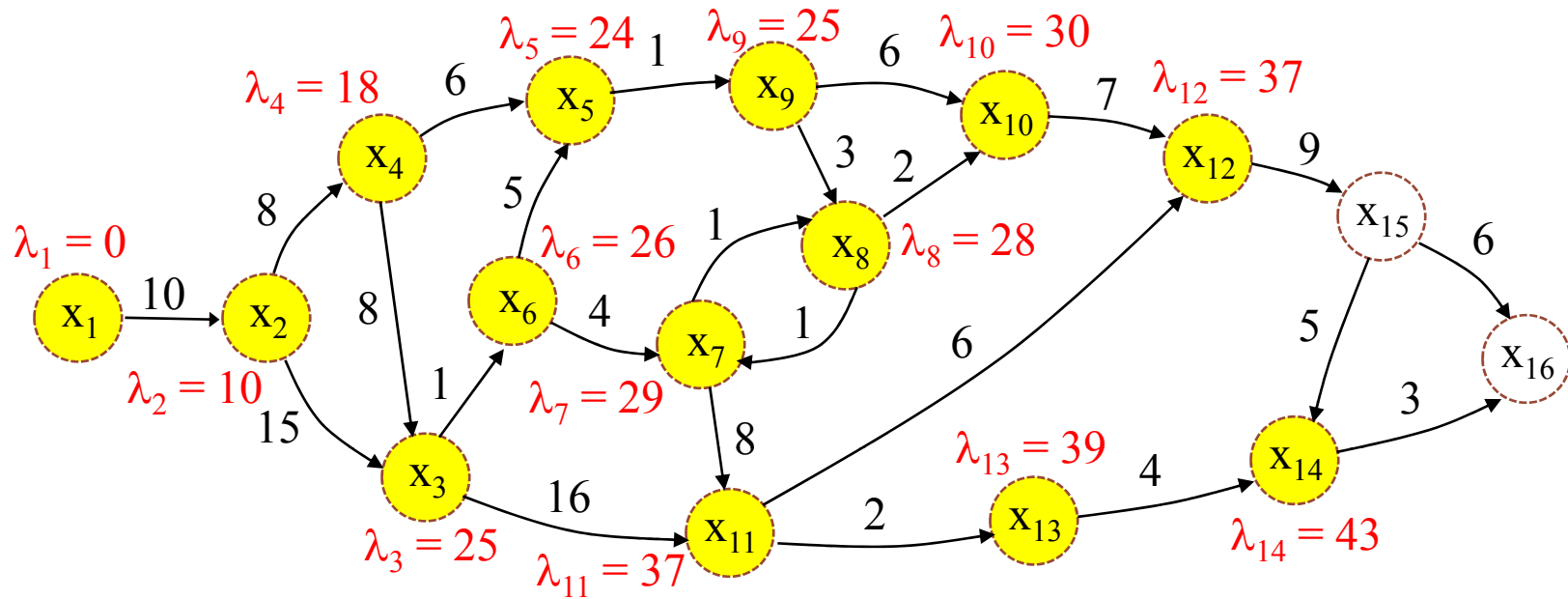
$$E_{12} = E_{11} \cup \{ x_{12} \}$$

k = 12 $v(x_{11}, x_{13}) \quad \lambda_{13} = \lambda_{11} + v(x_{11}, x_{13}) = 39$
 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$

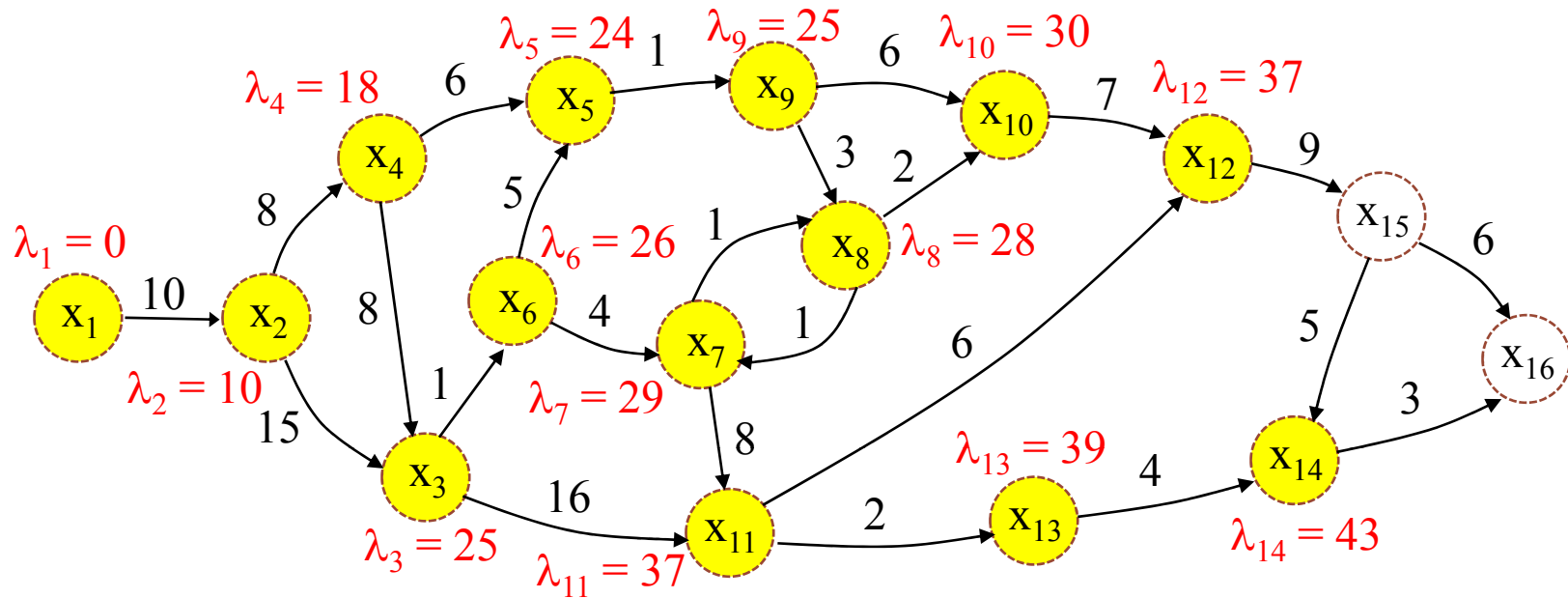
$$E_{13} = E_{12} \cup \{ x_{13} \}$$

k = 13 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{13}, x_{14}) \quad \lambda_{14} = \lambda_{13} + v(x_{13}, x_{14}) = 43$

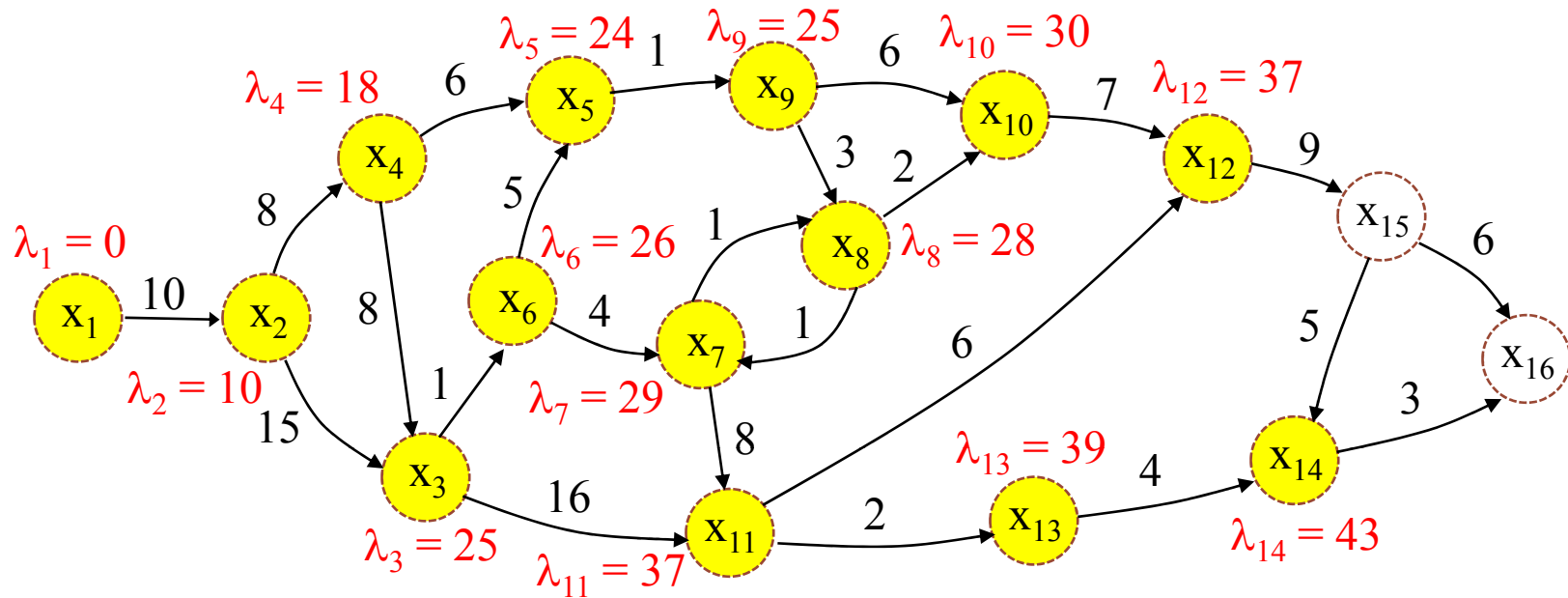
$$E_{14} = E_{13} \cup \{ x_{14} \}$$



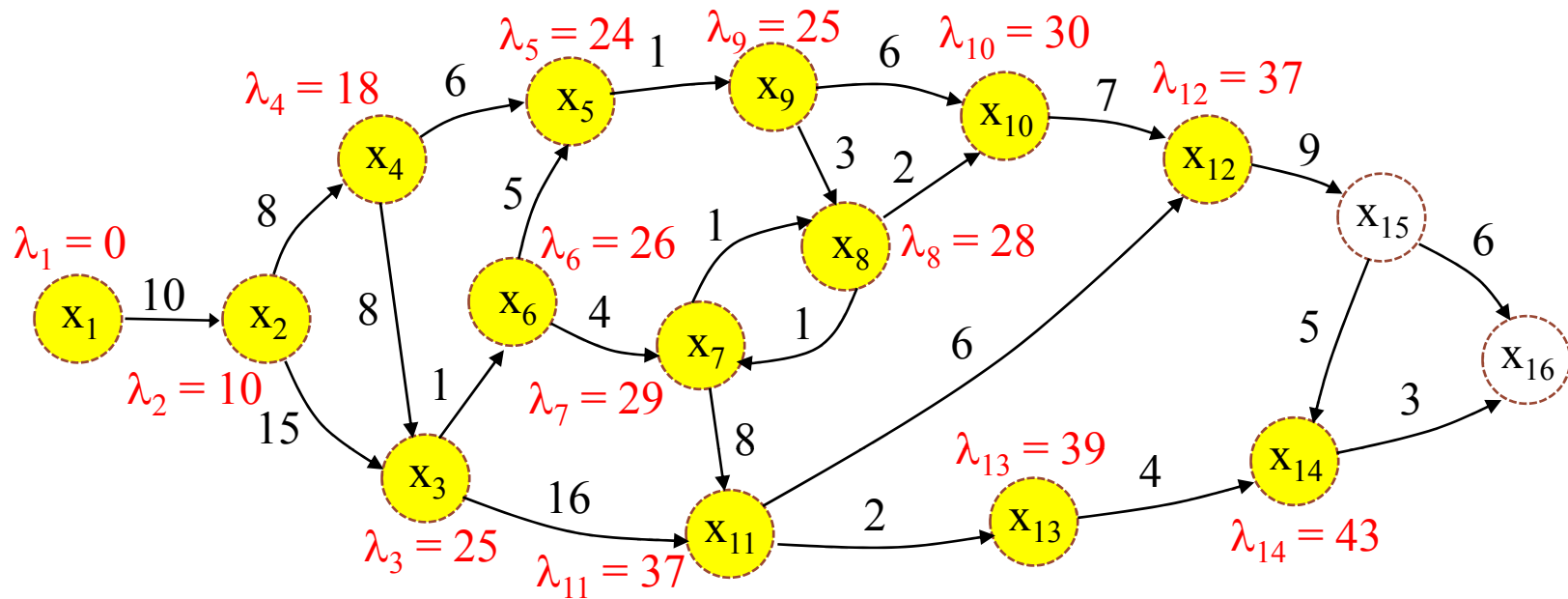
$k = 14$



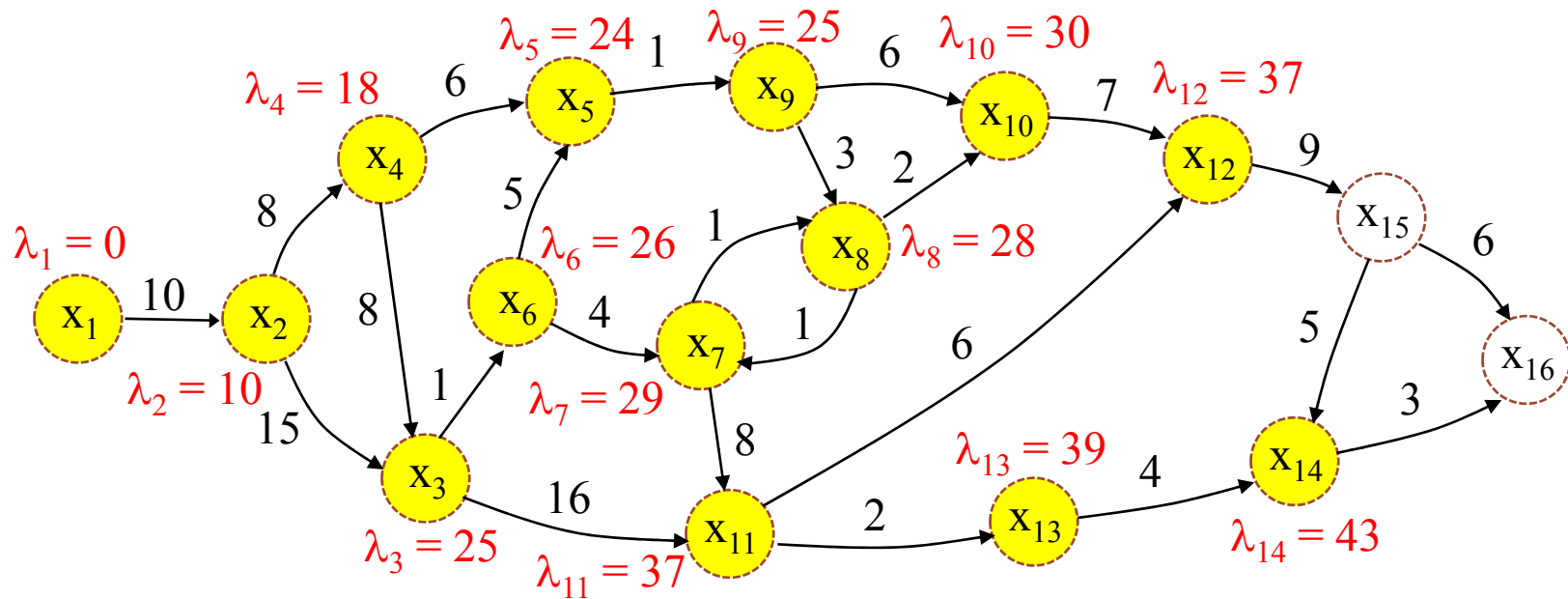
k = 14 $v(x_{12}, x_{15})$



k = 14 $v(x_{12}, x_{15})$
 $v(x_{14}, x_{16})$

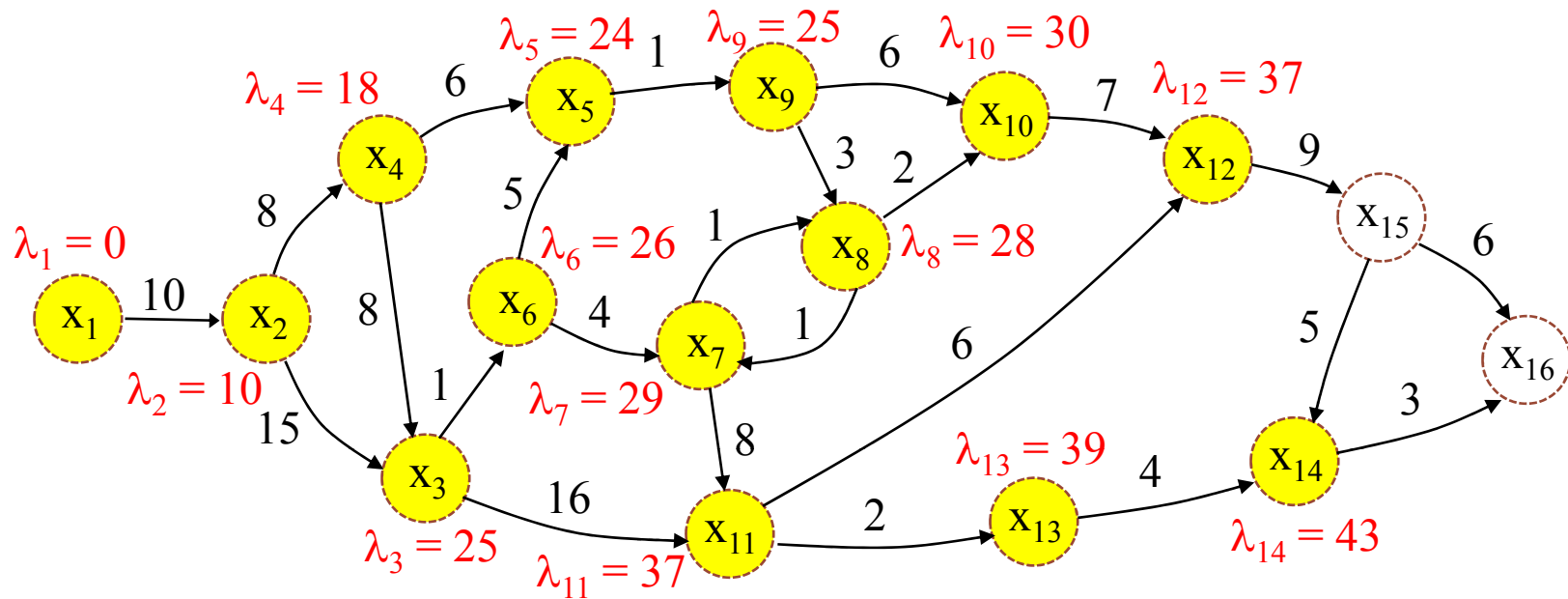


k = 14 $v(x_{12}, x_{15})$ $\lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{14}, x_{16})$

**k = 14**

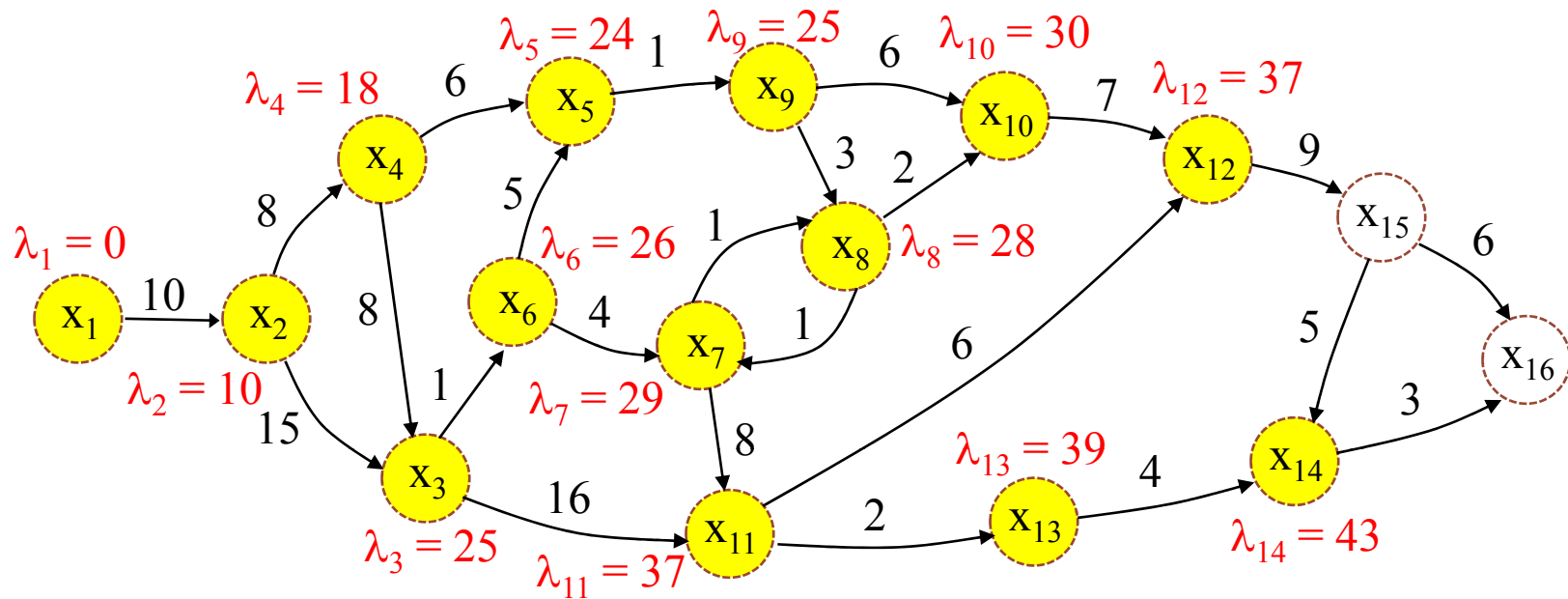
$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

**k = 14**

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

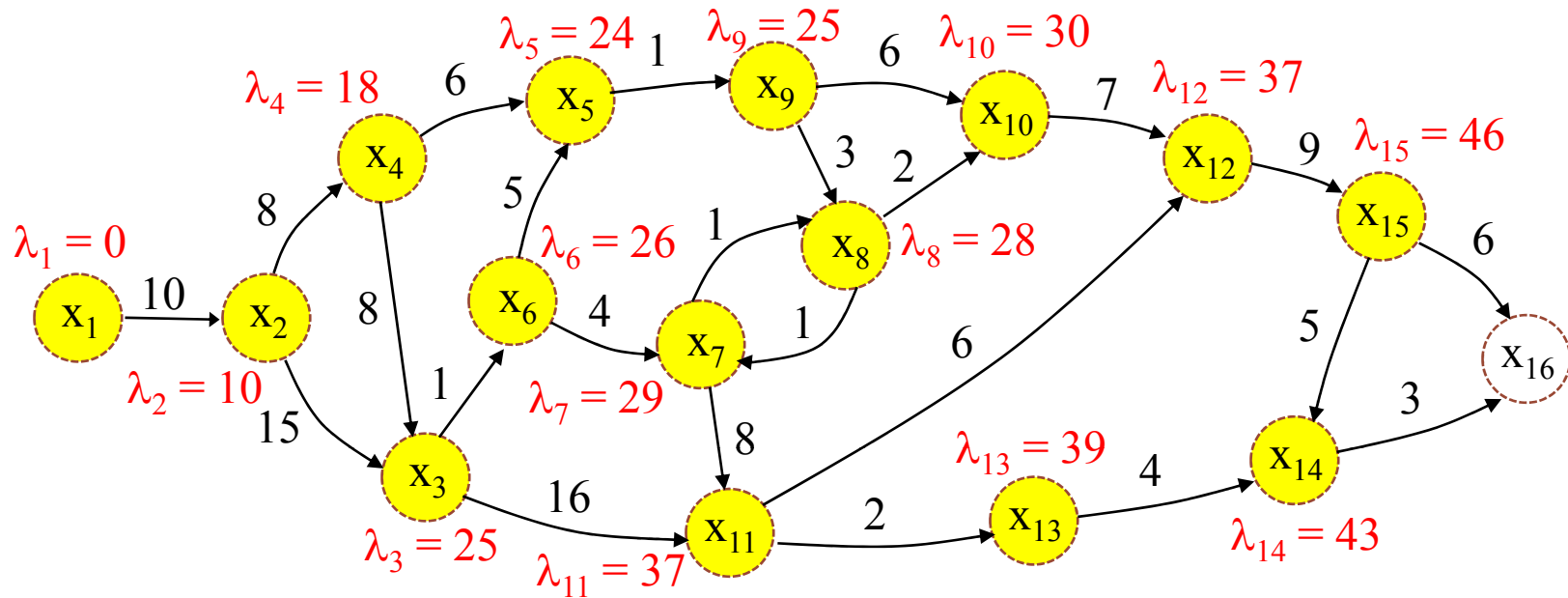
$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

**k = 14**

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

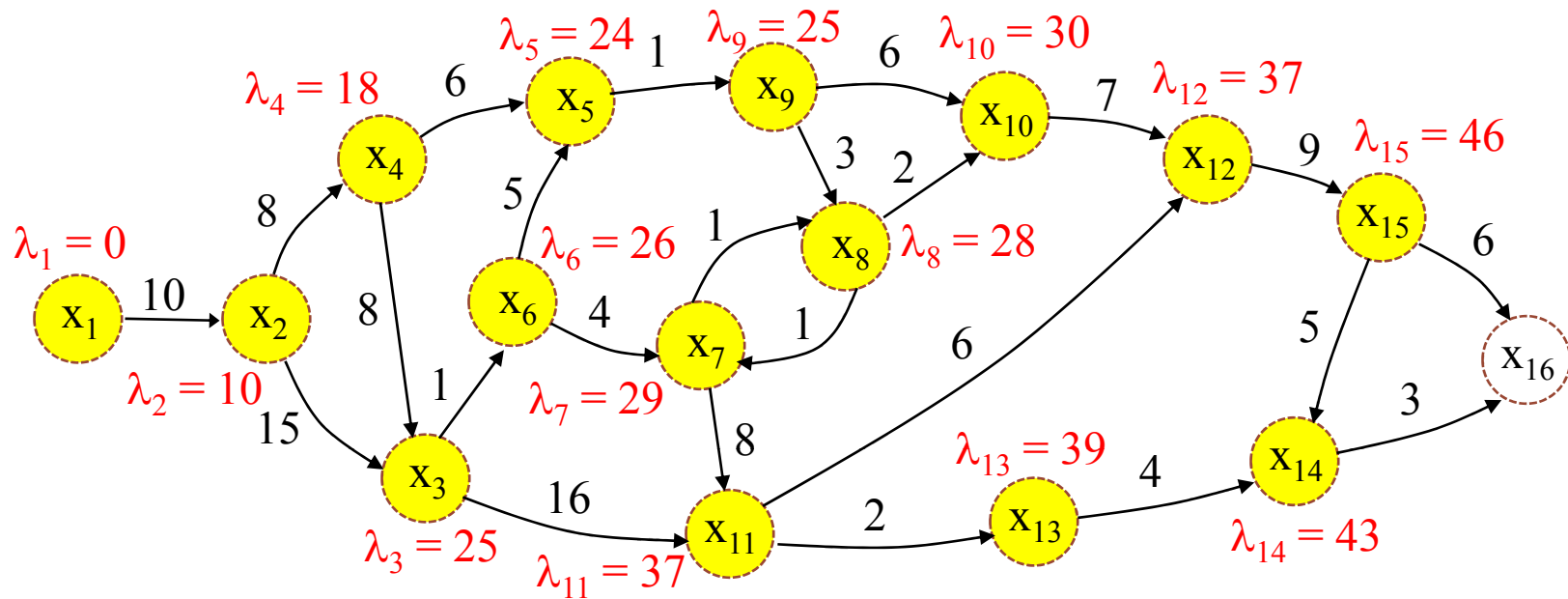


k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$



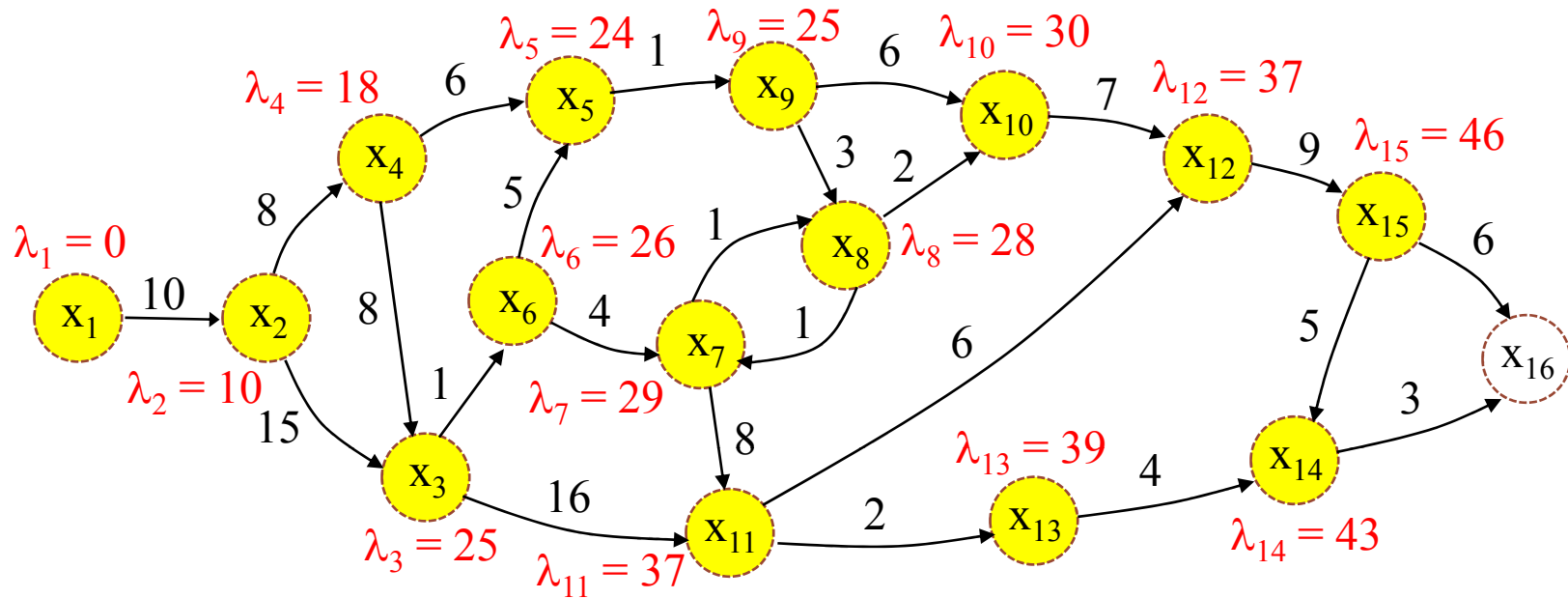
k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15



k = 14

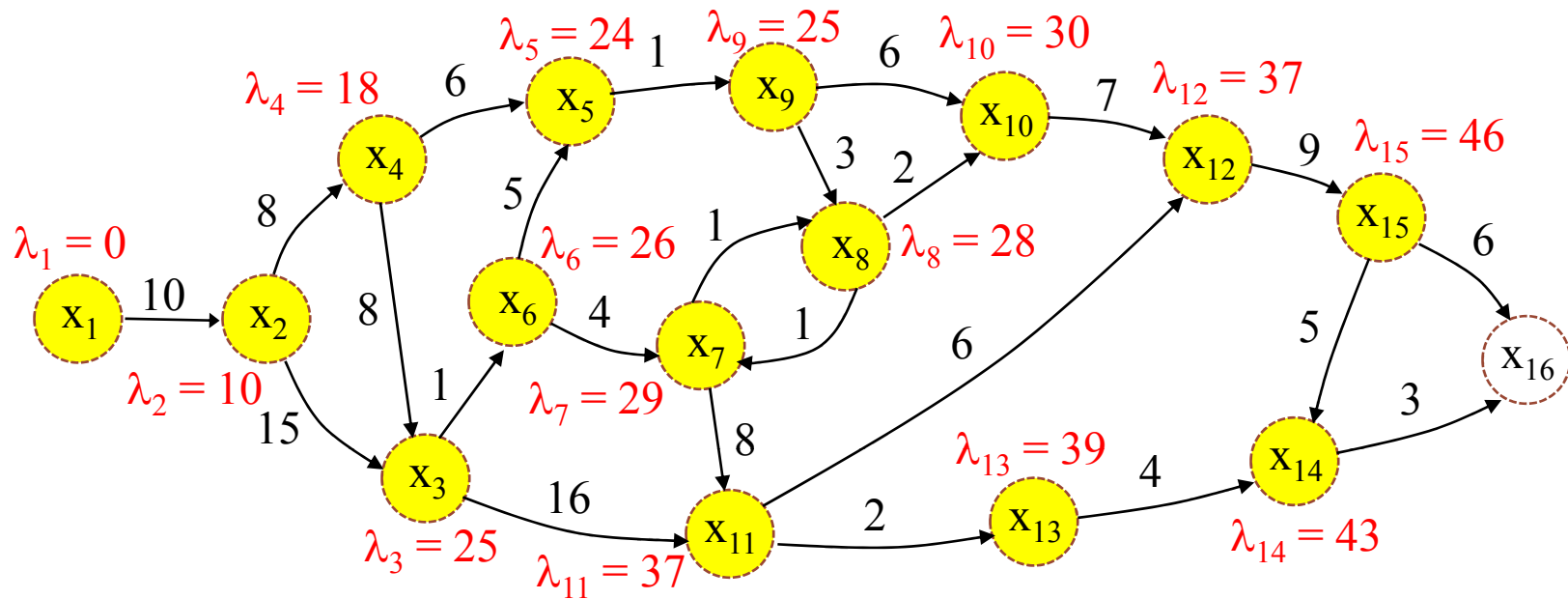
$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$$v(x_{14}, x_{16})$$



k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

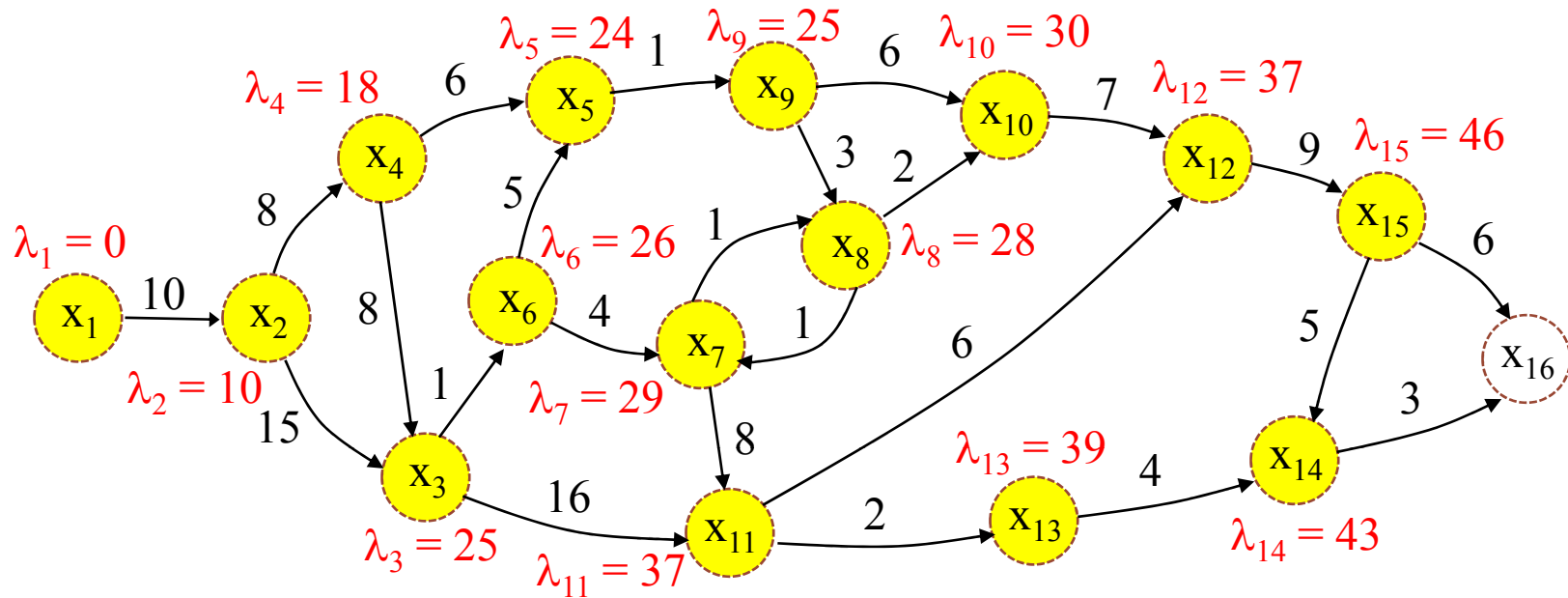
$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$$v(x_{14}, x_{16})$$

$$v(x_{15}, x_{16})$$



k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

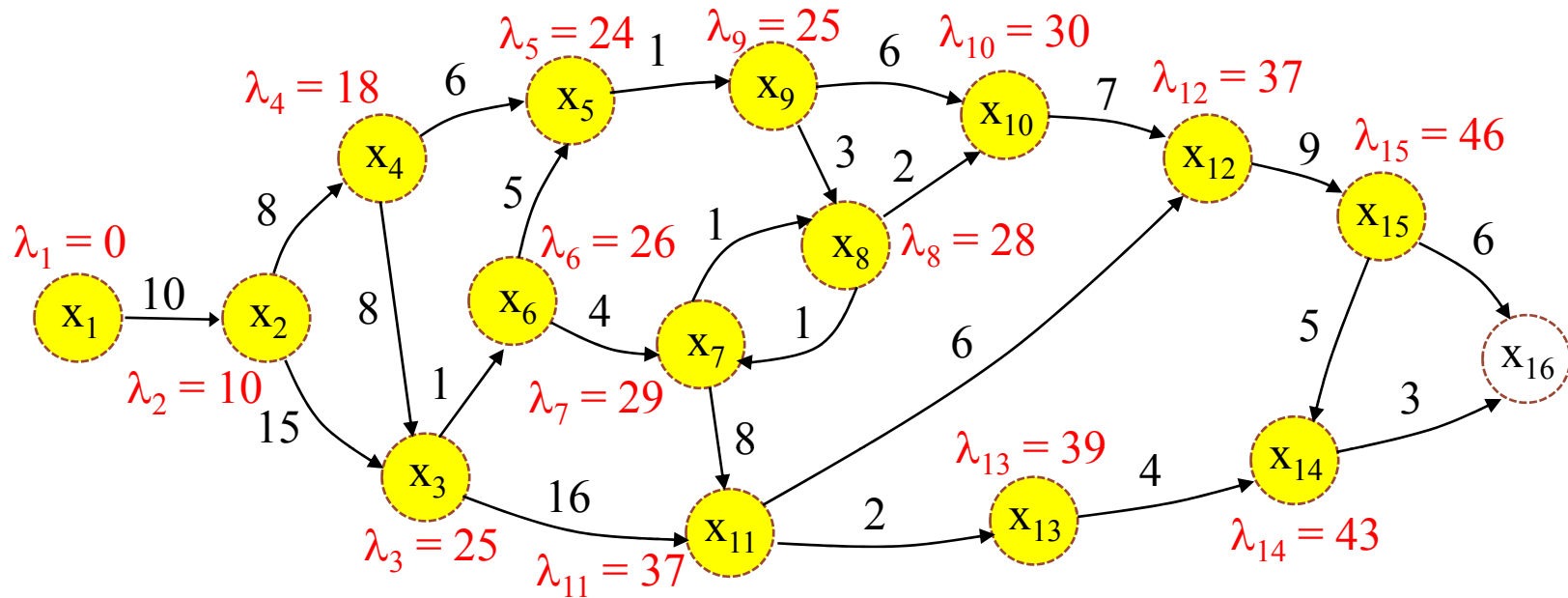
$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$v(x_{15}, x_{16})$$



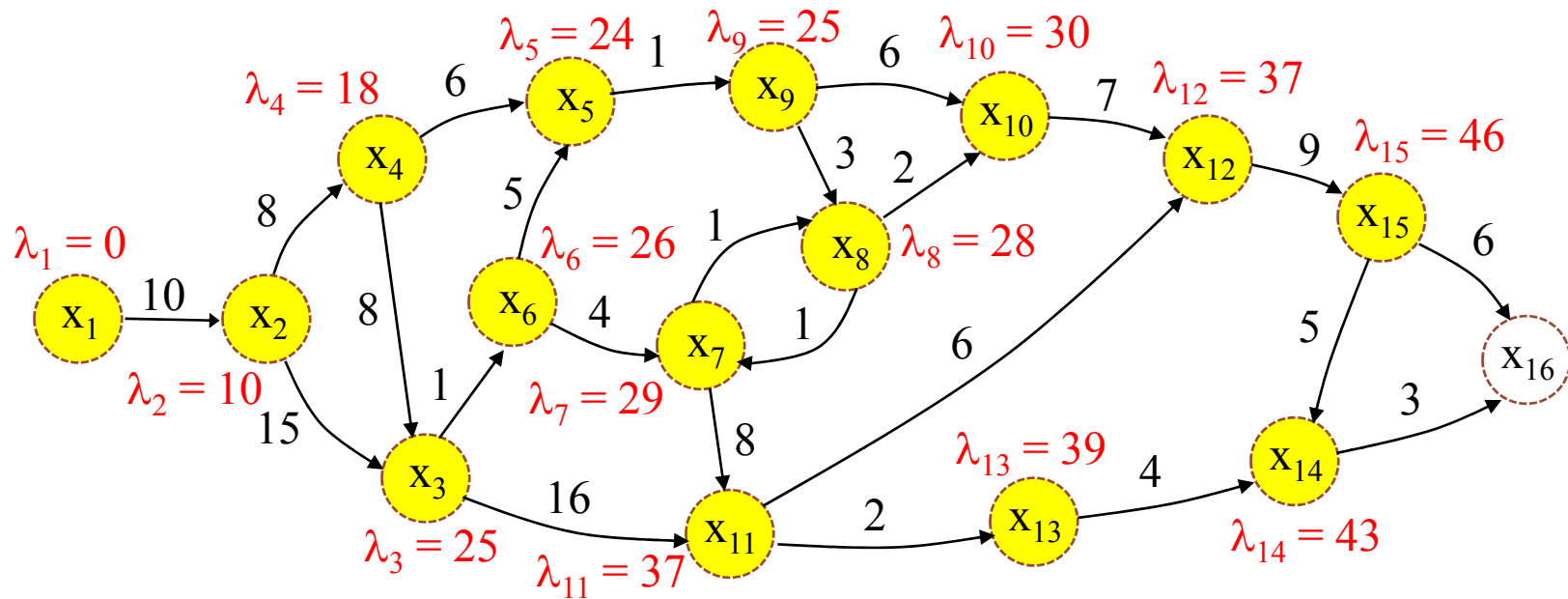
k = 14

$v(x_{12}, x_{15})$	$\lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
$v(x_{14}, x_{16})$	$\lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$v(x_{14}, x_{16})$	$\lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$
$v(x_{15}, x_{16})$	$\lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$



k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

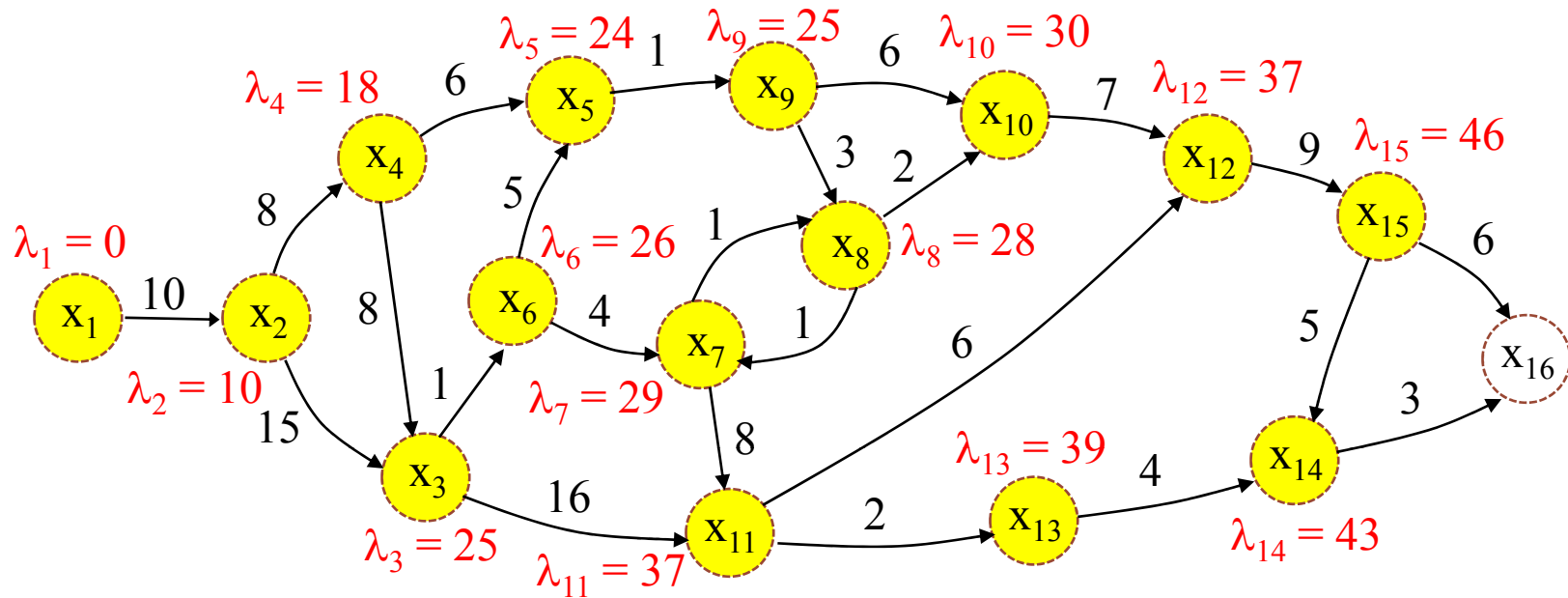
$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$v(x_{15}, x_{16}) \quad \lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$$

**k = 14**

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

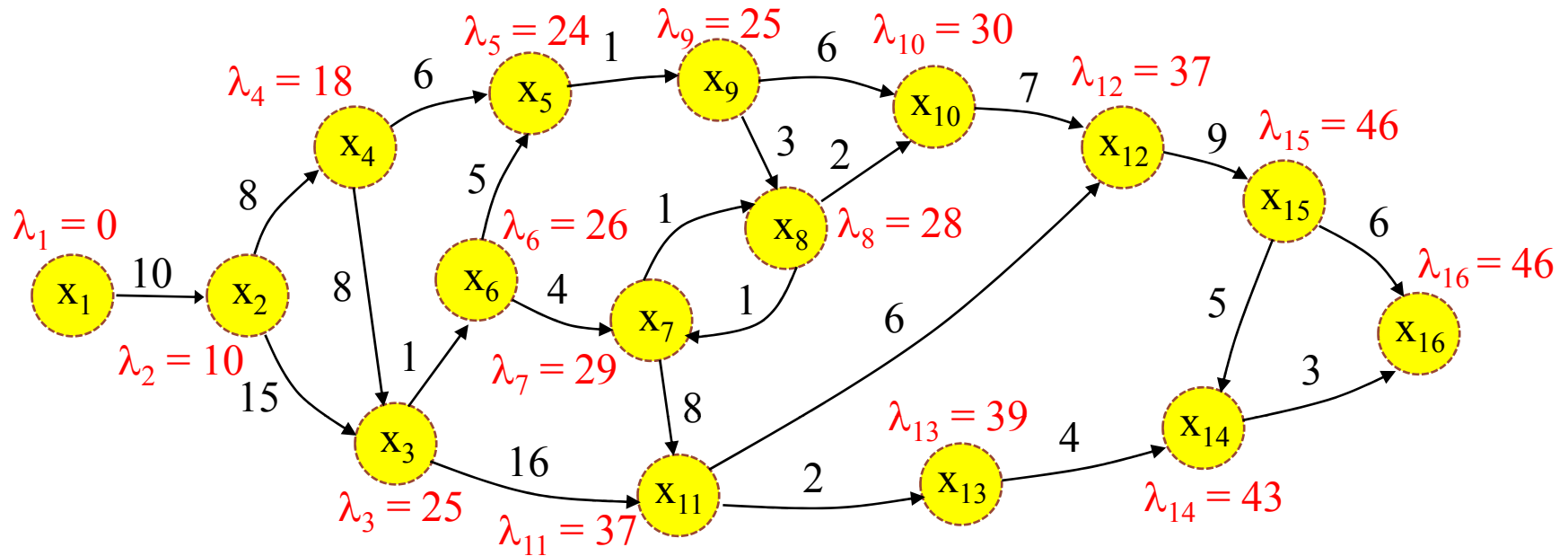
$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$v(x_{15}, x_{16}) \quad \lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$$

$$_{16} = E_{15} \cup \{x_{16}\}$$



k = 14

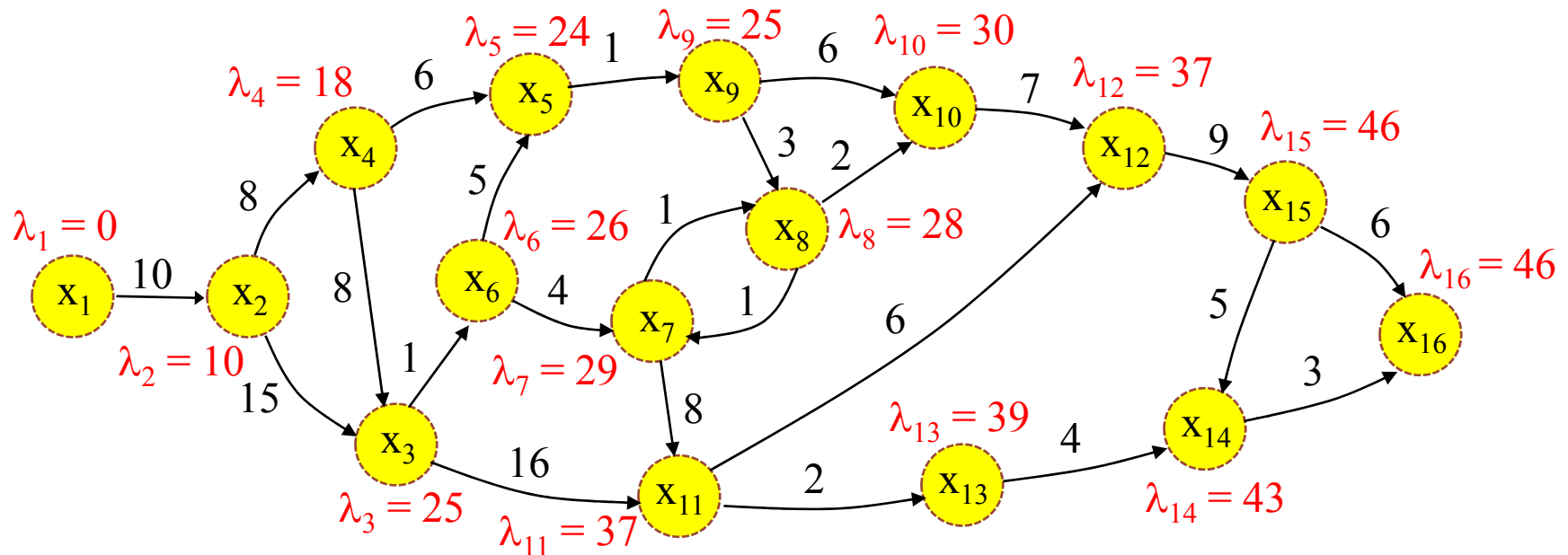
$v(x_{12}, x_{15})$	$\lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
$v(x_{14}, x_{16})$	$\lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15

$v(x_{14}, x_{16})$	$\lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$
$v(x_{15}, x_{16})$	$\lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$

$$_{16} = E_{15} \cup \{x_{16}\}$$



k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

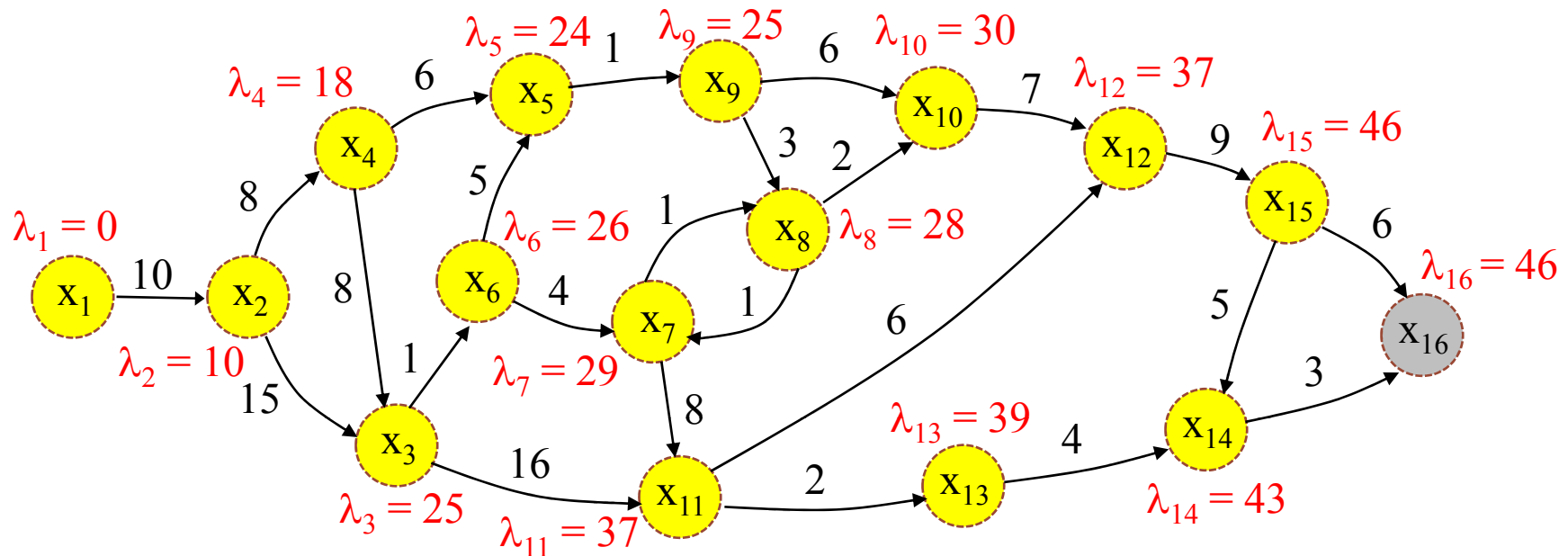
k = 15

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$v(x_{15}, x_{16}) \quad \lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$$

$$_{16} = E_{15} \cup \{x_{16}\}$$

La recherche de ce chemin minimal utilise le même principe que celle dans l'algorithme de FORD : on remonte dans le graphe, à partir de x_n , en cherchant au fur et à mesure, le ou les prédécesseurs(s) du sommet x_p , tels que $\lambda_p = \lambda_{p-1} + v(x_{p-1}, x_p)$.



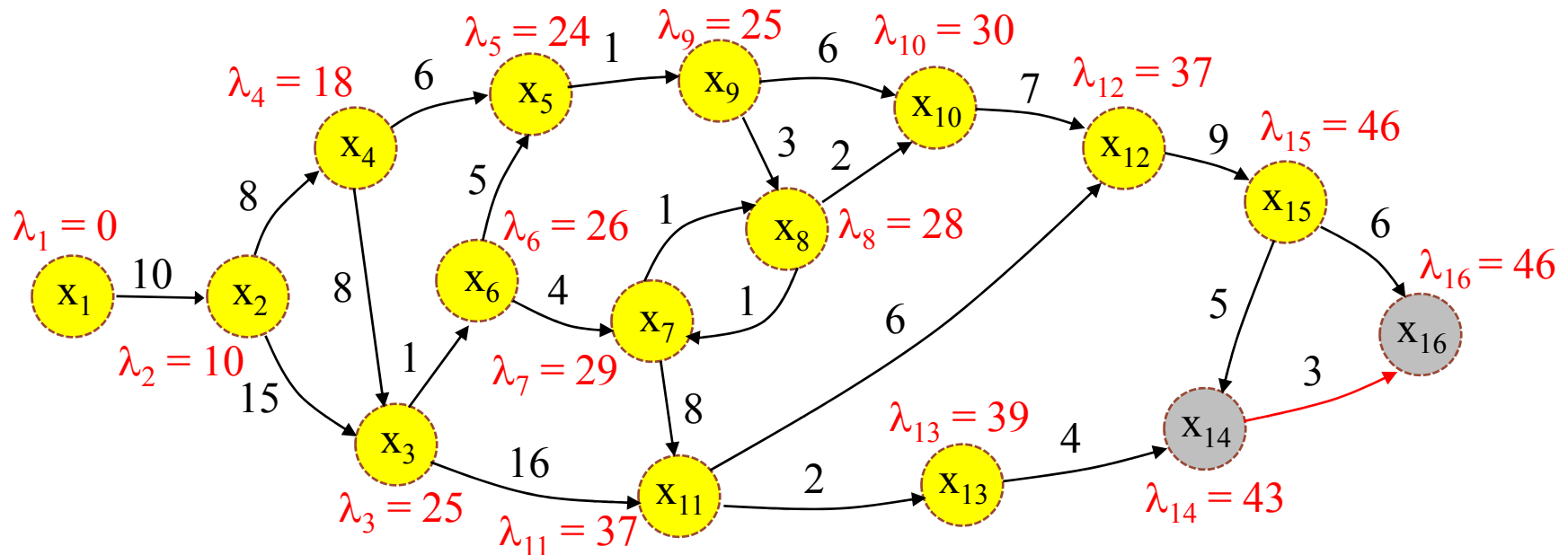
k = 14 $v(x_{12}, x_{15})$ $\lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{14}, x_{16})$ $\lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

k = 15 $v(x_{14}, x_{16})$ $\lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$
 $v(x_{15}, x_{16})$ $\lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$

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k = 14

$$v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$$

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$E_{15} = E_{14} \cup \{x_{15}\}$$

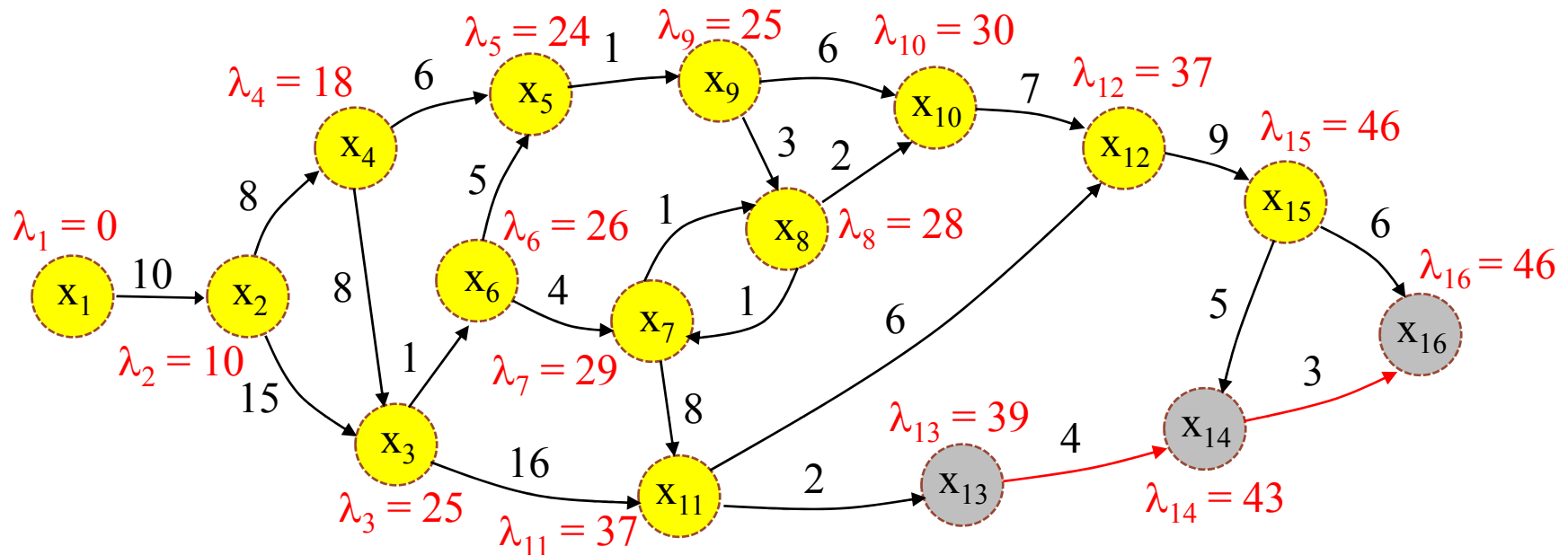
k = 15

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

$$v(x_{15}, x_{16}) \quad \lambda_{16} = \lambda_{15} + v(x_{15}, x_{16}) = 52$$

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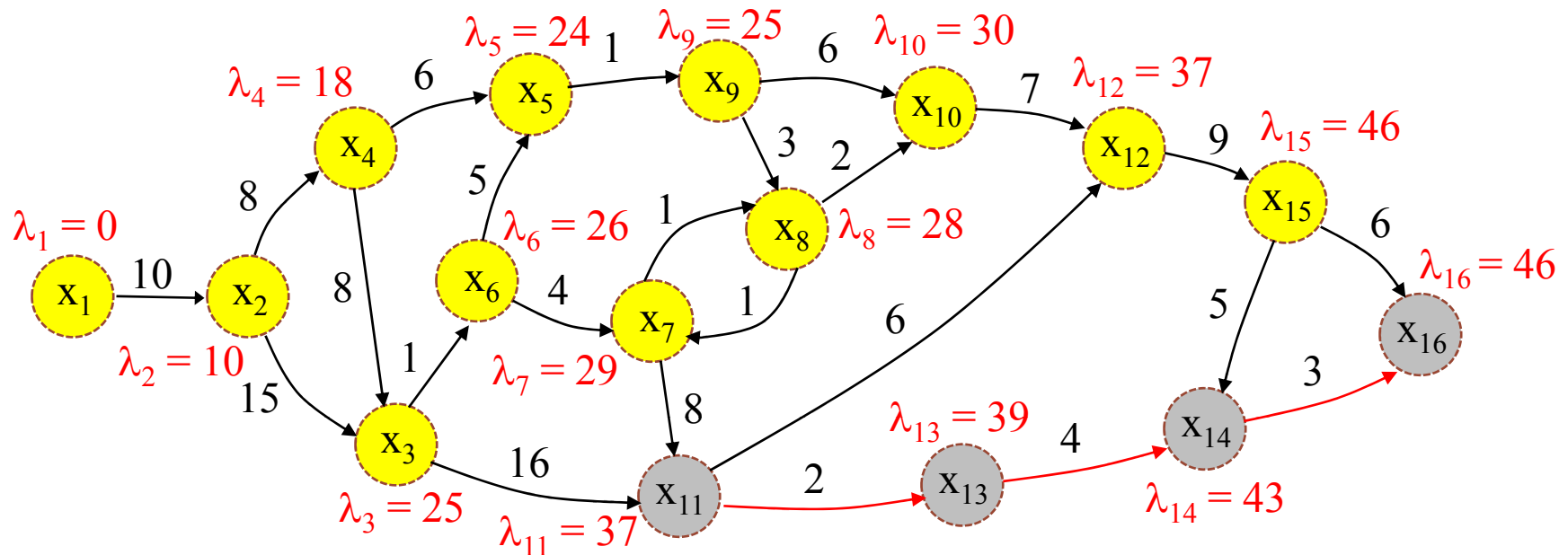
k = 14 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
 $v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$

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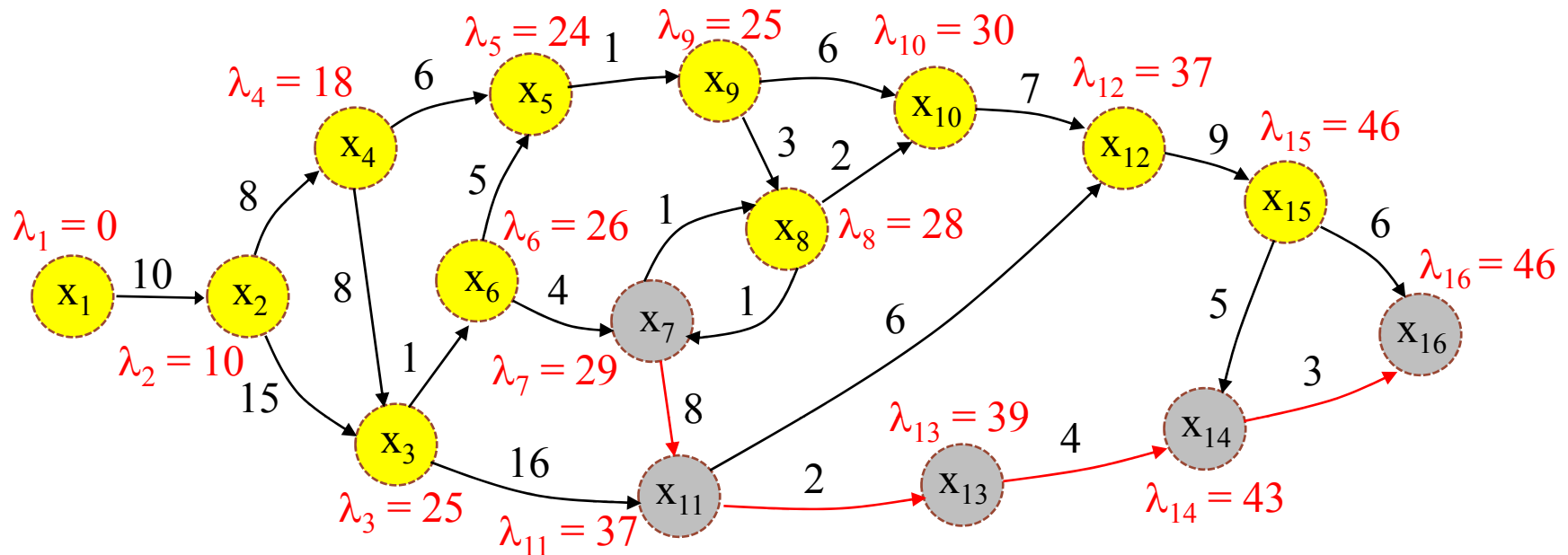
k = 15

$$v(x_{14}, x_{16}) \quad \lambda_{16} = \lambda_{14} + v(x_{14}, x_{16}) = 46$$

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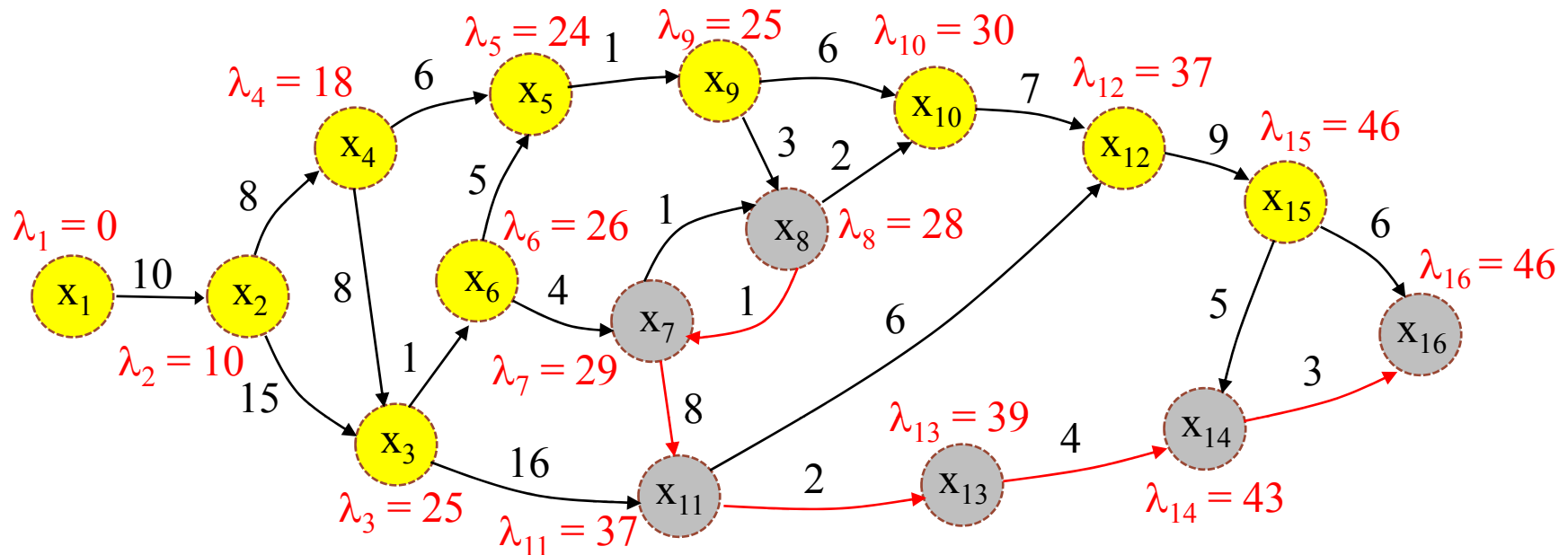
k = 14 $v(x_{12}, x_{15}) \quad \lambda_{15} = \lambda_{12} + v(x_{12}, x_{15}) = 46$
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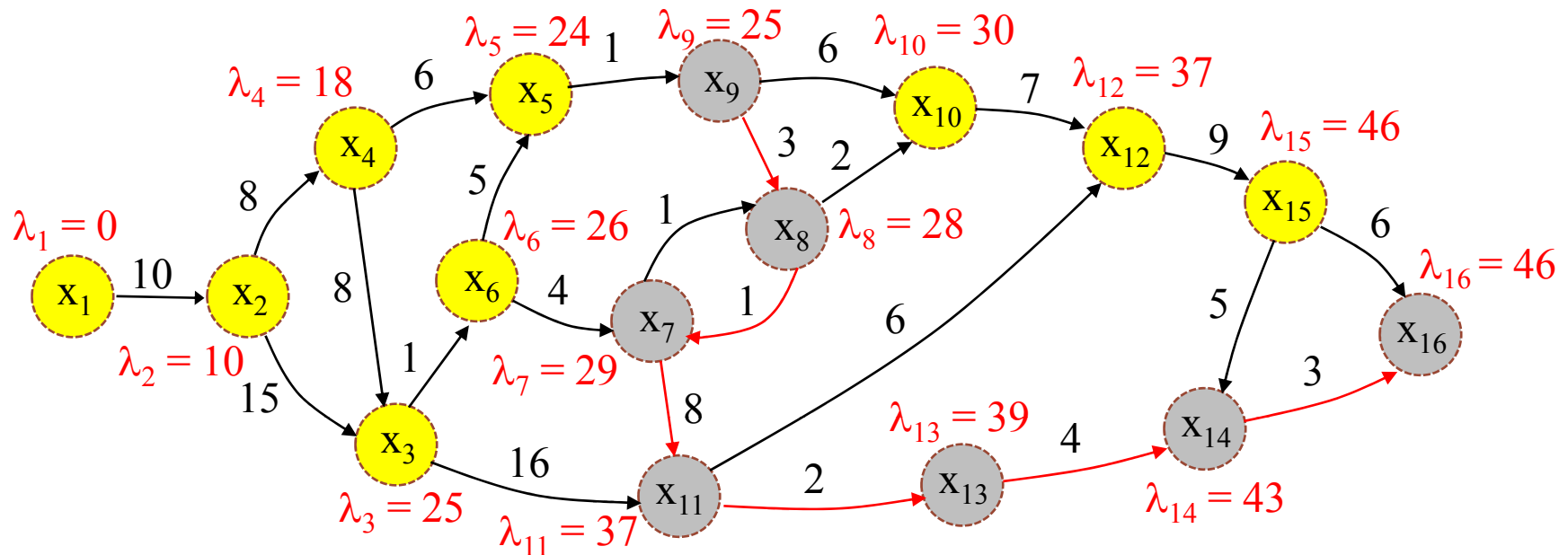
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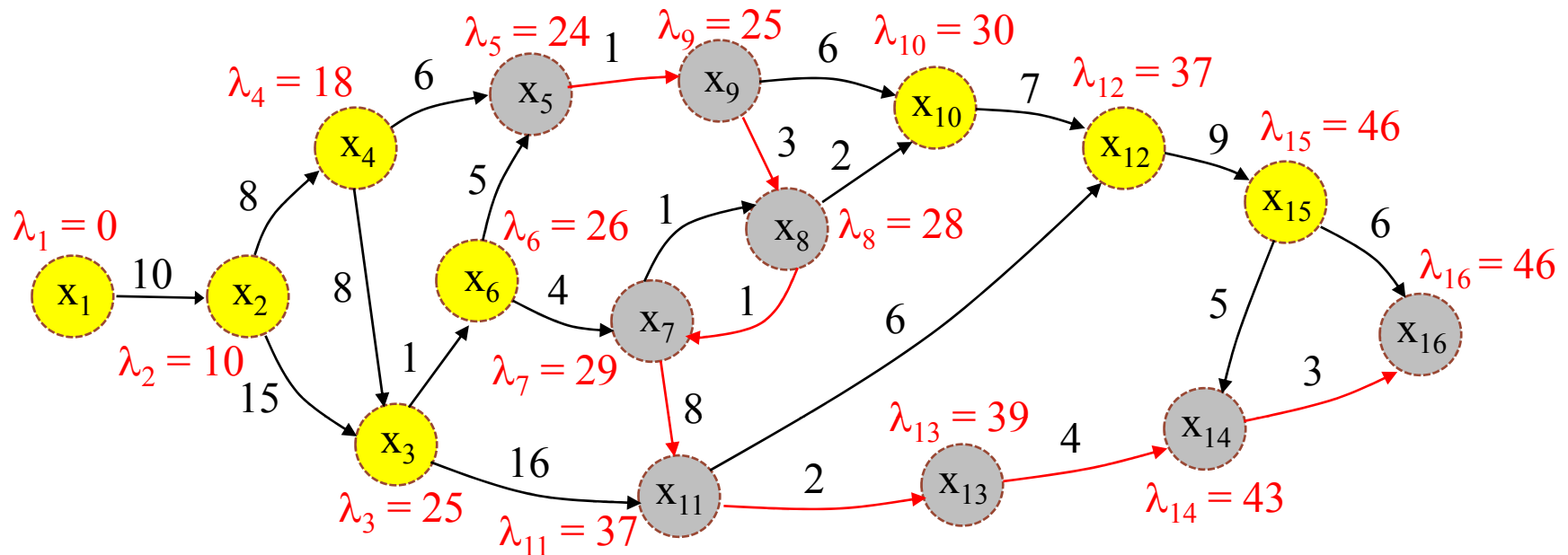
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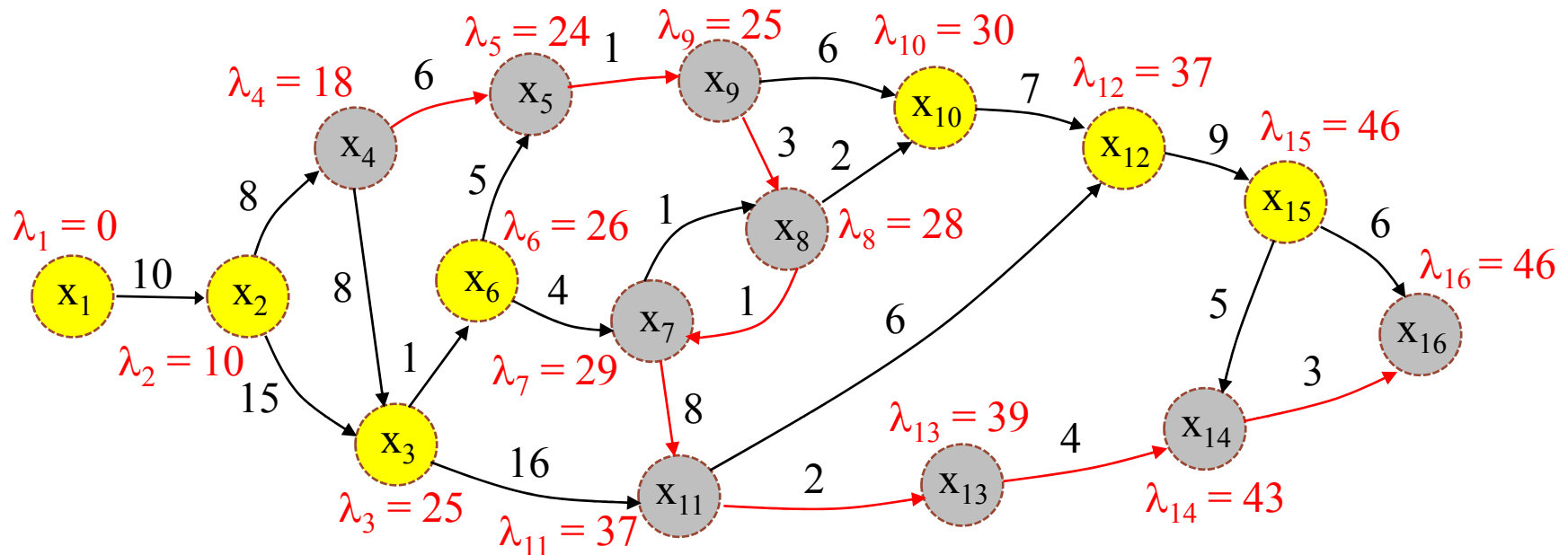
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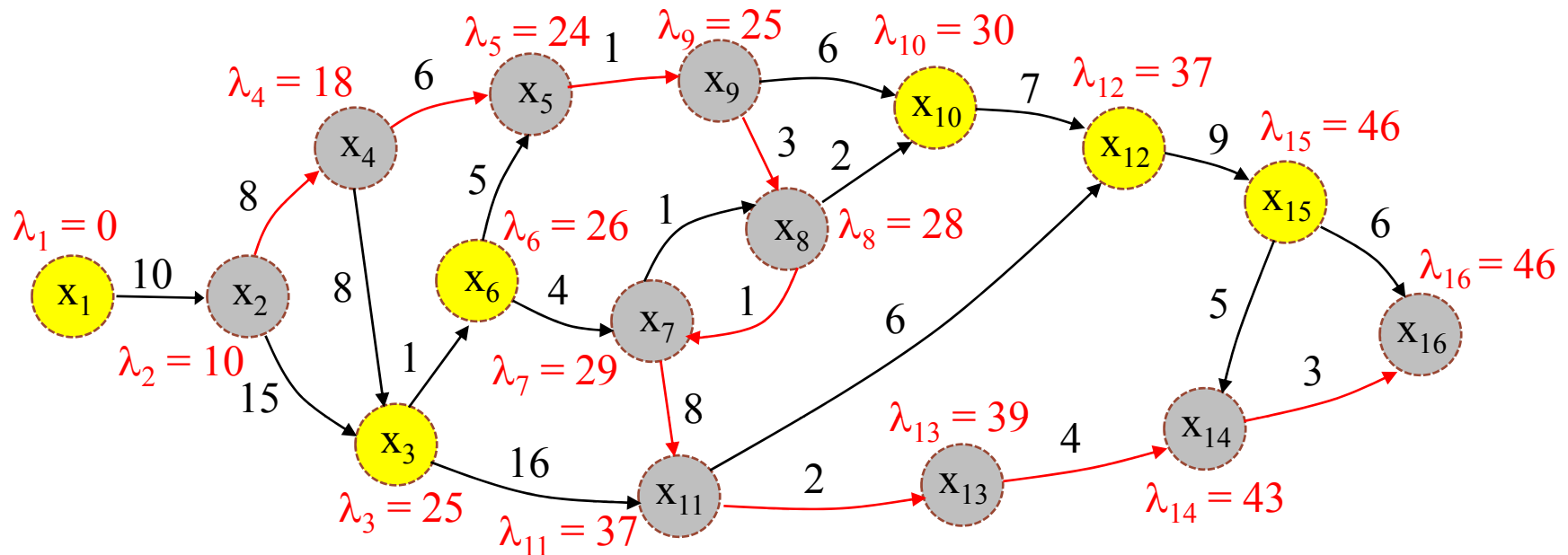
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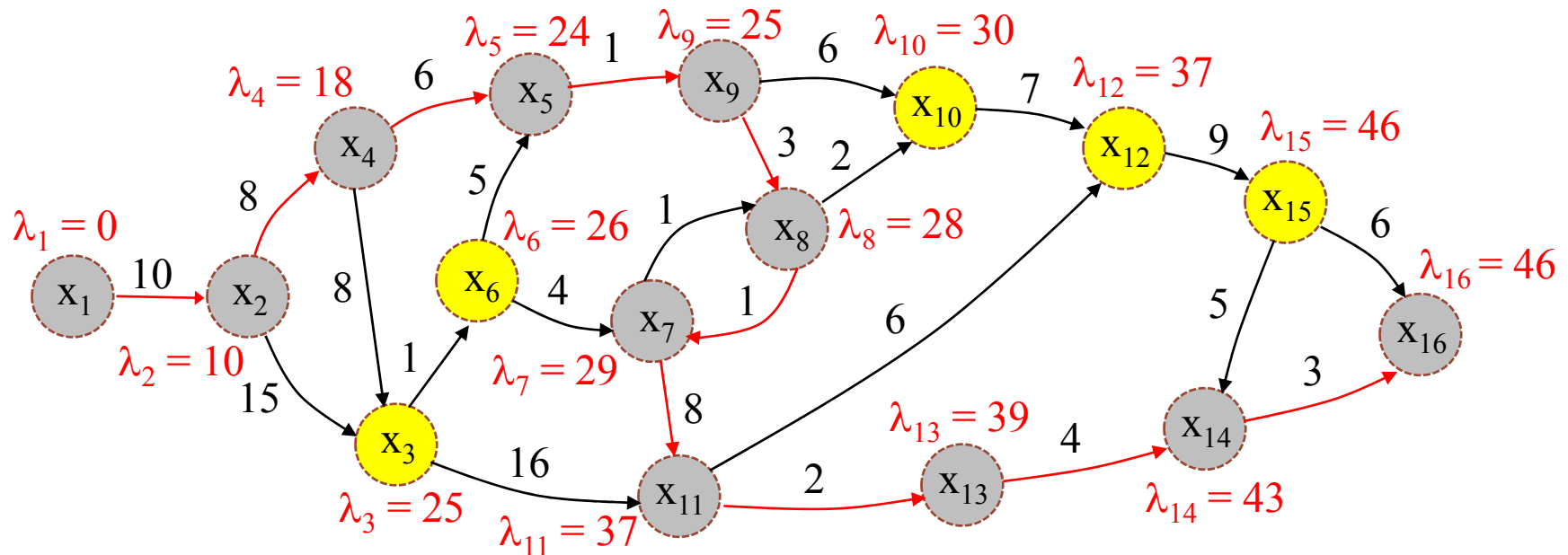
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